# Développement et mise à jour du modèle de dynamique de population du saumon de l'Allier

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# 1 Introduction

Entre 2010 et 2012, Guillaume Dauphin et Etienne PrÃ©vost (INRA – UMR ECOBIOP) ont dÃ©veloppÃ© un modÃ¨le de dynamique de population pour le saumon de l'Allier (? : voir le rapport en ligne). Le dÃ©veloppement de cet outil avait Ã©tÃ© demandÃ© de longue date par les acteurs de l'eau du bassin de la Loire et Ã©tait inscrit dans le plan de gestion des poissons migrateurs (PLAGEPOMI) du bassin de la Loire, des cÃ´tiers vendÃ©ens et de la SÃ¨vre niortaise 2009-2013 Ã  la mesure 62, intitulÃ©e « Comprendre les modalitÃ©s de renouvellement de la population : crÃ©ation d'un modÃ¨le de dynamique de populations ».
L'objectif de ce projet de modÃ©lisation est :

— amÃ©liorer la comprÃ©hension et quantifier les mÃ©canismes de renouvellement de la population de saumon de l'Allier ;
— fournir une analyse rÃ©trospective de la dynamique de population du saumon de l'allier des annÃ©es 70 Ã  nos jours ;
— Ã©valuer la capacitÃ© de la population de l'Allier Ã  se maintenir de faÃ§on autonome (sans repeuplement) ;
— identifier les conditions requises (d'ordre naturel ou anthropique) pour assurer la viabilitÃ© de la population « sauvage » de saumon atlantique dans le bassin de l'Allier.

La zone d'Ã©tude est situÃ©e sur l'Allier, des sources Ã  la station de Vichy (figure 1.1). Ce secteur est lui-mÃªme divisÃ© en 4 zones. Seuls les affluents principaux (Dore et Alagnon) sont pris en compte car nous ne disposons pas de donnÃ©es suffisantes sur les cours d'eau de moindre importance.

Depuis 2014, le travail de mise Ã  jour et de dÃ©veloppement de ce modÃ¨le se poursuit grÃ¢ce Ã  un partenariat entre l'INRAe et Logrami(? : voir le rapport en ligne ; ? : voir le rapport en ligne).

Afin que les dÃ©veloppements du modÃ¨le restent en adÃ©quation avec les besoins en connaissance des acteurs de l'eau du bassin de la Loire concernÃ©s par la gestion du saumon de l'Allier, un groupe de travail se rÃ©unit annuellement depuis 2014. Ce groupe[1] suit les avancÃ©s du projet et donne les orientations pour le travail Ã  venir (choix des thÃ©matiques).

1. 29 personnes de 21 structures diffÃ©rentes sont systÃ©matiquement invitÃ©es Ã  participer Ã  ces rÃ©unions. Les structures invitÃ©es sont : l'Agence de l'eau Loire-Bretagne, le Cnss, la Dreal de bassin Loire-Bretagne, Edf, l'Eptb Loire, la Fdaappma 03, la Fdaappma 15, la Fdaappma 42, la Fdaappma 43, la Fdaappma 48, la Fdaappma 63, l'Inra, Logrami, l'Onema, la Region Centre Val-de-Loire, le Sage Alagnon, le Sage Allier aval, le Sage Dore, le Sage Haut-Allier, le Sage Sioule et le Sigal

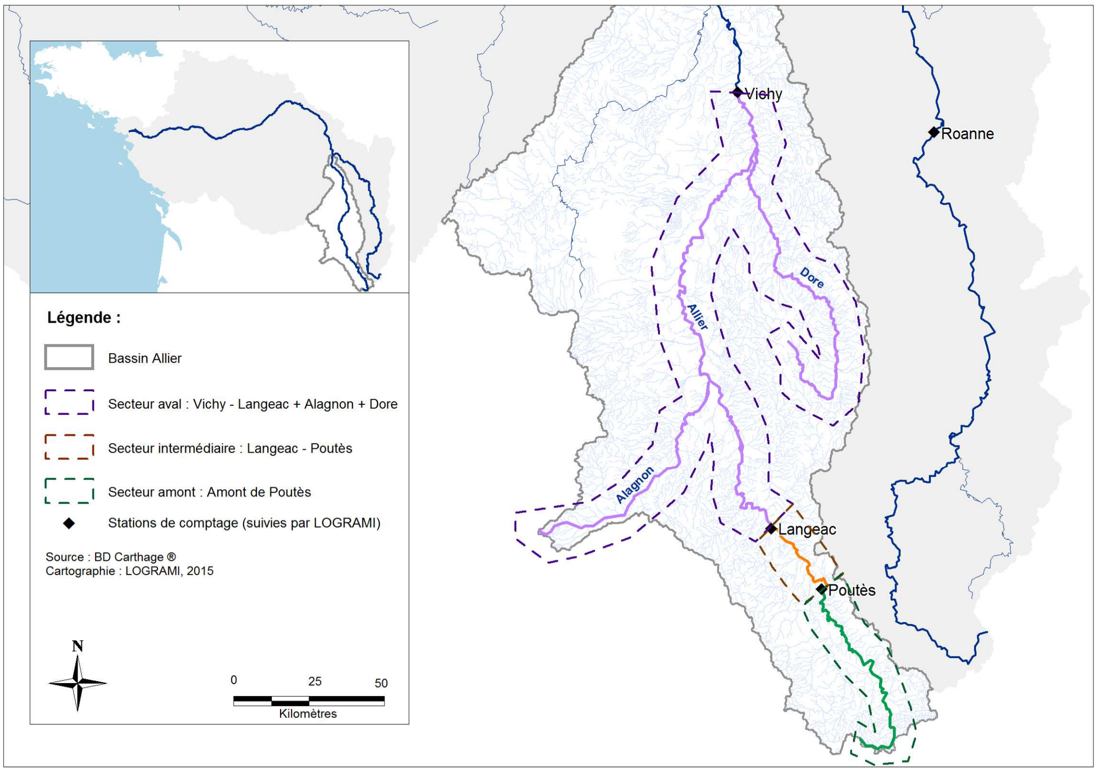


*Figure 1.1 – Localisation des diffÃ©rents secteurs du modÃ¨le*

# 2 chapitre 1

## 2.1 section 1

*Tableau 2.1*

| | |
|---|---|
| **2014** | |
| | Conversion des surfaces productives selon la formule des ERR développée sur l'Allier (Minster&Bomassi, 1999) |
| | Prise en compte plus fine des surfaces sous influence des déversements (jusqu'à l'année 2005) |
| | Mise à jour des données 2012 et 2013 |
| | Développement des projections liées au réaménagement de Poutès (50% d'amélioration et suppression de l'ouvrage) |
| **2015** | |
| | Différence de fitness entre les juvéniles issus de reproduction naturelle et les juvéniles déversés (bibliographie). Le travail n'a pas conclu sur l'utilité de modifier les hypothèses antérieures du modèle |
| | Mise à jour des données 2014 |
| | Développement d'un scénario de gestion (simulation à 20 ans) concernant la suppression des impacts à la dévalaison dans les ouvrages hydroélectriques situés dans le secteur du modèle dynamique de population |
| **2016** | |
| | Ajout d'une quatrième zone dans le modèle en extrayant l'Alagnon du secteur aval du modèle qui comprenait jusque là l'Allier en aval de Langeac + la Dore + l'Alagnon |
| | Mise à jour des données 2015 |

## 3 DensitÃ©s prÃ©dites pour un nombre de tacons 0+ capturÃ©s en 5 minutes

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | IA 5 min | Densité pédite 0+/m2 (quantiles) | | | | |
| 2 | | 5% | 25% | 50% | 75% | 95% |
| 3 | 1 | 0,002 | 0,007 | 0,012 | 0,019 | 0,036 |
| 4 | 2 | 0,004 | 0,012 | 0,018 | 0,026 | 0,046 |
| 5 | 3 | 0,007 | 0,016 | 0,023 | 0,031 | 0,053 |
| 6 | 4 | 0,010 | 0,021 | 0,029 | 0,038 | 0,060 |
| 7 | 5 | 0,014 | 0,026 | 0,034 | 0,044 | 0,069 |
| 8 | 6 | 0,017 | 0,030 | 0,039 | 0,050 | 0,076 |
| 9 | 7 | 0,021 | 0,035 | 0,045 | 0,057 | 0,083 |
| 10 | 8 | 0,024 | 0,040 | 0,050 | 0,062 | 0,090 |
| 11 | 9 | 0,028 | 0,044 | 0,056 | 0,069 | 0,097 |
| 12 | 10 | 0,032 | 0,049 | 0,061 | 0,074 | 0,103 |
| 13 | 11 | 0,035 | 0,054 | 0,066 | 0,080 | 0,111 |
| 14 | 12 | 0,039 | 0,059 | 0,072 | 0,087 | 0,118 |
| 15 | 13 | 0,044 | 0,064 | 0,077 | 0,091 | 0,127 |
| 16 | 14 | 0,049 | 0,069 | 0,082 | 0,098 | 0,132 |
| 17 | 15 | 0,052 | 0,074 | 0,087 | 0,103 | 0,138 |
| 18 | 16 | 0,055 | 0,078 | 0,093 | 0,109 | 0,146 |
| 19 | 17 | 0,059 | 0,084 | 0,099 | 0,116 | 0,153 |
| 20 | 18 | 0,063 | 0,089 | 0,105 | 0,121 | 0,159 |
| 21 | 19 | 0,068 | 0,094 | 0,110 | 0,128 | 0,167 |
| 22 | 20 | 0,074 | 0,099 | 0,115 | 0,133 | 0,172 |
| 23 | 21 | 0,078 | 0,104 | 0,121 | 0,140 | 0,180 |
| 24 | 22 | 0,082 | 0,109 | 0,125 | 0,145 | 0,186 |
| 25 | 23 | 0,087 | 0,115 | 0,132 | 0,151 | 0,192 |
| 26 | 24 | 0,091 | 0,120 | 0,137 | 0,157 | 0,199 |
| 27 | 25 | 0,095 | 0,124 | 0,142 | 0,162 | 0,206 |
| 28 | 26 | 0,097 | 0,129 | 0,147 | 0,168 | 0,212 |
| 29 | 27 | 0,103 | 0,135 | 0,154 | 0,174 | 0,220 |
| 30 | 28 | 0,107 | 0,139 | 0,158 | 0,180 | 0,227 |
| 31 | 29 | 0,111 | 0,145 | 0,165 | 0,187 | 0,232 |
| 32 | 30 | 0,117 | 0,150 | 0,170 | 0,192 | 0,240 |
| 33 | 31 | 0,121 | 0,154 | 0,175 | 0,197 | 0,244 |
| 34 | 32 | 0,124 | 0,159 | 0,180 | 0,203 | 0,251 |
| 35 | 33 | 0,129 | 0,165 | 0,186 | 0,209 | 0,258 |
| 36 | 34 | 0,133 | 0,169 | 0,190 | 0,213 | 0,261 |
| 37 | 35 | 0,138 | 0,174 | 0,196 | 0,218 | 0,266 |
| 38 | 36 | 0,140 | 0,180 | 0,202 | 0,226 | 0,278 |
| 39 | 37 | 0,147 | 0,185 | 0,207 | 0,231 | 0,281 |
| 40 | 38 | 0,152 | 0,192 | 0,214 | 0,237 | 0,288 |
| 41 | 39 | 0,155 | 0,195 | 0,218 | 0,243 | 0,294 |
| 42 | 40 | 0,161 | 0,200 | 0,223 | 0,248 | 0,302 |
| 43 | 41 | 0,166 | 0,206 | 0,228 | 0,253 | 0,309 |
| 44 | 42 | 0,169 | 0,209 | 0,233 | 0,259 | 0,313 |
| 45 | 43 | 0,174 | 0,215 | 0,240 | 0,266 | 0,323 |
| 46 | 44 | 0,181 | 0,221 | 0,245 | 0,270 | 0,326 |
| 47 | 45 | 0,186 | 0,227 | 0,250 | 0,276 | 0,332 |
| 48 | 46 | 0,188 | 0,231 | 0,257 | 0,284 | 0,343 |
| 49 | 47 | 0,194 | 0,238 | 0,263 | 0,290 | 0,345 |

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | IA 5 min | Densité pédite 0+/m2 (quantiles) | | | | |
| 2 | | 5% | 25% | 50% | 75% | 95% |
| 50 | 48 | 0,198 | 0,241 | 0,267 | 0,294 | 0,350 |
| 51 | 49 | 0,203 | 0,247 | 0,274 | 0,301 | 0,359 |
| 52 | 50 | 0,206 | 0,251 | 0,278 | 0,305 | 0,365 |
| 53 | 51 | 0,213 | 0,257 | 0,284 | 0,312 | 0,374 |
| 54 | 52 | 0,215 | 0,262 | 0,289 | 0,319 | 0,375 |
| 55 | 53 | 0,223 | 0,268 | 0,295 | 0,323 | 0,384 |
| 56 | 54 | 0,227 | 0,273 | 0,300 | 0,329 | 0,390 |
| 57 | 55 | 0,230 | 0,277 | 0,306 | 0,334 | 0,396 |
| 58 | 56 | 0,235 | 0,282 | 0,309 | 0,339 | 0,401 |
| 59 | 57 | 0,238 | 0,287 | 0,316 | 0,346 | 0,409 |
| 60 | 58 | 0,246 | 0,294 | 0,322 | 0,352 | 0,412 |
| 61 | 59 | 0,246 | 0,297 | 0,327 | 0,357 | 0,421 |
| 62 | 60 | 0,251 | 0,304 | 0,332 | 0,363 | 0,427 |
| 63 | 61 | 0,260 | 0,310 | 0,338 | 0,369 | 0,431 |
| 64 | 62 | 0,264 | 0,315 | 0,343 | 0,373 | 0,439 |
| 65 | 63 | 0,269 | 0,318 | 0,347 | 0,378 | 0,444 |
| 66 | 64 | 0,274 | 0,325 | 0,354 | 0,386 | 0,453 |
| 67 | 65 | 0,276 | 0,329 | 0,359 | 0,391 | 0,458 |
| 68 | 66 | 0,279 | 0,334 | 0,364 | 0,395 | 0,461 |
| 69 | 67 | 0,287 | 0,340 | 0,371 | 0,403 | 0,467 |
| 70 | 68 | 0,292 | 0,344 | 0,376 | 0,409 | 0,476 |
| 71 | 69 | 0,294 | 0,350 | 0,382 | 0,415 | 0,484 |
| 72 | 70 | 0,300 | 0,356 | 0,387 | 0,419 | 0,489 |
| 73 | 71 | 0,305 | 0,360 | 0,391 | 0,425 | 0,491 |
| 74 | 72 | 0,311 | 0,364 | 0,396 | 0,430 | 0,499 |
| 75 | 73 | 0,315 | 0,372 | 0,403 | 0,438 | 0,508 |
| 76 | 74 | 0,320 | 0,375 | 0,408 | 0,442 | 0,512 |
| 77 | 75 | 0,326 | 0,381 | 0,413 | 0,447 | 0,521 |
| 78 | 76 | 0,330 | 0,387 | 0,419 | 0,453 | 0,528 |
| 79 | 77 | 0,334 | 0,391 | 0,424 | 0,458 | 0,528 |
| 80 | 78 | 0,338 | 0,397 | 0,430 | 0,465 | 0,535 |
| 81 | 79 | 0,343 | 0,401 | 0,434 | 0,471 | 0,544 |
| 82 | 80 | 0,348 | 0,407 | 0,441 | 0,476 | 0,554 |
| 83 | 81 | 0,355 | 0,411 | 0,445 | 0,482 | 0,556 |
| 84 | 82 | 0,358 | 0,419 | 0,451 | 0,487 | 0,560 |
| 85 | 83 | 0,361 | 0,422 | 0,457 | 0,494 | 0,569 |
| 86 | 84 | 0,362 | 0,427 | 0,461 | 0,498 | 0,574 |
| 87 | 85 | 0,372 | 0,434 | 0,469 | 0,505 | 0,581 |
| 88 | 86 | 0,377 | 0,438 | 0,472 | 0,508 | 0,584 |
| 89 | 87 | 0,380 | 0,443 | 0,477 | 0,515 | 0,591 |
| 90 | 88 | 0,385 | 0,448 | 0,482 | 0,520 | 0,598 |
| 91 | 89 | 0,391 | 0,453 | 0,489 | 0,527 | 0,603 |
| 92 | 90 | 0,397 | 0,460 | 0,496 | 0,533 | 0,612 |
| 93 | 91 | 0,400 | 0,464 | 0,500 | 0,537 | 0,616 |
| 94 | 92 | 0,406 | 0,470 | 0,506 | 0,545 | 0,623 |
| 95 | 93 | 0,408 | 0,474 | 0,510 | 0,550 | 0,626 |
| 96 | 94 | 0,415 | 0,480 | 0,518 | 0,557 | 0,633 |

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | IA 5 min | Densité pédite 0+/m2 (quantiles) | | | | |
| 2 | | 5% | 25% | 50% | 75% | 95% |
| 97 | 95 | 0,420 | 0,486 | 0,524 | 0,562 | 0,643 |
| 98 | 96 | 0,426 | 0,490 | 0,526 | 0,566 | 0,643 |
| 99 | 97 | 0,428 | 0,496 | 0,534 | 0,574 | 0,650 |
| 100 | 98 | 0,432 | 0,499 | 0,538 | 0,577 | 0,658 |
| 101 | 99 | 0,439 | 0,505 | 0,542 | 0,583 | 0,661 |
| 102 | 100 | 0,445 | 0,512 | 0,550 | 0,589 | 0,671 |
| 103 | 101 | 0,451 | 0,515 | 0,554 | 0,596 | 0,680 |
| 104 | 102 | 0,451 | 0,521 | 0,560 | 0,600 | 0,684 |
| 105 | 103 | 0,457 | 0,526 | 0,565 | 0,607 | 0,693 |
| 106 | 104 | 0,465 | 0,532 | 0,570 | 0,612 | 0,700 |
| 107 | 105 | 0,468 | 0,536 | 0,575 | 0,614 | 0,701 |
| 108 | 106 | 0,473 | 0,542 | 0,582 | 0,622 | 0,707 |
| 109 | 107 | 0,476 | 0,547 | 0,588 | 0,629 | 0,718 |
| 110 | 108 | 0,480 | 0,552 | 0,592 | 0,634 | 0,718 |
| 111 | 109 | 0,488 | 0,559 | 0,599 | 0,641 | 0,728 |
| 112 | 110 | 0,495 | 0,563 | 0,602 | 0,645 | 0,730 |
| 113 | 111 | 0,500 | 0,566 | 0,608 | 0,650 | 0,734 |
| 114 | 112 | 0,501 | 0,573 | 0,614 | 0,656 | 0,744 |
| 115 | 113 | 0,506 | 0,578 | 0,620 | 0,662 | 0,749 |
| 116 | 114 | 0,511 | 0,584 | 0,624 | 0,667 | 0,755 |
| 117 | 115 | 0,514 | 0,589 | 0,630 | 0,672 | 0,757 |
| 118 | 116 | 0,518 | 0,594 | 0,635 | 0,680 | 0,773 |
| 119 | 117 | 0,526 | 0,599 | 0,638 | 0,681 | 0,770 |
| 120 | 118 | 0,532 | 0,604 | 0,647 | 0,692 | 0,782 |
| 121 | 119 | 0,537 | 0,611 | 0,651 | 0,696 | 0,781 |
| 122 | 120 | 0,539 | 0,614 | 0,657 | 0,701 | 0,791 |
| 123 | 121 | 0,547 | 0,619 | 0,663 | 0,708 | 0,796 |
| 124 | 122 | 0,553 | 0,626 | 0,669 | 0,714 | 0,802 |
| 125 | 123 | 0,551 | 0,628 | 0,670 | 0,716 | 0,808 |
| 126 | 124 | 0,560 | 0,636 | 0,678 | 0,724 | 0,819 |
| 127 | 125 | 0,562 | 0,643 | 0,686 | 0,732 | 0,822 |
| 128 | 126 | 0,570 | 0,646 | 0,689 | 0,734 | 0,825 |
| 129 | 127 | 0,577 | 0,652 | 0,697 | 0,743 | 0,838 |
| 130 | 128 | 0,577 | 0,655 | 0,699 | 0,744 | 0,839 |
| 131 | 129 | 0,584 | 0,662 | 0,706 | 0,753 | 0,849 |
| 132 | 130 | 0,589 | 0,667 | 0,713 | 0,757 | 0,848 |
| 133 | 131 | 0,597 | 0,673 | 0,718 | 0,764 | 0,854 |
| 134 | 132 | 0,596 | 0,677 | 0,721 | 0,770 | 0,864 |
| 135 | 133 | 0,605 | 0,682 | 0,726 | 0,774 | 0,868 |
| 136 | 134 | 0,609 | 0,688 | 0,734 | 0,780 | 0,883 |
| 137 | 135 | 0,613 | 0,693 | 0,737 | 0,783 | 0,878 |
| 138 | 136 | 0,622 | 0,697 | 0,743 | 0,791 | 0,890 |
| 139 | 137 | 0,624 | 0,701 | 0,747 | 0,793 | 0,895 |
| 140 | 138 | 0,630 | 0,710 | 0,755 | 0,802 | 0,905 |
| 141 | 139 | 0,636 | 0,713 | 0,759 | 0,810 | 0,911 |
| 142 | 140 | 0,640 | 0,720 | 0,765 | 0,812 | 0,911 |
| 143 | 141 | 0,646 | 0,725 | 0,772 | 0,819 | 0,916 |

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | IA 5 min | Densité pédite 0+/m2 (quantiles) | | | | |
| 2 | | 5% | 25% | 50% | 75% | 95% |
| 144 | 142 | 0,645 | 0,730 | 0,775 | 0,823 | 0,924 |
| 145 | 143 | 0,654 | 0,736 | 0,782 | 0,831 | 0,931 |
| 146 | 144 | 0,657 | 0,742 | 0,787 | 0,837 | 0,935 |
| 147 | 145 | 0,663 | 0,747 | 0,795 | 0,842 | 0,942 |
| 148 | 146 | 0,664 | 0,750 | 0,798 | 0,848 | 0,946 |
| 149 | 147 | 0,673 | 0,755 | 0,802 | 0,853 | 0,956 |
| 150 | 148 | 0,681 | 0,762 | 0,810 | 0,859 | 0,963 |
| 151 | 149 | 0,682 | 0,768 | 0,814 | 0,862 | 0,966 |
| 152 | 150 | 0,684 | 0,770 | 0,820 | 0,870 | 0,969 |
| 153 | 151 | 0,690 | 0,777 | 0,824 | 0,872 | 0,982 |
| 154 | 152 | 0,696 | 0,783 | 0,831 | 0,883 | 0,981 |
| 155 | 153 | 0,704 | 0,788 | 0,838 | 0,887 | 0,986 |
| 156 | 154 | 0,706 | 0,796 | 0,844 | 0,893 | 0,998 |
| 157 | 155 | 0,712 | 0,799 | 0,846 | 0,898 | 1,002 |
| 158 | 156 | 0,722 | 0,805 | 0,854 | 0,906 | 1,012 |
| 159 | 157 | 0,718 | 0,809 | 0,857 | 0,908 | 1,017 |
| 160 | 158 | 0,724 | 0,814 | 0,865 | 0,917 | 1,023 |
| 161 | 159 | 0,730 | 0,820 | 0,870 | 0,921 | 1,031 |
| 162 | 160 | 0,736 | 0,824 | 0,874 | 0,927 | 1,035 |
| 163 | 161 | 0,743 | 0,829 | 0,878 | 0,932 | 1,041 |
| 164 | 162 | 0,747 | 0,833 | 0,883 | 0,937 | 1,047 |
| 165 | 163 | 0,752 | 0,840 | 0,889 | 0,942 | 1,052 |
| 166 | 164 | 0,757 | 0,846 | 0,896 | 0,949 | 1,056 |
| 167 | 165 | 0,760 | 0,850 | 0,901 | 0,953 | 1,060 |
| 168 | 166 | 0,763 | 0,856 | 0,906 | 0,959 | 1,075 |
| 169 | 167 | 0,770 | 0,863 | 0,912 | 0,963 | 1,072 |
| 170 | 168 | 0,771 | 0,866 | 0,916 | 0,973 | 1,084 |
| 171 | 169 | 0,782 | 0,872 | 0,925 | 0,980 | 1,087 |
| 172 | 170 | 0,782 | 0,878 | 0,928 | 0,982 | 1,089 |
| 173 | 171 | 0,791 | 0,881 | 0,934 | 0,988 | 1,102 |
| 174 | 172 | 0,793 | 0,887 | 0,938 | 0,993 | 1,105 |
| 175 | 173 | 0,801 | 0,892 | 0,944 | 1,000 | 1,112 |
| 176 | 174 | 0,808 | 0,898 | 0,950 | 1,005 | 1,113 |
| 177 | 175 | 0,810 | 0,904 | 0,953 | 1,009 | 1,116 |
| 178 | 176 | 0,818 | 0,911 | 0,963 | 1,017 | 1,129 |
| 179 | 177 | 0,815 | 0,913 | 0,966 | 1,021 | 1,135 |
| 180 | 178 | 0,828 | 0,918 | 0,970 | 1,028 | 1,143 |
| 181 | 179 | 0,824 | 0,923 | 0,979 | 1,033 | 1,143 |
| 182 | 180 | 0,834 | 0,927 | 0,982 | 1,038 | 1,158 |
| 183 | 181 | 0,841 | 0,934 | 0,987 | 1,043 | 1,160 |
| 184 | 182 | 0,841 | 0,939 | 0,994 | 1,051 | 1,159 |
| 185 | 183 | 0,848 | 0,946 | 0,999 | 1,057 | 1,173 |
| 186 | 184 | 0,857 | 0,951 | 1,003 | 1,060 | 1,173 |
| 187 | 185 | 0,860 | 0,957 | 1,010 | 1,070 | 1,182 |
| 188 | 186 | 0,864 | 0,961 | 1,014 | 1,073 | 1,192 |
| 189 | 187 | 0,871 | 0,966 | 1,021 | 1,079 | 1,195 |
| 190 | 188 | 0,873 | 0,970 | 1,024 | 1,084 | 1,197 |

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | IA 5 min | Densité pédite 0+/m2 (quantiles) | | | | |
| 2 | | 5% | 25% | 50% | 75% | 95% |
| 191 | 189 | 0,877 | 0,976 | 1,030 | 1,088 | 1,199 |
| 192 | 190 | 0,884 | 0,982 | 1,038 | 1,094 | 1,211 |
| 193 | 191 | 0,892 | 0,987 | 1,042 | 1,099 | 1,219 |
| 194 | 192 | 0,895 | 0,990 | 1,046 | 1,104 | 1,225 |
| 195 | 193 | 0,897 | 0,999 | 1,053 | 1,111 | 1,232 |
| 196 | 194 | 0,906 | 1,005 | 1,059 | 1,115 | 1,237 |
| 197 | 195 | 0,902 | 1,007 | 1,066 | 1,123 | 1,248 |
| 198 | 196 | 0,911 | 1,013 | 1,070 | 1,128 | 1,254 |
| 199 | 197 | 0,919 | 1,019 | 1,074 | 1,134 | 1,254 |
| 200 | 198 | 0,923 | 1,020 | 1,078 | 1,138 | 1,261 |
| 201 | 199 | 0,925 | 1,028 | 1,083 | 1,142 | 1,267 |
| 202 | 200 | 0,934 | 1,032 | 1,088 | 1,148 | 1,275 |
| 203 | 201 | 0,940 | 1,041 | 1,098 | 1,158 | 1,285 |
| 204 | 202 | 0,942 | 1,041 | 1,102 | 1,162 | 1,280 |
| 205 | 203 | 0,952 | 1,049 | 1,107 | 1,170 | 1,291 |
| 206 | 204 | 0,953 | 1,055 | 1,111 | 1,172 | 1,294 |
| 207 | 205 | 0,953 | 1,060 | 1,118 | 1,178 | 1,299 |
| 208 | 206 | 0,965 | 1,064 | 1,121 | 1,184 | 1,300 |
| 209 | 207 | 0,967 | 1,070 | 1,130 | 1,191 | 1,308 |
| 210 | 208 | 0,974 | 1,077 | 1,135 | 1,195 | 1,319 |
| 211 | 209 | 0,975 | 1,081 | 1,138 | 1,201 | 1,324 |
| 212 | 210 | 0,980 | 1,087 | 1,146 | 1,206 | 1,338 |
| 213 | 211 | 0,987 | 1,091 | 1,151 | 1,212 | 1,333 |
| 214 | 212 | 0,988 | 1,096 | 1,154 | 1,215 | 1,338 |
| 215 | 213 | 0,999 | 1,103 | 1,161 | 1,224 | 1,348 |
| 216 | 214 | 0,999 | 1,107 | 1,167 | 1,229 | 1,353 |
| 217 | 215 | 1,006 | 1,114 | 1,172 | 1,232 | 1,359 |
| 218 | 216 | 1,009 | 1,116 | 1,176 | 1,241 | 1,364 |
| 219 | 217 | 1,018 | 1,124 | 1,185 | 1,247 | 1,374 |
| 220 | 218 | 1,018 | 1,130 | 1,188 | 1,248 | 1,375 |
| 221 | 219 | 1,025 | 1,134 | 1,195 | 1,258 | 1,384 |
| 222 | 220 | 1,032 | 1,138 | 1,202 | 1,261 | 1,391 |
| 223 | 221 | 1,035 | 1,142 | 1,205 | 1,267 | 1,398 |
| 224 | 222 | 1,037 | 1,147 | 1,209 | 1,274 | 1,398 |
| 225 | 223 | 1,049 | 1,155 | 1,215 | 1,280 | 1,411 |
| 226 | 224 | 1,050 | 1,160 | 1,221 | 1,283 | 1,409 |
| 227 | 225 | 1,054 | 1,166 | 1,226 | 1,291 | 1,421 |
| 228 | 226 | 1,058 | 1,170 | 1,233 | 1,294 | 1,419 |
| 229 | 227 | 1,067 | 1,174 | 1,234 | 1,297 | 1,436 |
| 230 | 228 | 1,070 | 1,182 | 1,244 | 1,309 | 1,438 |
| 231 | 229 | 1,073 | 1,185 | 1,247 | 1,312 | 1,448 |
| 232 | 230 | 1,082 | 1,192 | 1,254 | 1,320 | 1,446 |
| 233 | 231 | 1,090 | 1,195 | 1,260 | 1,323 | 1,455 |
| 234 | 232 | 1,092 | 1,204 | 1,265 | 1,330 | 1,456 |
| 235 | 233 | 1,096 | 1,207 | 1,269 | 1,335 | 1,467 |
| 236 | 234 | 1,100 | 1,213 | 1,274 | 1,339 | 1,472 |
| 237 | 235 | 1,110 | 1,217 | 1,282 | 1,348 | 1,477 |

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | IA 5 min | Densité pédite 0+/m2 (quantiles) | | | | |
| 2 | | 5% | 25% | 50% | 75% | 95% |
| 238 | 236 | 1,113 | 1,221 | 1,286 | 1,351 | 1,477 |
| 239 | 237 | 1,112 | 1,228 | 1,292 | 1,355 | 1,496 |
| 240 | 238 | 1,124 | 1,234 | 1,298 | 1,363 | 1,498 |
| 241 | 239 | 1,124 | 1,240 | 1,302 | 1,368 | 1,504 |
| 242 | 240 | 1,129 | 1,244 | 1,308 | 1,374 | 1,502 |
| 243 | 241 | 1,138 | 1,249 | 1,313 | 1,379 | 1,511 |
| 244 | 242 | 1,134 | 1,253 | 1,318 | 1,385 | 1,523 |
| 245 | 243 | 1,144 | 1,262 | 1,326 | 1,391 | 1,527 |
| 246 | 244 | 1,146 | 1,263 | 1,327 | 1,396 | 1,532 |
| 247 | 245 | 1,154 | 1,274 | 1,339 | 1,404 | 1,541 |
| 248 | 246 | 1,154 | 1,276 | 1,341 | 1,408 | 1,545 |
| 249 | 247 | 1,166 | 1,279 | 1,343 | 1,415 | 1,553 |
| 250 | 248 | 1,165 | 1,285 | 1,351 | 1,417 | 1,552 |
| 251 | 249 | 1,180 | 1,293 | 1,359 | 1,424 | 1,565 |
| 252 | 250 | 1,177 | 1,299 | 1,362 | 1,428 | 1,555 |
| 253 | 251 | 1,184 | 1,302 | 1,365 | 1,431 | 1,568 |
| 254 | 252 | 1,189 | 1,308 | 1,373 | 1,441 | 1,587 |
| 255 | 253 | 1,196 | 1,314 | 1,378 | 1,447 | 1,587 |
| 256 | 254 | 1,200 | 1,319 | 1,384 | 1,452 | 1,592 |
| 257 | 255 | 1,206 | 1,321 | 1,389 | 1,458 | 1,599 |
| 258 | 256 | 1,209 | 1,331 | 1,396 | 1,462 | 1,607 |
| 259 | 257 | 1,213 | 1,330 | 1,396 | 1,468 | 1,611 |
| 260 | 258 | 1,222 | 1,340 | 1,405 | 1,473 | 1,614 |
| 261 | 259 | 1,223 | 1,342 | 1,410 | 1,478 | 1,619 |
| 262 | 260 | 1,231 | 1,347 | 1,413 | 1,481 | 1,627 |
| 263 | 261 | 1,233 | 1,354 | 1,421 | 1,490 | 1,634 |
| 264 | 262 | 1,247 | 1,360 | 1,427 | 1,496 | 1,643 |
| 265 | 263 | 1,244 | 1,367 | 1,433 | 1,503 | 1,648 |
| 266 | 264 | 1,247 | 1,368 | 1,436 | 1,506 | 1,648 |
| 267 | 265 | 1,249 | 1,372 | 1,439 | 1,512 | 1,658 |
| 268 | 266 | 1,257 | 1,384 | 1,450 | 1,518 | 1,664 |
| 269 | 267 | 1,260 | 1,385 | 1,454 | 1,525 | 1,663 |
| 270 | 268 | 1,263 | 1,392 | 1,463 | 1,531 | 1,675 |
| 271 | 269 | 1,271 | 1,399 | 1,465 | 1,537 | 1,688 |
| 272 | 270 | 1,275 | 1,402 | 1,470 | 1,543 | 1,692 |
| 273 | 271 | 1,287 | 1,408 | 1,473 | 1,547 | 1,694 |
| 274 | 272 | 1,286 | 1,410 | 1,480 | 1,551 | 1,694 |
| 275 | 273 | 1,294 | 1,415 | 1,486 | 1,558 | 1,707 |
| 276 | 274 | 1,298 | 1,423 | 1,494 | 1,567 | 1,711 |
| 277 | 275 | 1,305 | 1,425 | 1,498 | 1,571 | 1,722 |
| 278 | 276 | 1,307 | 1,433 | 1,503 | 1,574 | 1,719 |
| 279 | 277 | 1,317 | 1,443 | 1,510 | 1,583 | 1,734 |
| 280 | 278 | 1,321 | 1,444 | 1,514 | 1,584 | 1,727 |
| 281 | 279 | 1,323 | 1,448 | 1,517 | 1,592 | 1,740 |
| 282 | 280 | 1,325 | 1,454 | 1,524 | 1,596 | 1,743 |
| 283 | 281 | 1,343 | 1,459 | 1,526 | 1,600 | 1,750 |
| 284 | 282 | 1,343 | 1,465 | 1,535 | 1,607 | 1,759 |

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | IA 5 min | Densité pédite 0+/m2 (quantiles) | | | | |
| 2 | | 5% | 25% | 50% | 75% | 95% |
| 285 | 283 | 1,338 | 1,470 | 1,538 | 1,615 | 1,764 |
| 286 | 284 | 1,348 | 1,476 | 1,546 | 1,619 | 1,767 |
| 287 | 285 | 1,350 | 1,478 | 1,552 | 1,626 | 1,780 |
| 288 | 286 | 1,364 | 1,486 | 1,558 | 1,630 | 1,780 |
| 289 | 287 | 1,365 | 1,494 | 1,565 | 1,636 | 1,792 |
| 290 | 288 | 1,369 | 1,493 | 1,566 | 1,639 | 1,787 |
| 291 | 289 | 1,374 | 1,501 | 1,570 | 1,643 | 1,802 |
| 292 | 290 | 1,379 | 1,506 | 1,581 | 1,656 | 1,805 |
| 293 | 291 | 1,386 | 1,515 | 1,585 | 1,663 | 1,818 |
| 294 | 292 | 1,389 | 1,518 | 1,591 | 1,666 | 1,809 |
| 295 | 293 | 1,395 | 1,523 | 1,597 | 1,672 | 1,822 |
| 296 | 294 | 1,396 | 1,525 | 1,597 | 1,677 | 1,832 |
| 297 | 295 | 1,402 | 1,536 | 1,607 | 1,685 | 1,829 |
| 298 | 296 | 1,409 | 1,540 | 1,615 | 1,687 | 1,840 |
| 299 | 297 | 1,412 | 1,544 | 1,616 | 1,691 | 1,845 |
| 300 | 298 | 1,419 | 1,547 | 1,621 | 1,698 | 1,844 |
| 301 | 299 | 1,424 | 1,556 | 1,626 | 1,702 | 1,861 |
| 302 | 300 | 1,427 | 1,563 | 1,636 | 1,708 | 1,865 |

# 4 PrÃ©sentation des paramÃ¨tres estimÃ©s par le modÃ¨le

Sortie des paramètres Openbugs - Modèle 2016_12_19_4zones

marion.legrand

13 septembre 2017

# 1 sigma_juv_moy

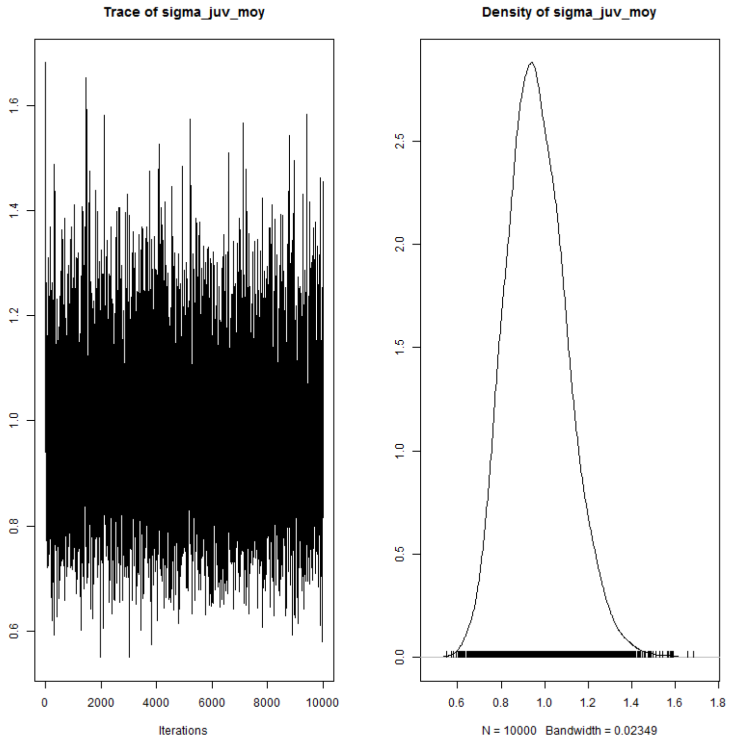


Figure 1 – sigma_juv_moy

Table 1 – Statistiques de sigma_juv

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.72 | 0.87 | 0.96 | 1.06 | 1.27 | 0.97 | 0.14 |



# 2 sigma_wild_moy

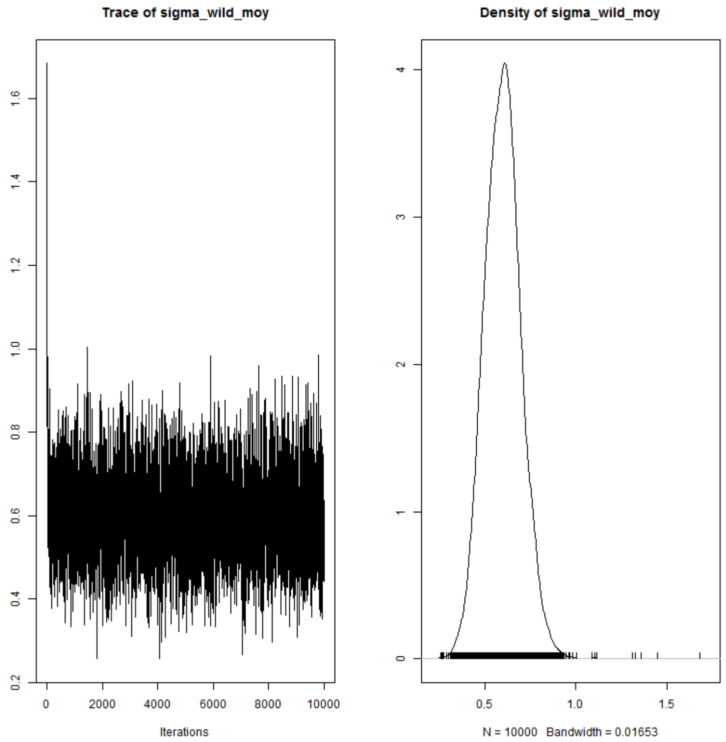


Figure 2 – sigma_wild_moy

Table 2 – Statistiques de sigma_wild

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.41 | 0.53 | 0.60 | 0.66 | 0.80 | 0.60 | 0.10 |

# 3 sigma_egg_moy

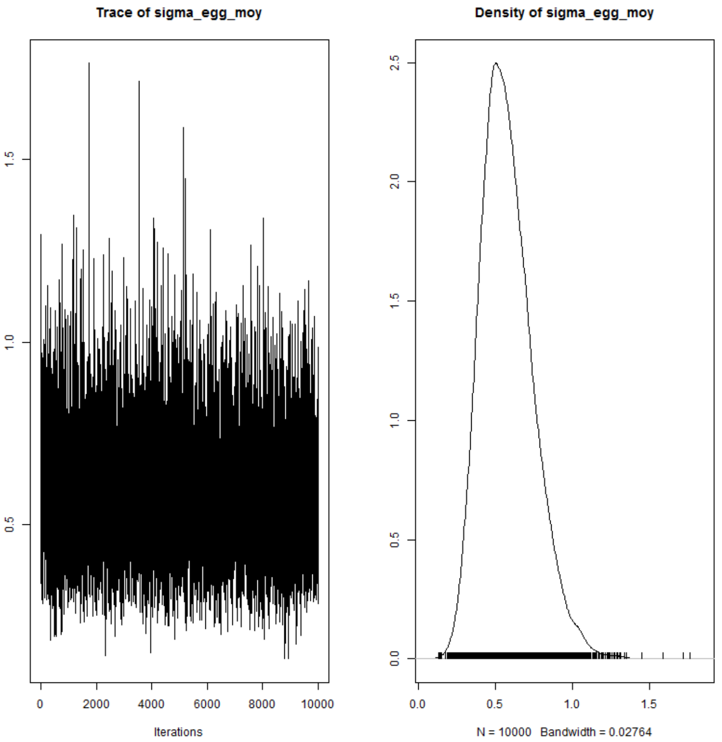


FIGURE 3 – sigma_egg_moy

TABLE 3 – Statistiques de sigma_egg

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.30 | 0.46 | 0.56 | 0.68 | 0.95 | 0.58 | 0.17 |

# 4 nu_wild

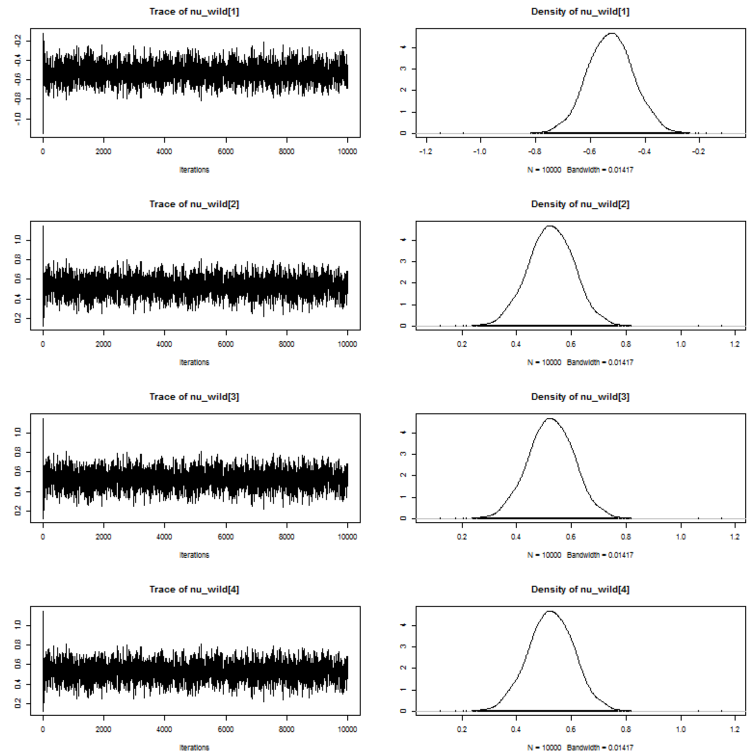


FIGURE 4 – nu_wild

TABLE 4 – Statistiques de nu_d

| | 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|---|
| nu_wild_V | -0.69 | -0.58 | -0.53 | -0.47 | -0.36 | -0.53 | 0.08 |
| nu_wild_A | 0.36 | 0.47 | 0.53 | 0.58 | 0.69 | 0.53 | 0.08 |
| nu_wild_L | 0.36 | 0.47 | 0.53 | 0.58 | 0.69 | 0.53 | 0.08 |
| nu_wild_P | 0.36 | 0.47 | 0.53 | 0.58 | 0.69 | 0.53 | 0.08 |

## 5 a

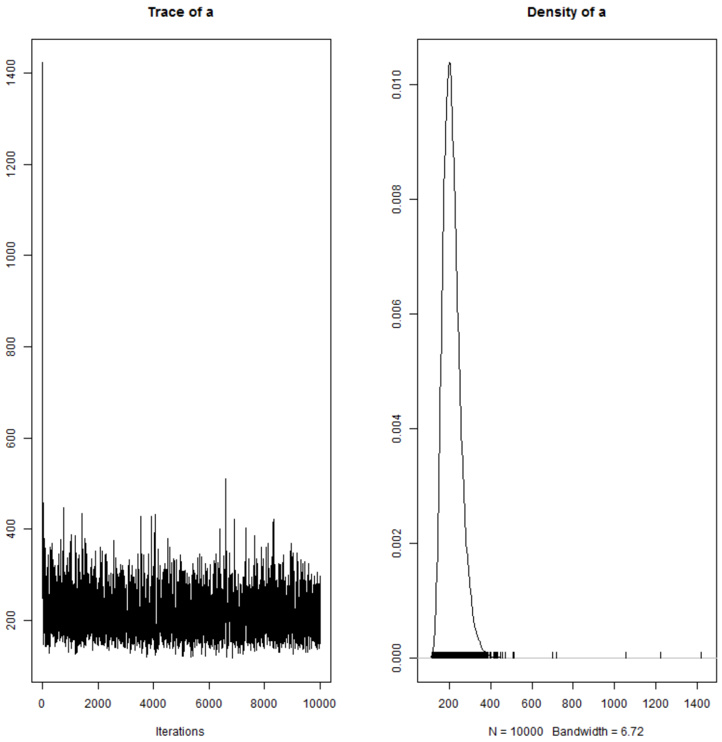


Figure 5 – a

Table 5 – Statistiques de a

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 146.80 | 181.60 | 205.80 | 235.20 | 312.10 | 212.37 | 46.96 |

## 6 a_juv

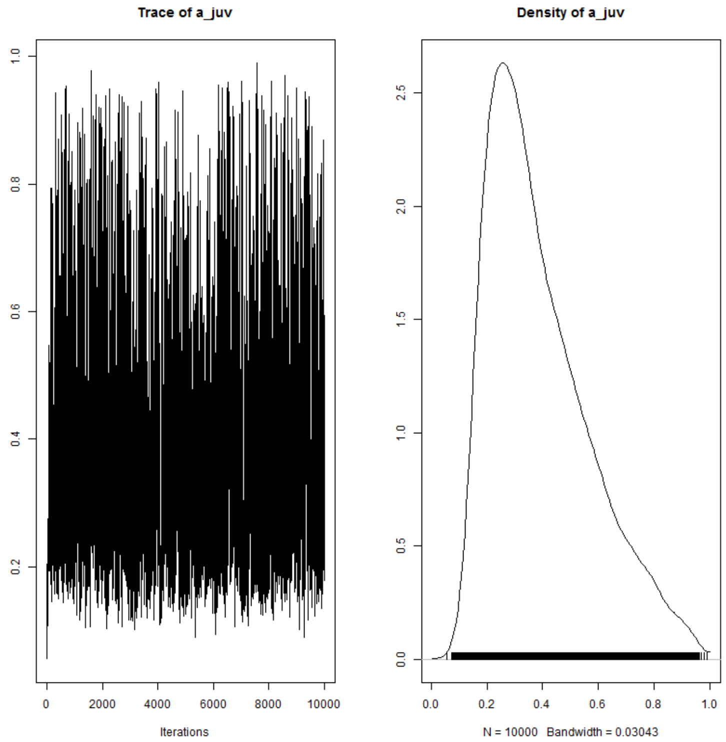


Figure 6 – a_juv

Table 6 – Statistiques de a_juv

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.15 | 0.25 | 0.35 | 0.50 | 0.82 | 0.39 | 0.18 |

# 7 Rmax

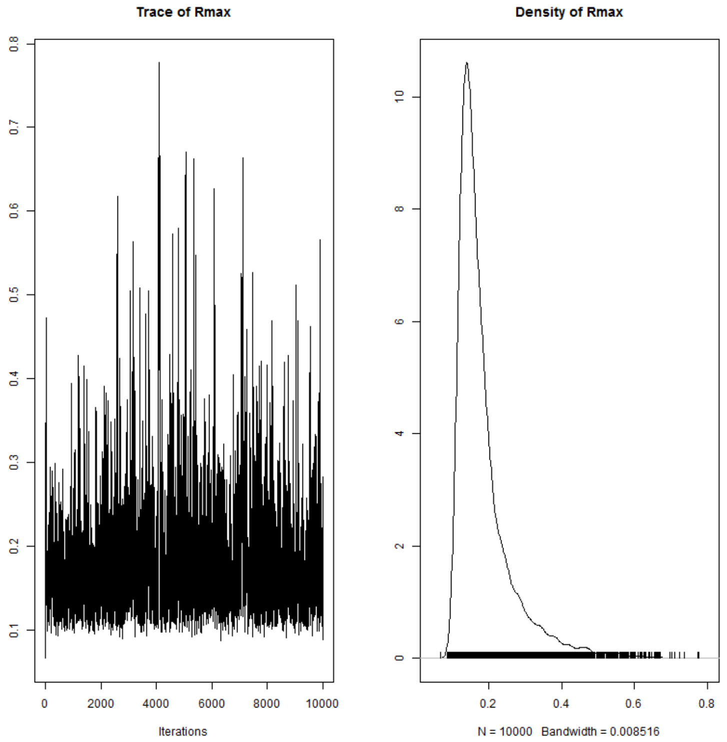


FIGURE 7 – Rmax

TABLE 7 – Statistiques de Rmax

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.11 | 0.13 | 0.16 | 0.20 | 0.42 | 0.18 | 0.08 |

# 8 sigma_juv_site

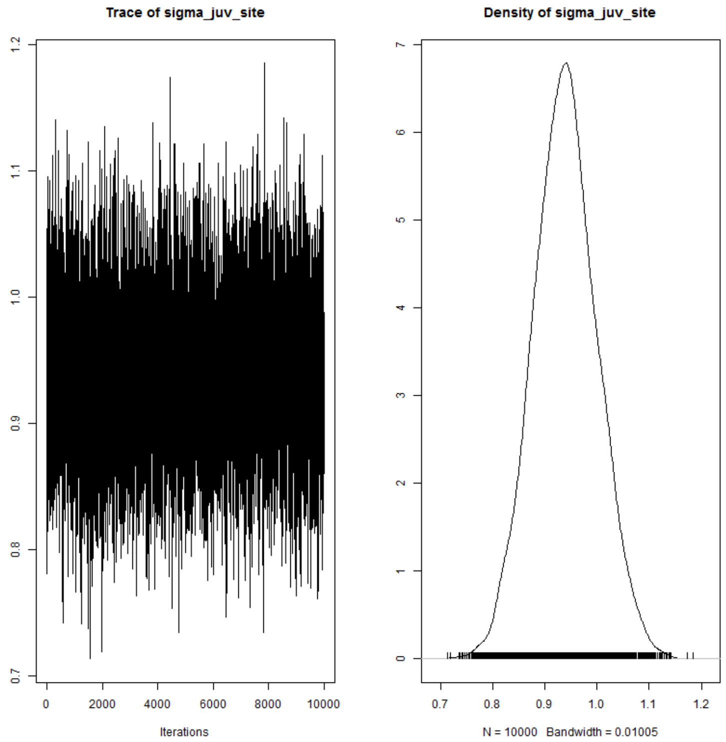


FIGURE 8 – sigma_juv_site

TABLE 8 – Statistiques de sigma_juv_site

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.82 | 0.90 | 0.94 | 0.98 | 1.06 | 0.94 | 0.06 |

# 9 sigma_wild_site

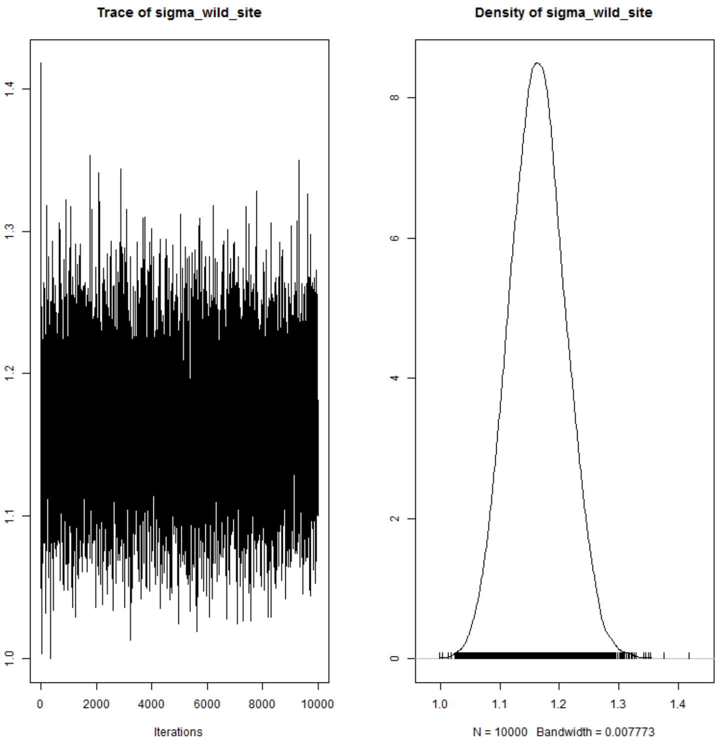


Figure 9 – sigma_wild_site

Table 9 – Statistiques de sigma_wild_site

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 1.07 | 1.13 | 1.16 | 1.19 | 1.26 | 1.16 | 0.05 |

# 10 sigma_egg_site

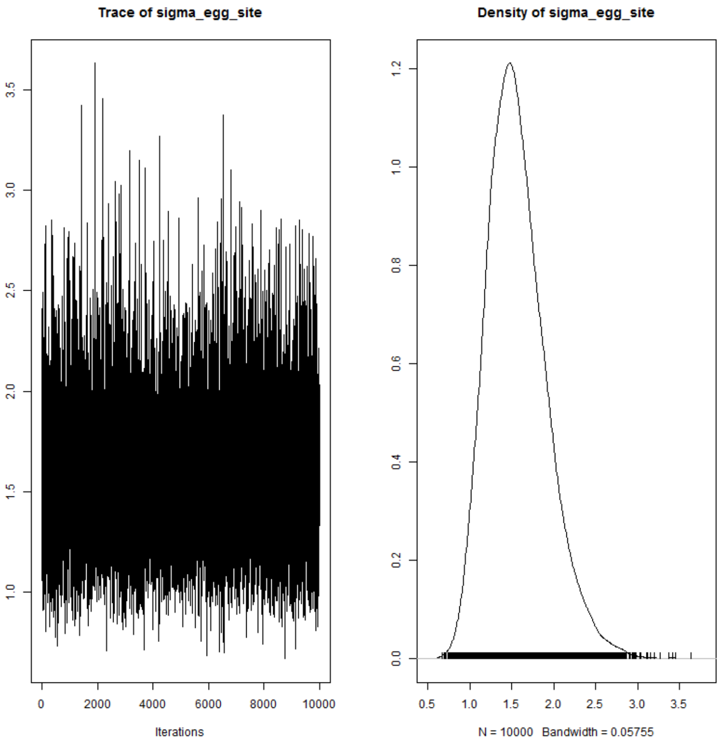


Figure 10 – sigma_egg_site

Table 10 – Statistiques de sigma_egg_site

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.99 | 1.31 | 1.53 | 1.77 | 2.38 | 1.57 | 0.35 |

## 11 adjust_p_A

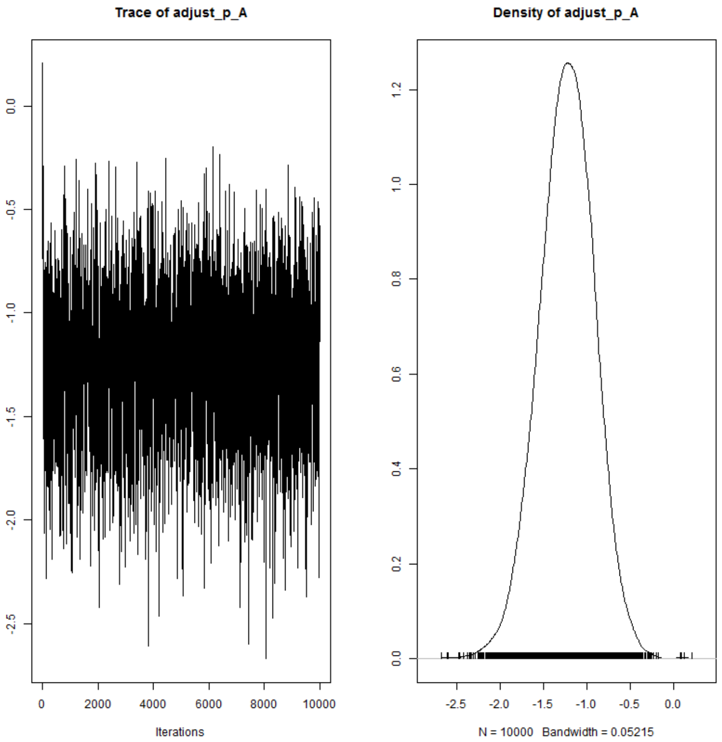


FIGURE 11 – adjust_p_A

TABLE 11 – Statistiques de adjust_p_A

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| -1.86 | -1.43 | -1.22 | -1.02 | -0.63 | -1.23 | 0.32 |

## 12 adjust_p_L

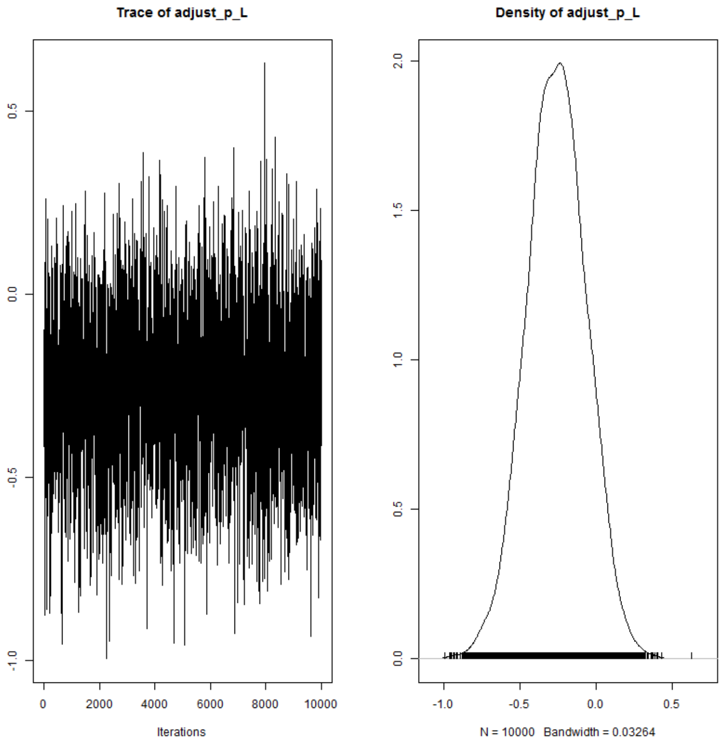


FIGURE 12 – adjust_p_L

TABLE 12 – Statistiques de adjust_p_L

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| -0.65 | -0.39 | -0.26 | -0.13 | 0.12 | -0.26 | 0.20 |

# 13 adjust_p_P

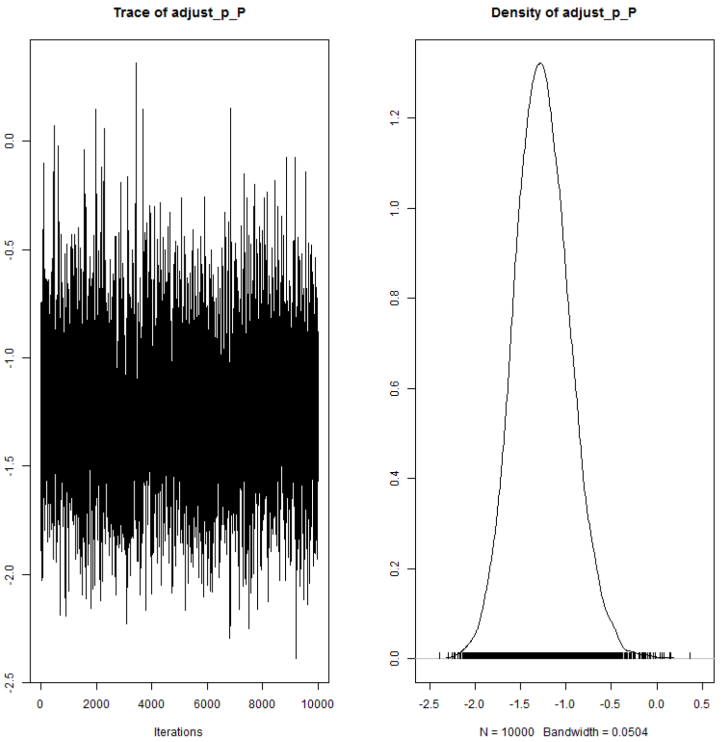


FIGURE 13 – adjust_p_P

TABLE 13 – Statistiques de adjust_p_P

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| -1.83 | -1.46 | -1.27 | -1.06 | -0.63 | -1.26 | 0.31 |

# 14 sigma_p_alagnon

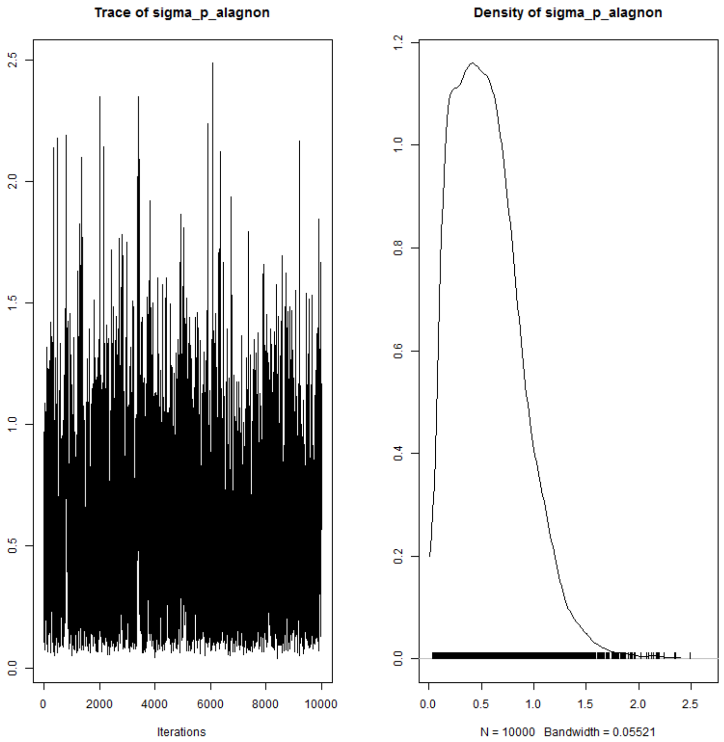


FIGURE 14 – sigma_p_alagnon

TABLE 14 – Statistiques de sigma_p_alagnon

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.10 | 0.31 | 0.52 | 0.76 | 1.30 | 0.56 | 0.33 |

## 15 sigma_p_langeac

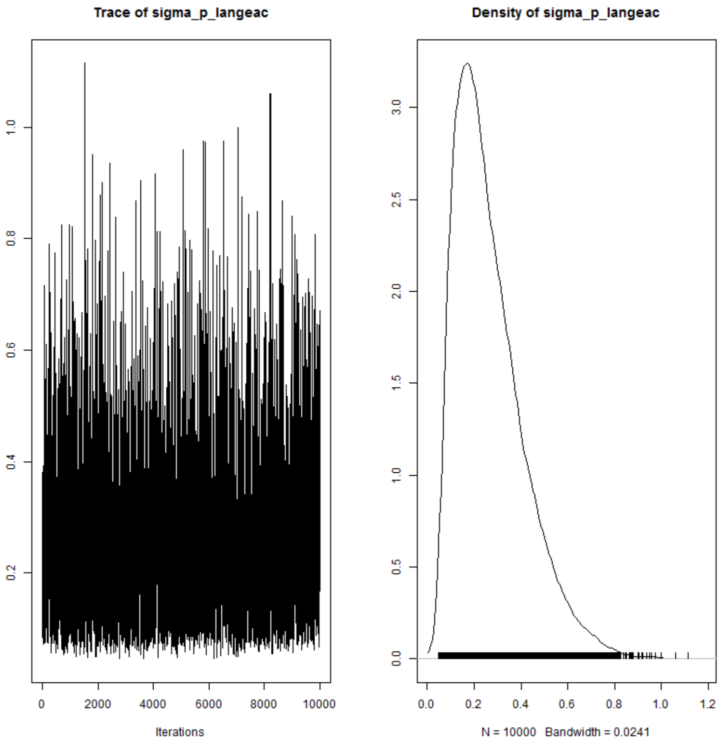


Figure 15 – sigma_p_langeac

Table 15 – Statistiques de sigma_p_langeac

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.08 | 0.15 | 0.23 | 0.35 | 0.63 | 0.27 | 0.15 |

## 16 sigma_p_poutes

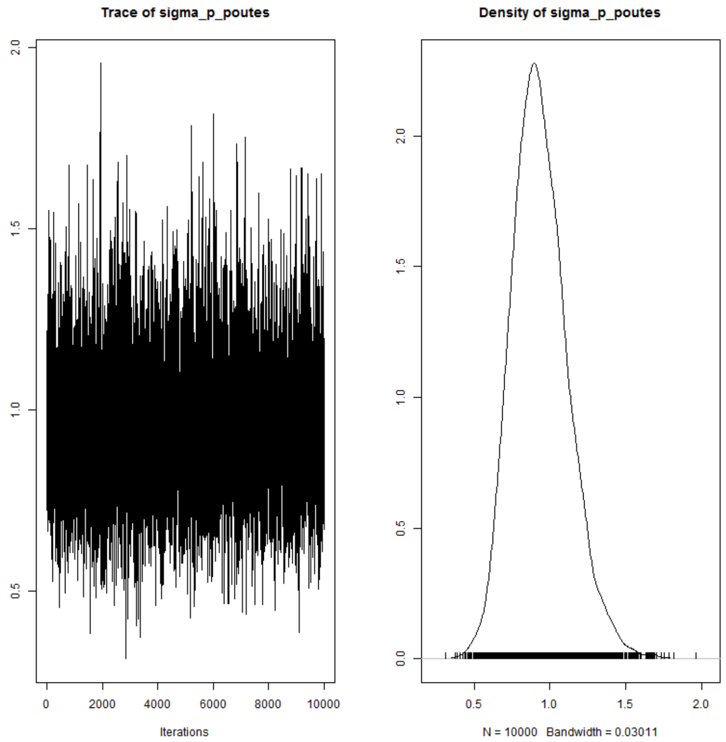


Figure 16 – sigma_p_poutes

Table 16 – Statistiques de sigma_p_poutes

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.62 | 0.81 | 0.92 | 1.05 | 1.34 | 0.94 | 0.18 |

## 17 rho_station

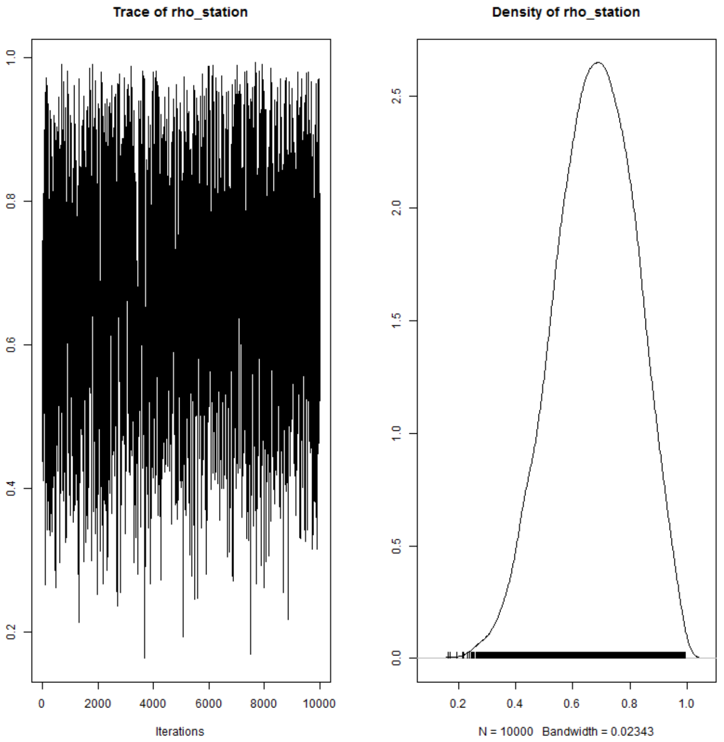


FIGURE 17 – rho_station

TABLE 17 – Statistiques de rho_station

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.40 | 0.58 | 0.68 | 0.78 | 0.93 | 0.68 | 0.14 |

## 18 hel_effect

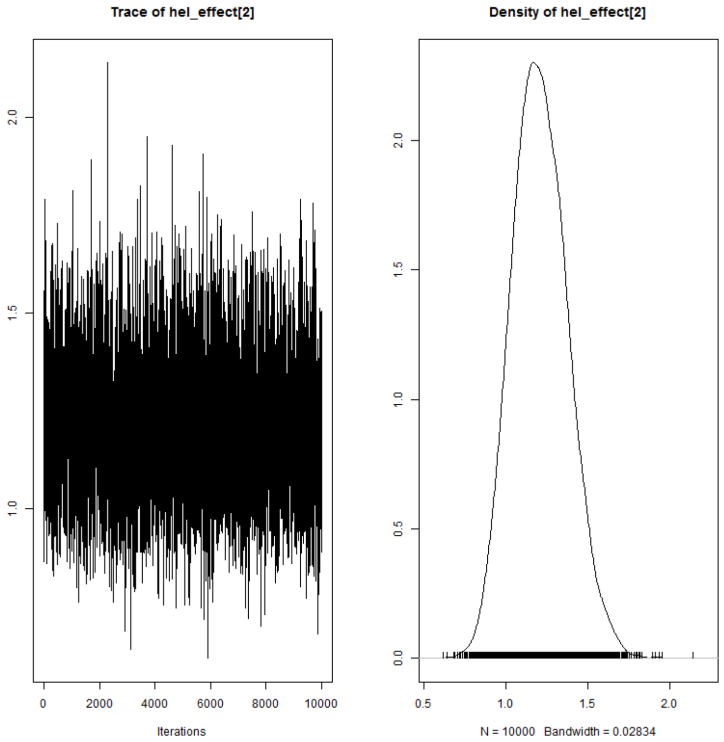


FIGURE 18 – hel_effect

TABLE 18 – Statistiques de hel_effect

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.90 | 1.09 | 1.20 | 1.32 | 1.56 | 1.21 | 0.17 |

## 19 mu_tau

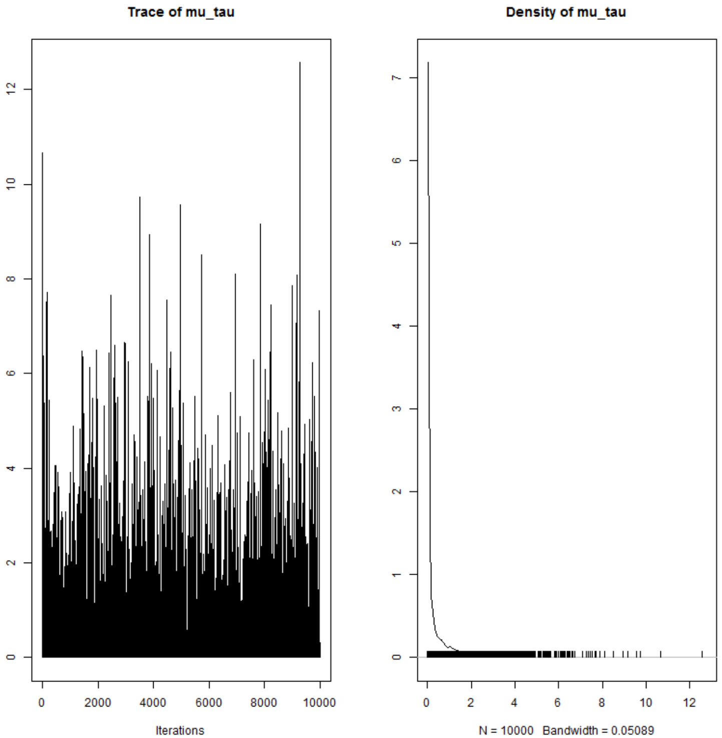


FIGURE 19 – mu_tau

TABLE 19 – Statistiques de mu_tau

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.000002 | 0.000654 | 0.032860 | 0.406600 | 3.223200 | 0.431616 | 0.925831 |

## 20 beta_tau

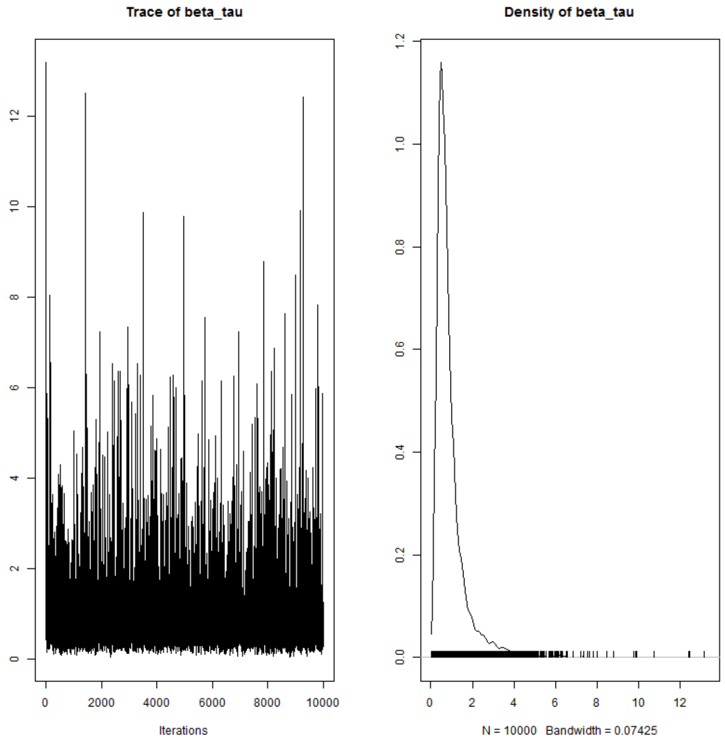


FIGURE 20 – beta_tau

TABLE 20 – Statistiques de beta_tau

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.21 | 0.46 | 0.68 | 1.05 | 3.24 | 0.92 | 0.84 |

## 21 s_juv2ad

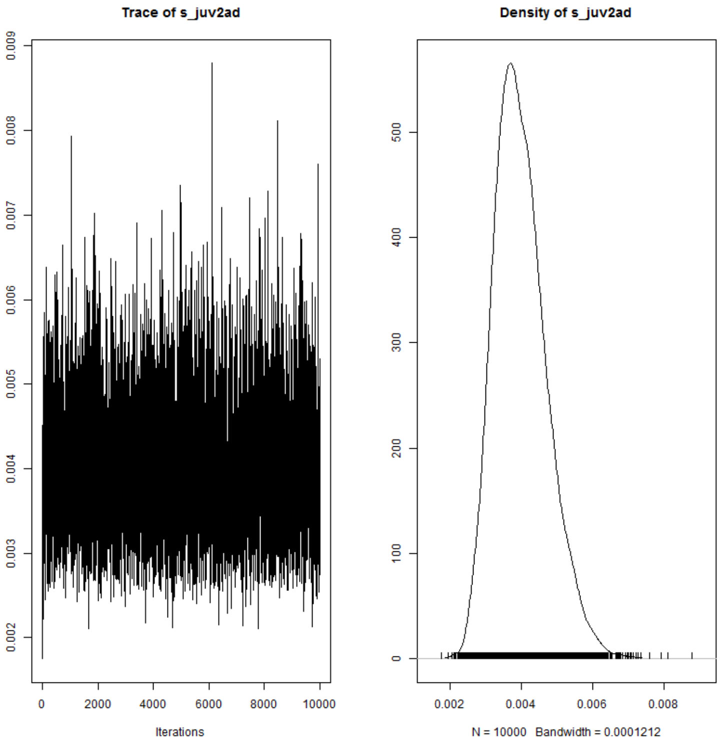


Figure 21 – s_juv2ad

Table 21 – Statistiques de s_juv2ad

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.0028 | 0.0035 | 0.0039 | 0.0044 | 0.0056 | 0.0040 | 0.0007 |

## 22 level_s

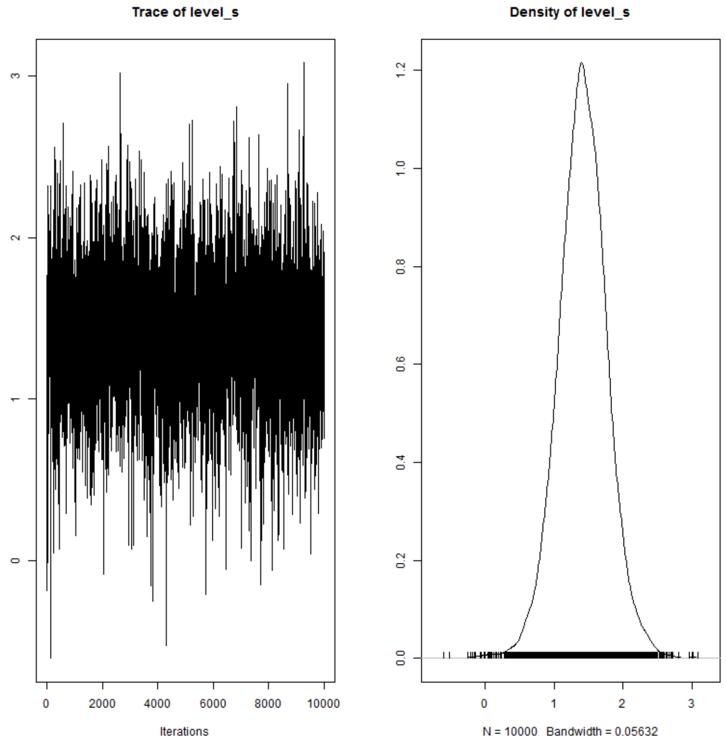


Figure 22 – level_s

Table 22 – Statistiques de level_s

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.70 | 1.19 | 1.41 | 1.64 | 2.13 | 1.41 | 0.36 |

## 23 rho_poutes

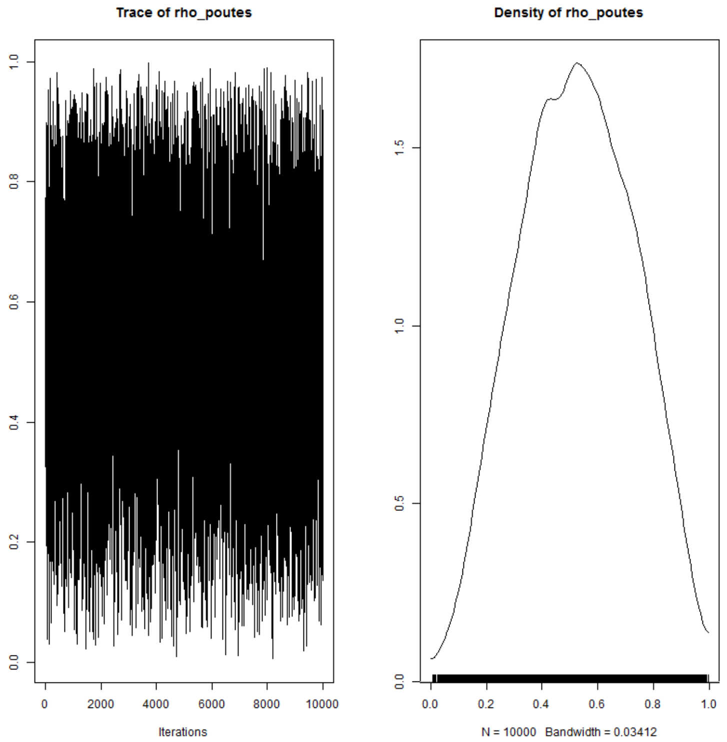


FIGURE 23 – rho_poutes

TABLE 23 – Statistiques de rho_poutes

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.14 | 0.37 | 0.52 | 0.68 | 0.90 | 0.52 | 0.20 |

## 24 sigma_vichy

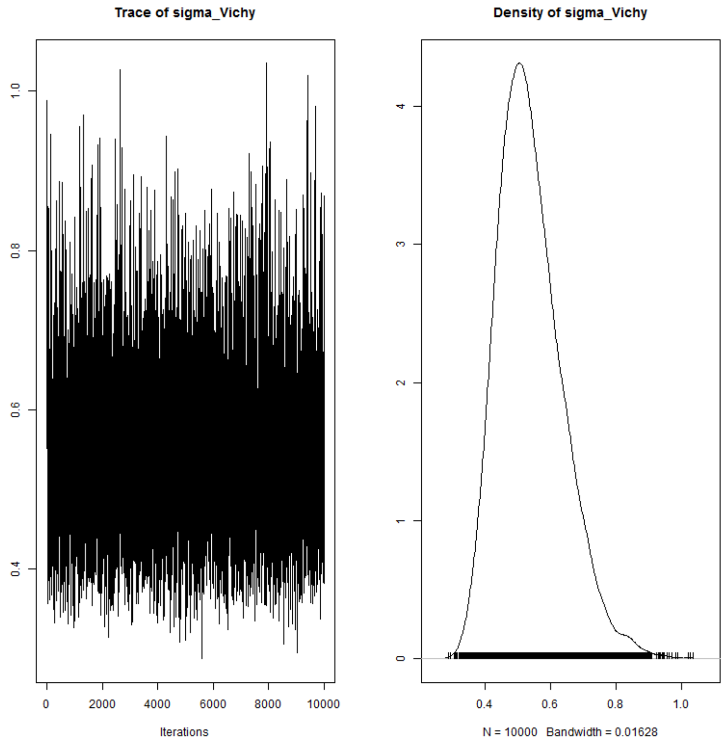


FIGURE 24 – sigma_vichy

TABLE 24 – Statistiques de sigma_vichy

| 2.5% | 25% | 50% | 75% | 97.5% | Mean | SD |
|---|---|---|---|---|---|---|
| 0.38 | 0.47 | 0.53 | 0.60 | 0.76 | 0.54 | 0.10 |

## 25 res_p_alagnon

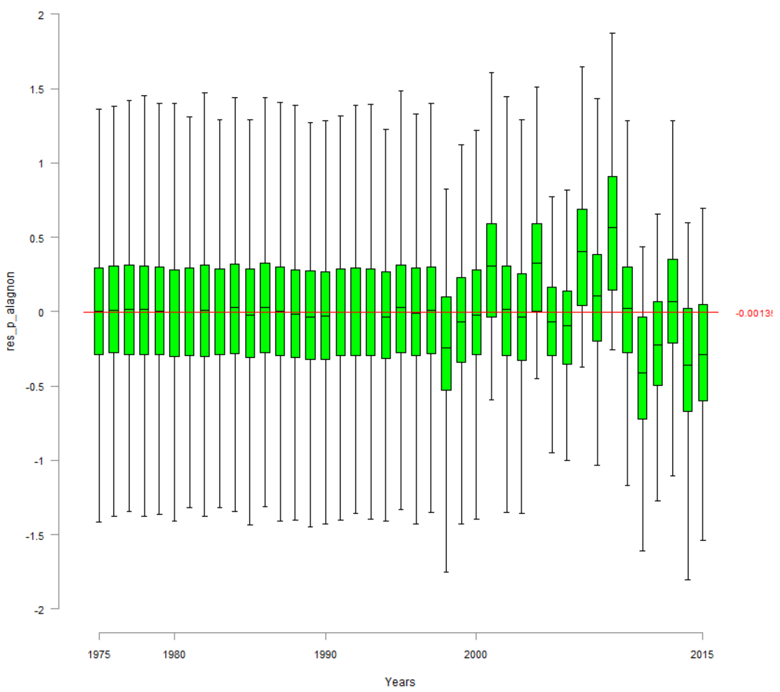


FIGURE 25 – res_p_alagnon

## 26 res_p_langeac

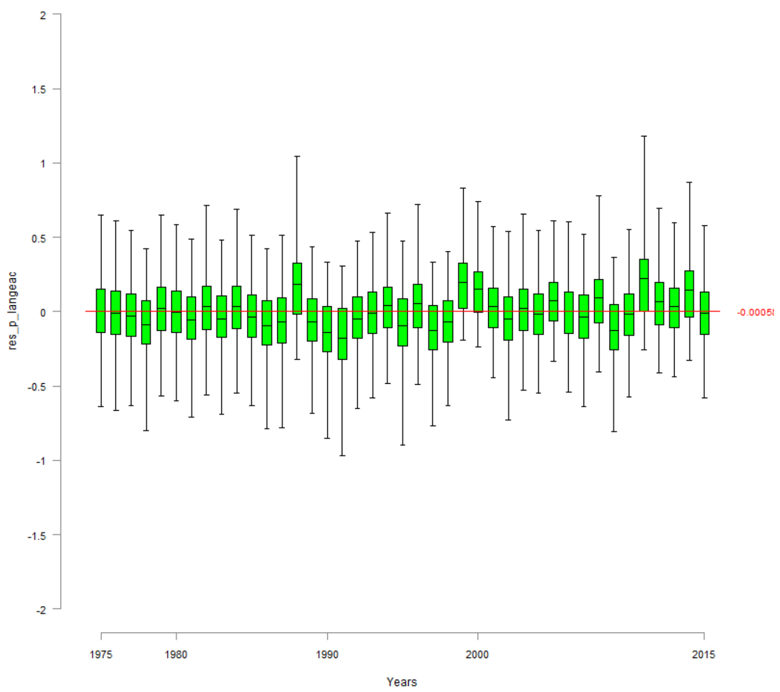


FIGURE 26 – res_p_langeac

## 27 res_p_poutes

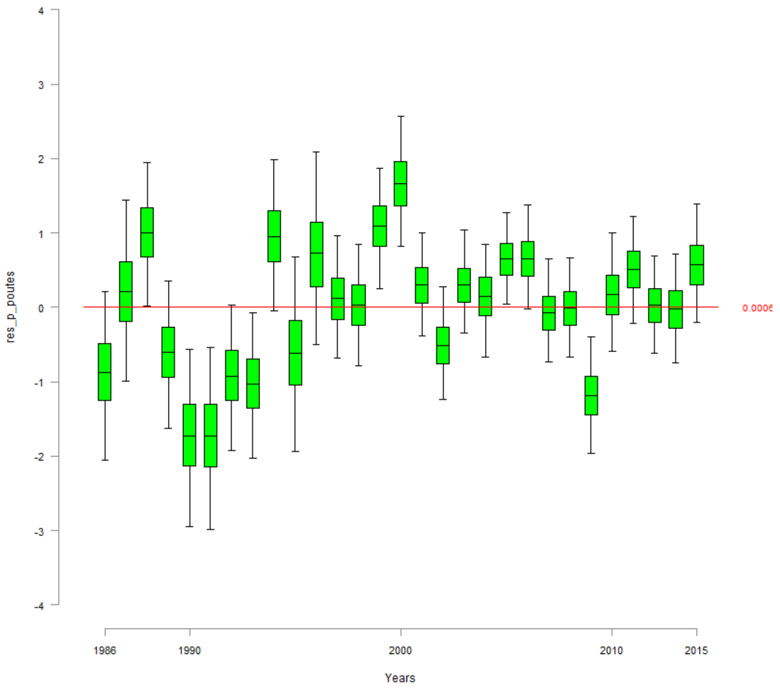


FIGURE 27 – res_p_poutes

## 28 res_vichy

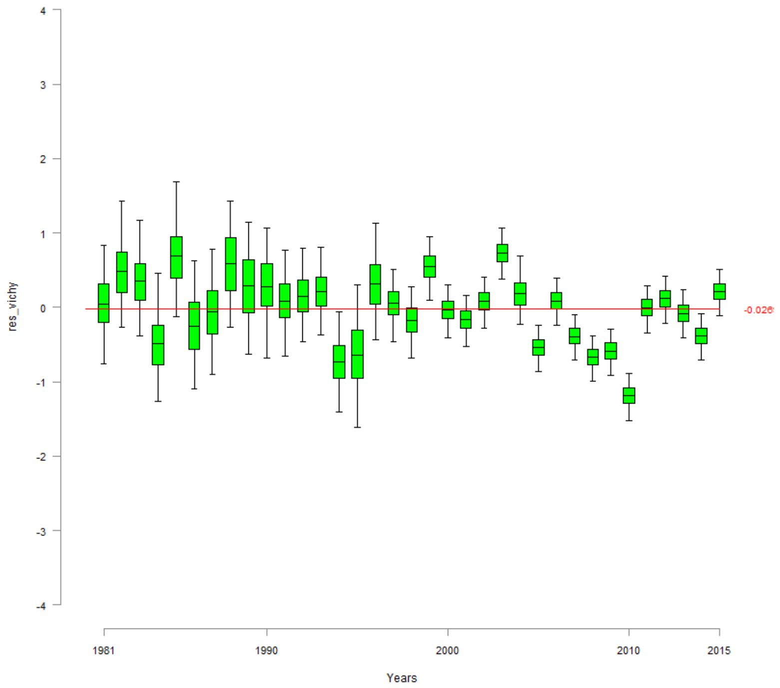


FIGURE 28 – res_vichy

# 29 zone_effect

## 29.1 zone_effect_Vichy

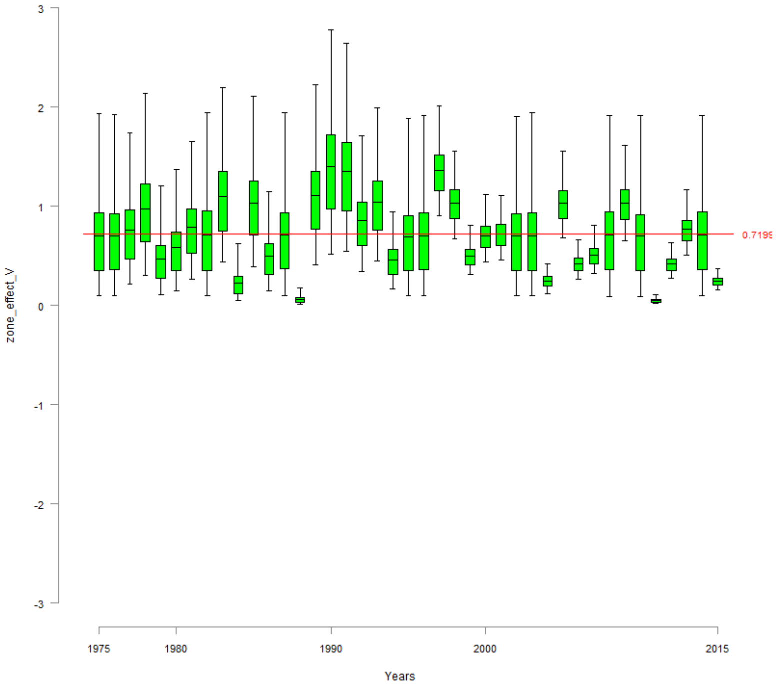


Figure 29 – zone_effect_V

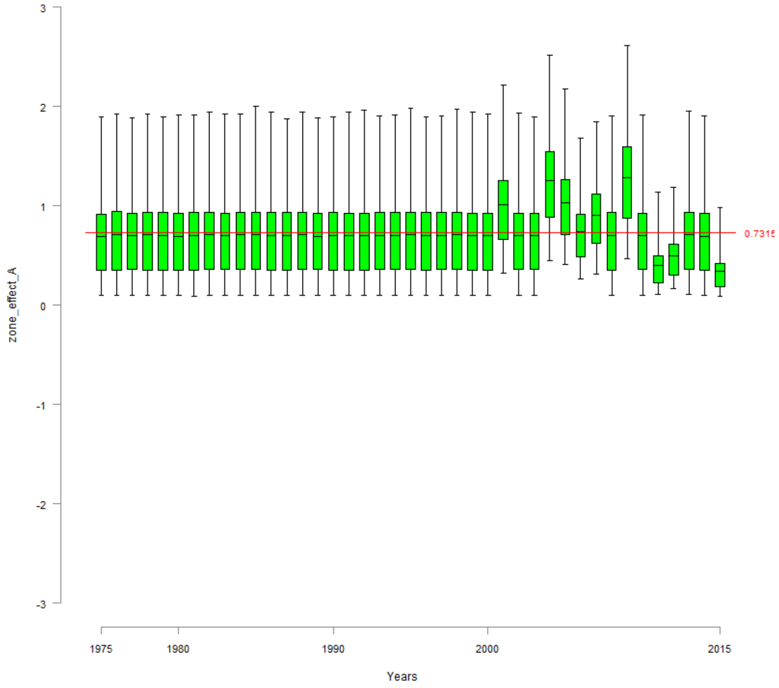


Figure 30 – zone_effect_A

### 29.2 zone_effect_Langeac

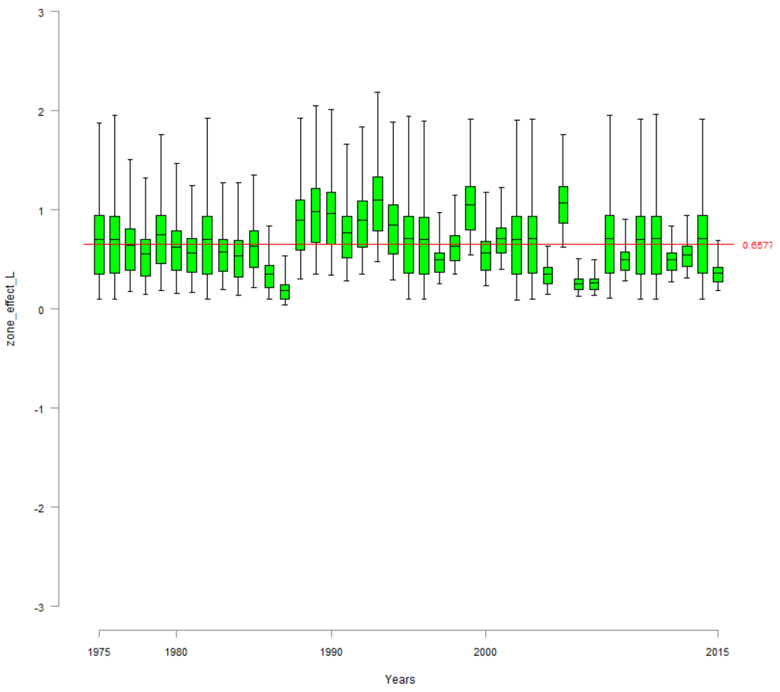


FIGURE 31 – zone_effect_L

### 29.3 zone_effect_Poutes

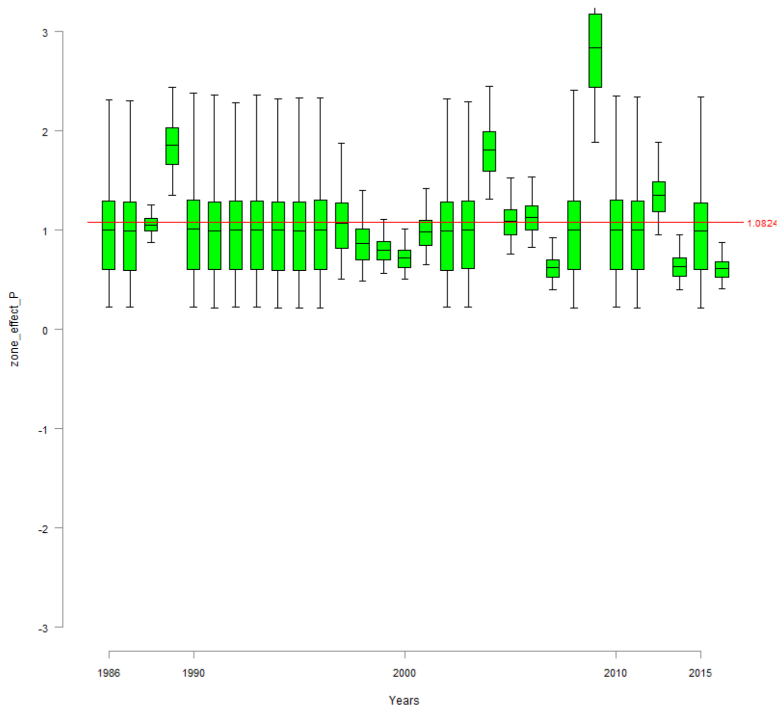


FIGURE 32 – zone_effect_P

## 30 N_Vichy

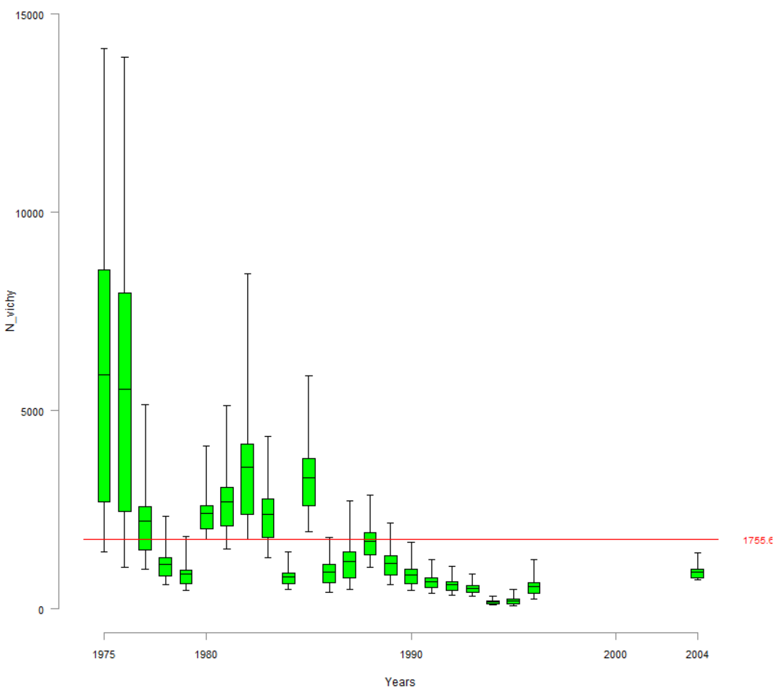


FIGURE 33 – N_vichy

## 31 N_Alagnon

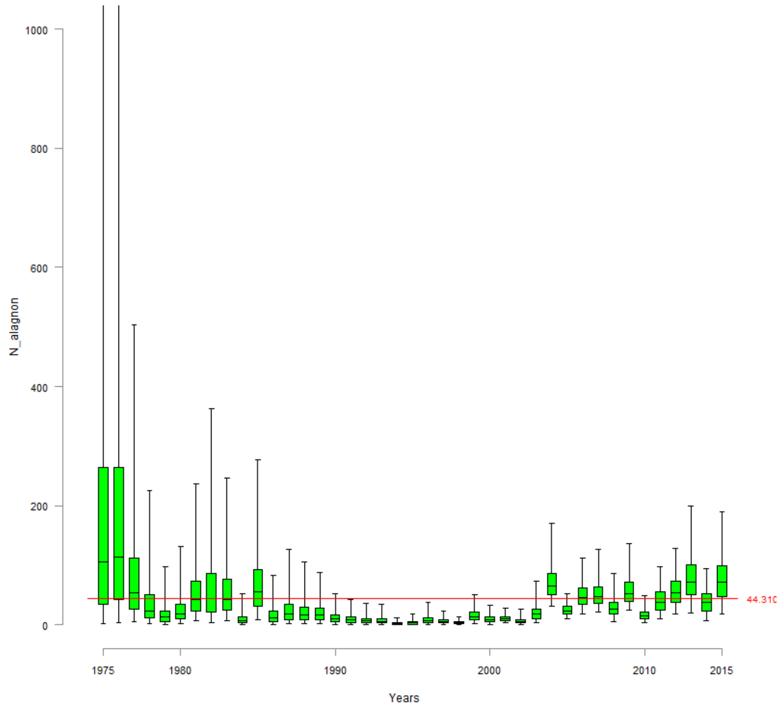


FIGURE 34 – N_alagnon

# 32 N_Langeac

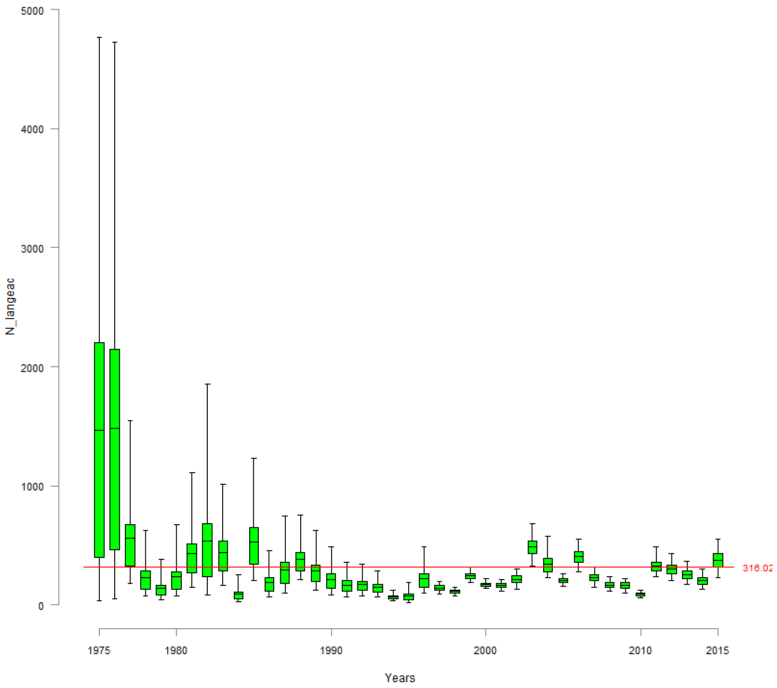


FIGURE 35 – N_langeac

# 33 d_wild_moy

## 33.1 d_wild_moy_Vichy

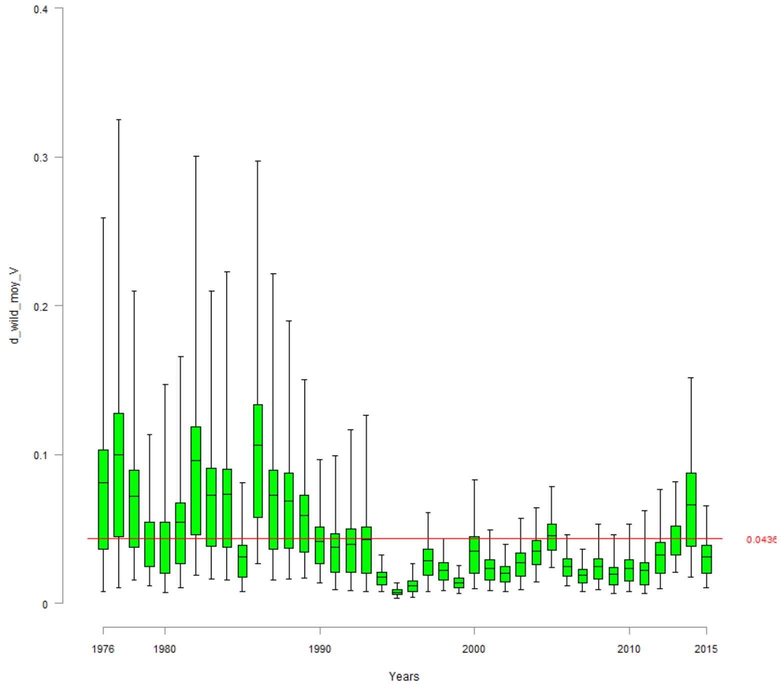


FIGURE 36 – d_wild_moy_V

### 33.2 d_wild_moy_Alagnon

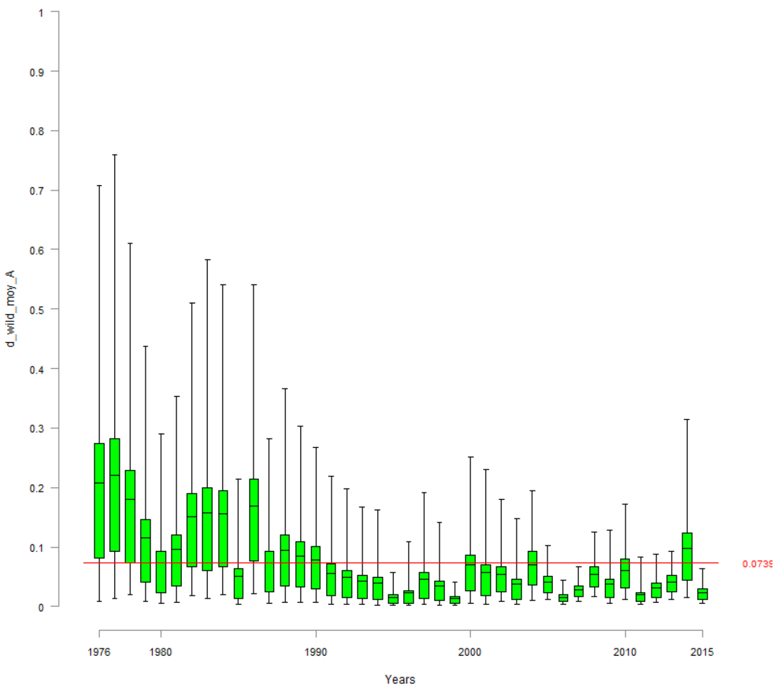


FIGURE 37 – d_wild_moy_A

### 33.3 d_wild_moy_Langeac

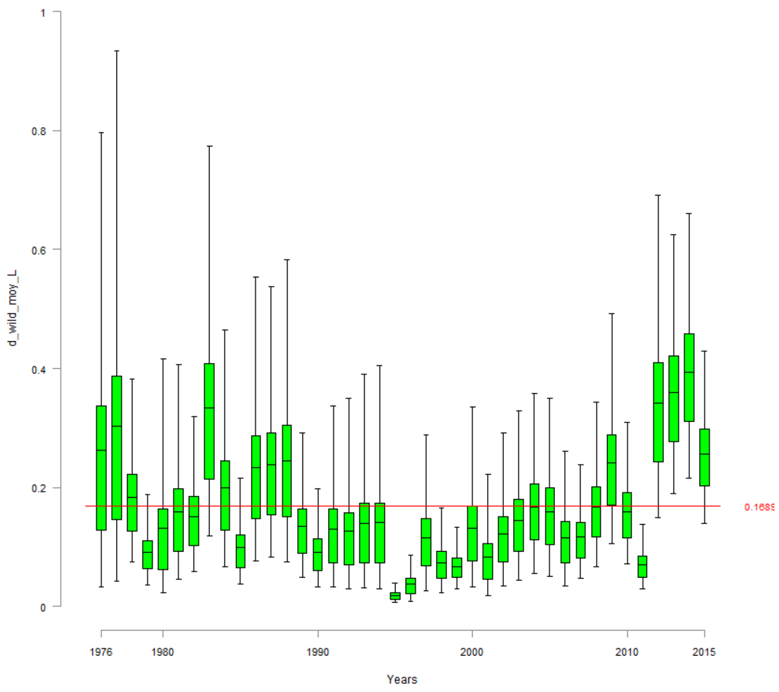


FIGURE 38 – d_wild_moy_L

### 33.4 d_wild_moy_Poutes

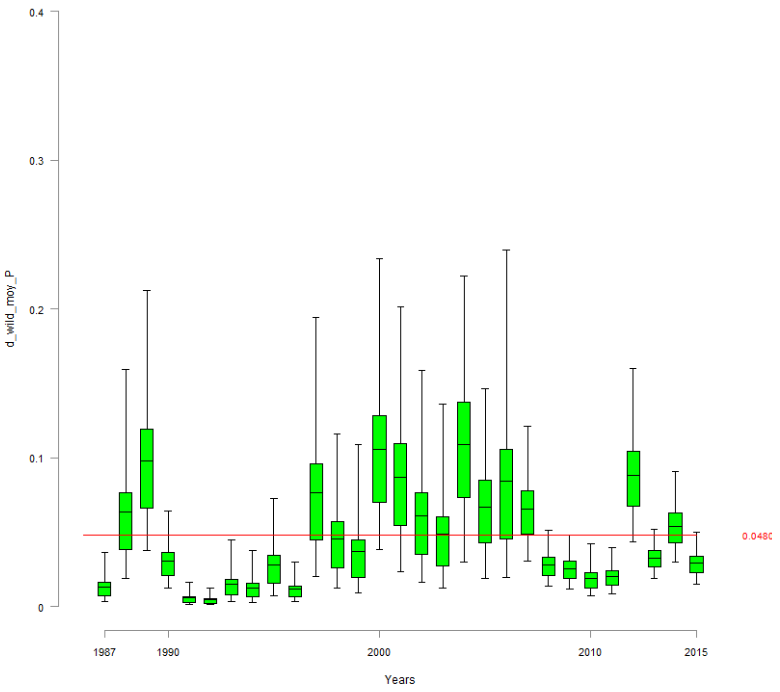


FIGURE 39 – d_wild_moy_P

## 34 d_juv_moy

### 34.1 d_juv_moy_Vichy

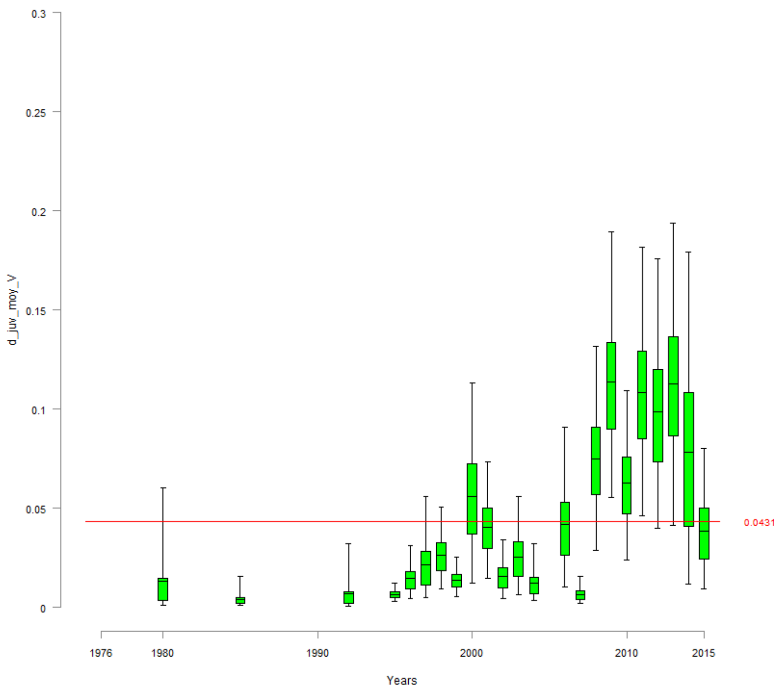


FIGURE 40 – d_juv_moy_V

### 34.2 d_juv_moy_Alagnon

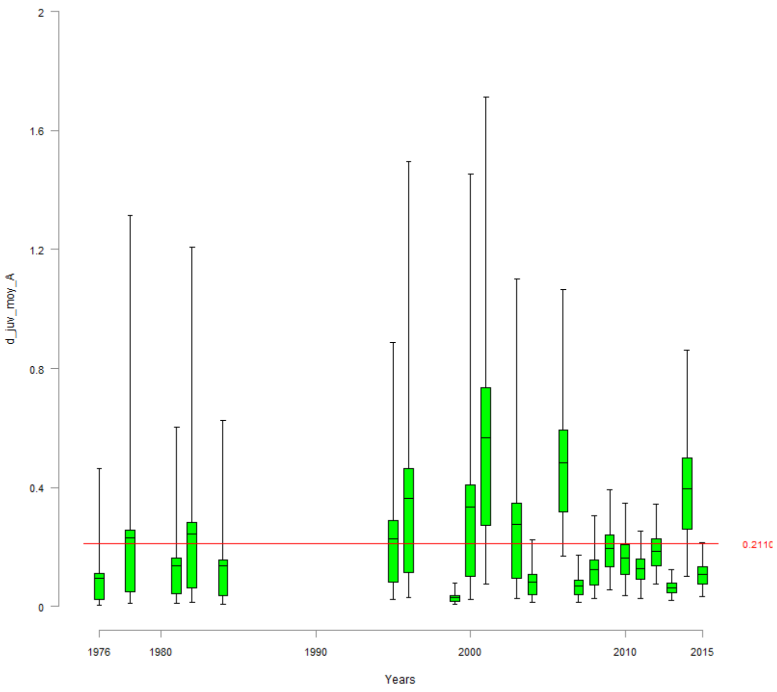


Figure 41 – d_juv_moy_A

### 34.3 d_juv_moy_Langeac

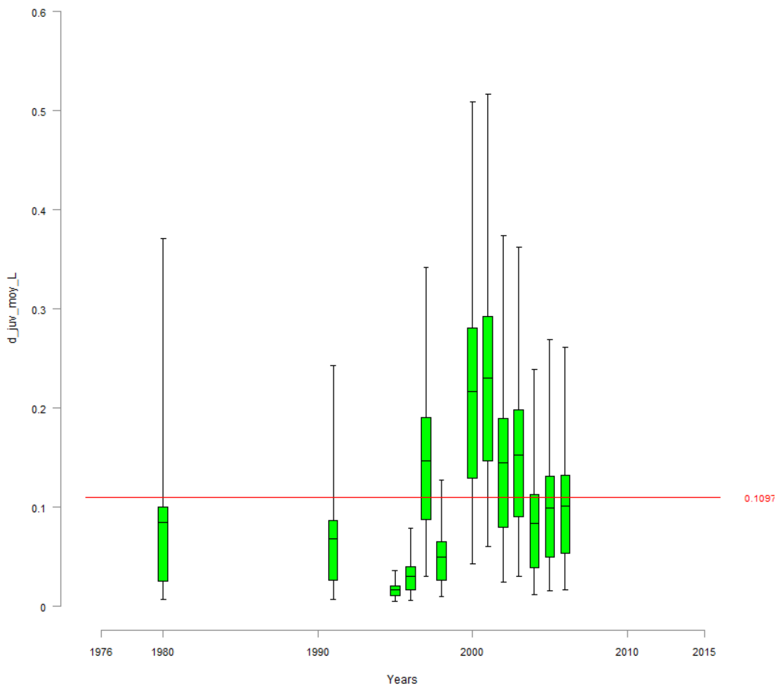


Figure 42 – d_juv_moy_L

### 34.4 d_juv_moy_Poutes

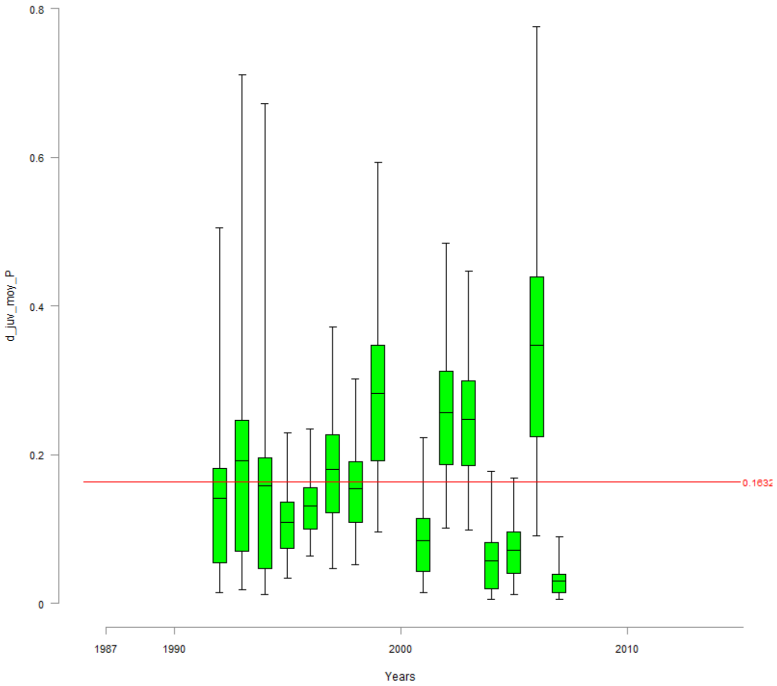


FIGURE 43 – d_juv_moy_P

## 35 d_egg_moy

### 35.1 d_egg_moy_Vichy

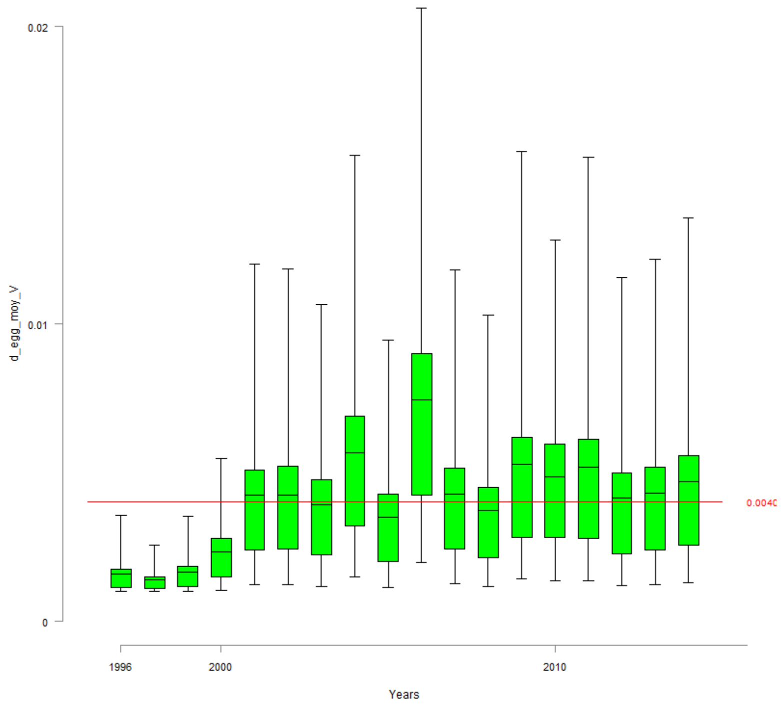


FIGURE 44 – d_egg_moy_V

### 35.2 d_egg_moy_Langeac

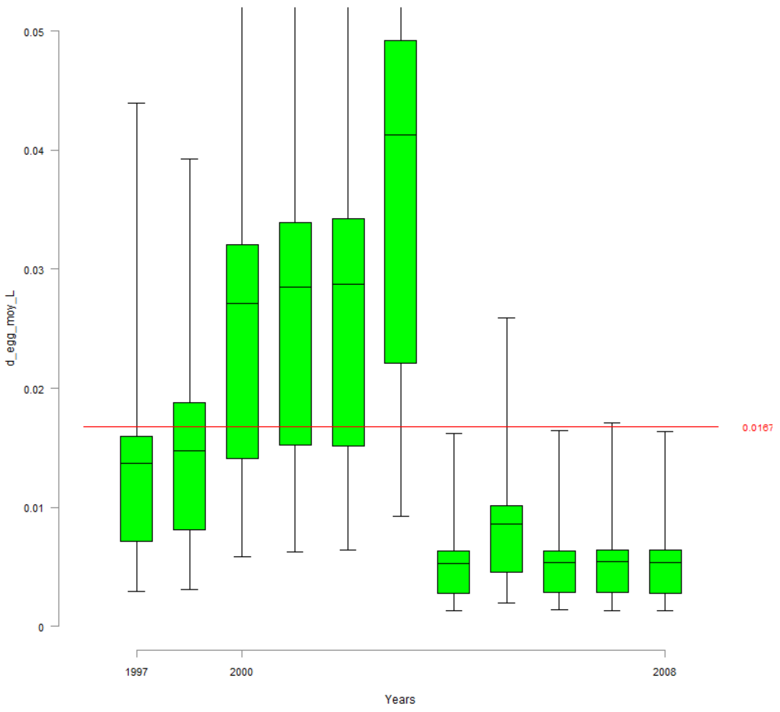


FIGURE 45 – d_egg_moy_L

## 36 res_wild_moy

### 36.1 res_wild_moy_Vichy

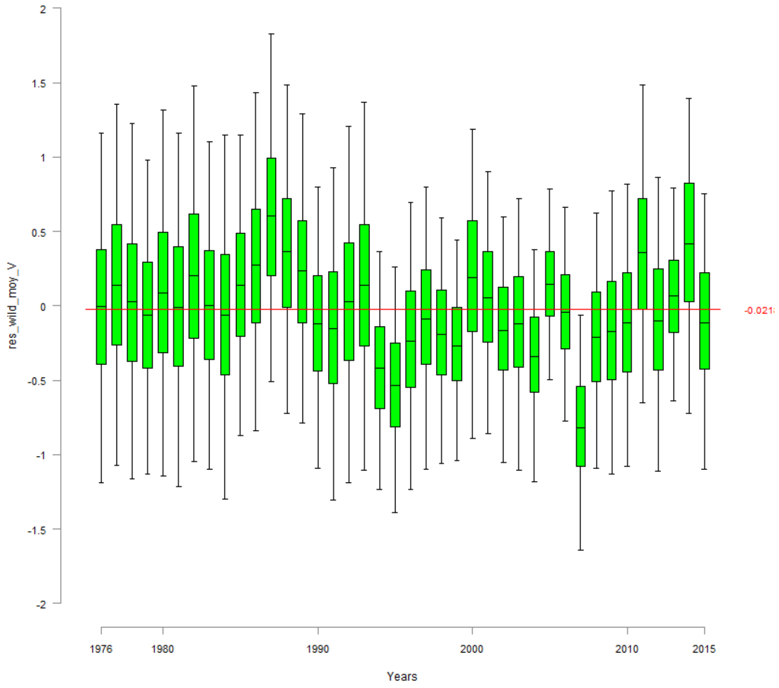


FIGURE 46 – res_wild_moy_V

### 36.2 res_wild_moy_Alagnon

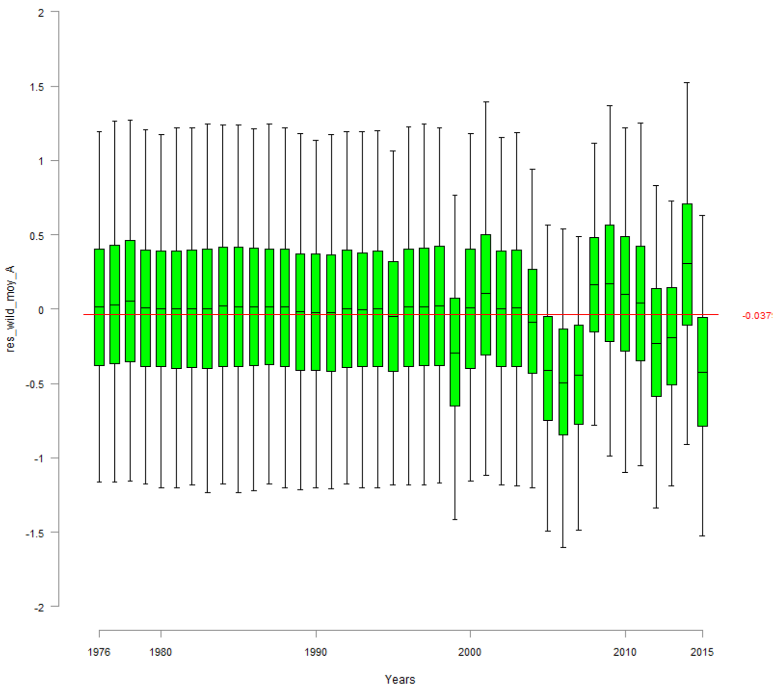


Figure 47 – res_wild_moy_A

### 36.3 res_wild_moy_Langeac

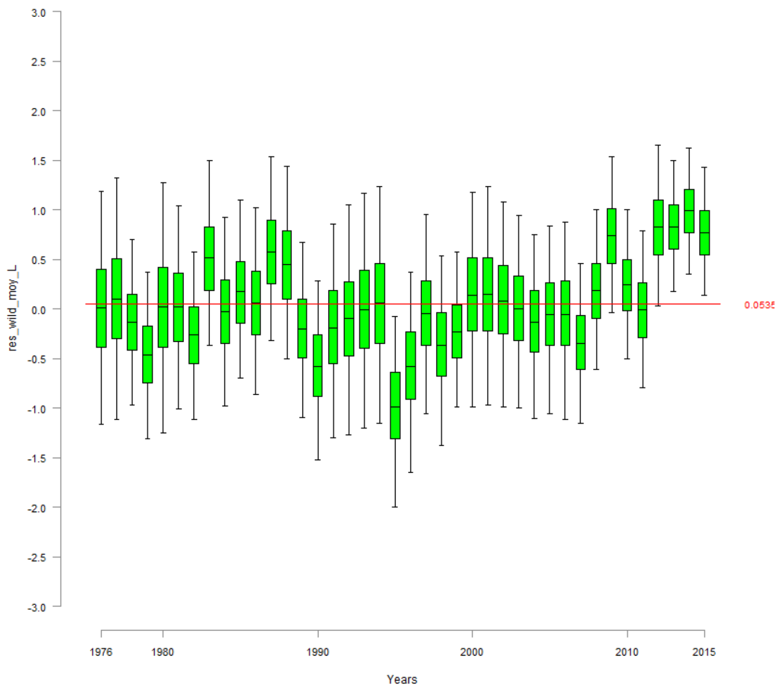


Figure 48 – res_wild_moy_L

### 36.4 res_wild_moy_Poutes

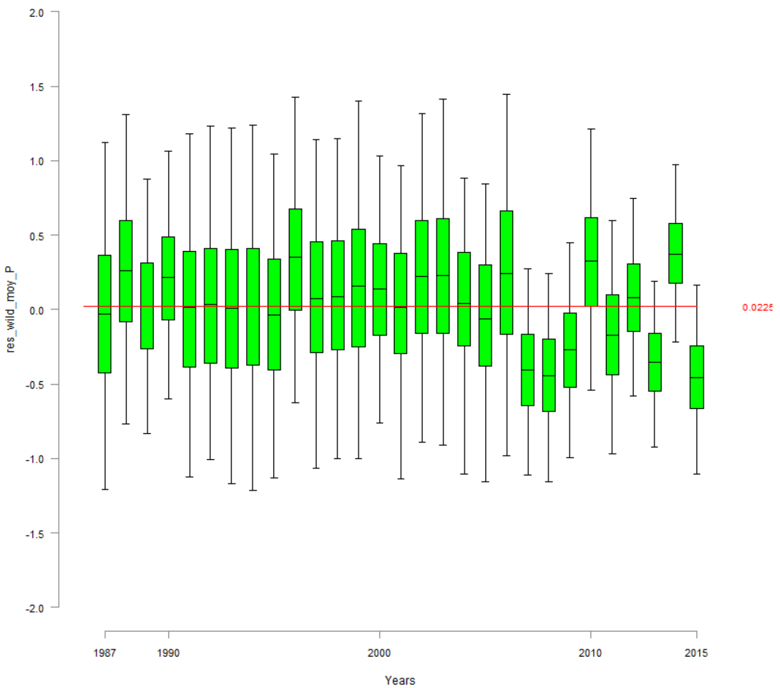


FIGURE 49 – res_wild_moy_P

## 37 res_juv_moy

### 37.1 res_juv_moy_Vichy

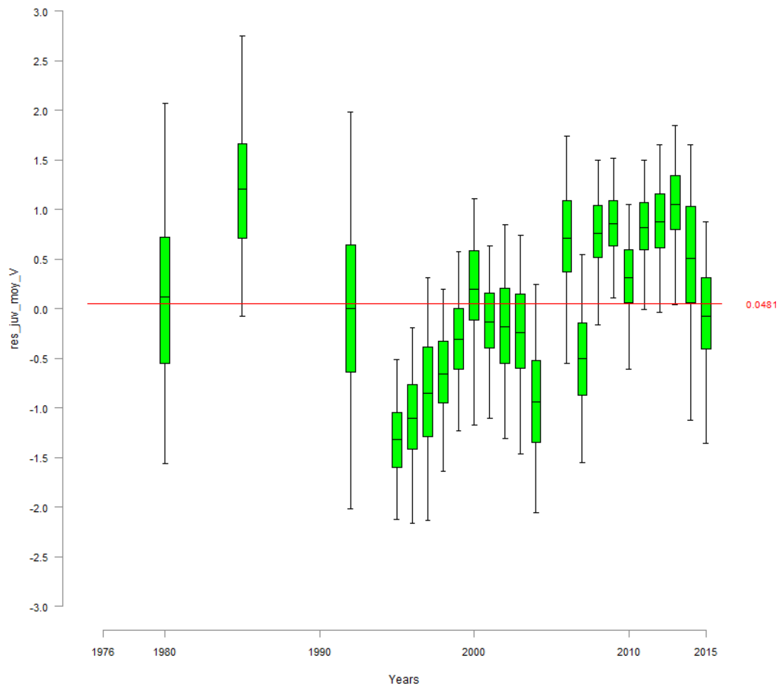


FIGURE 50 – res_juv_moy_V

### 37.2 res_juv_moy_Alagnon

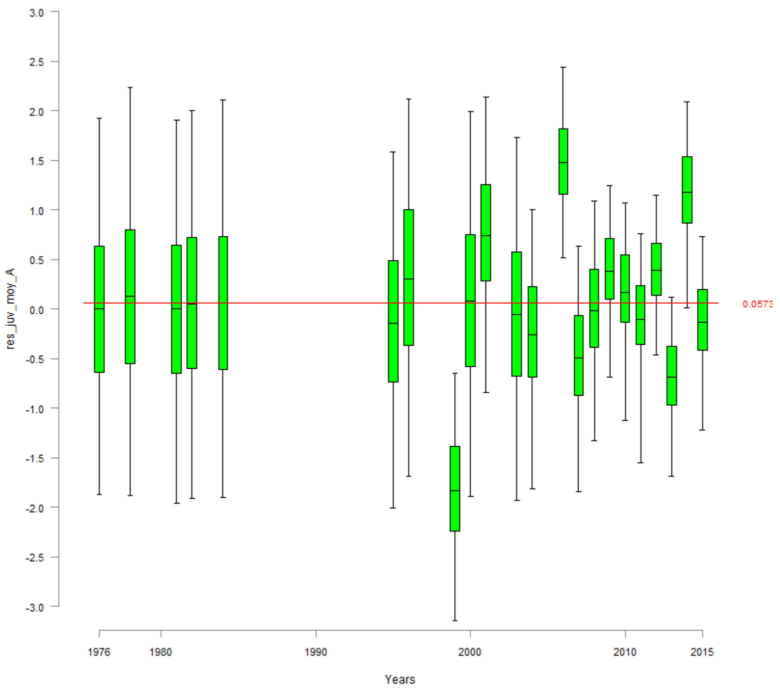


FIGURE 51 – res_juv_moy_A

### 37.3 res_juv_moy_Langeac

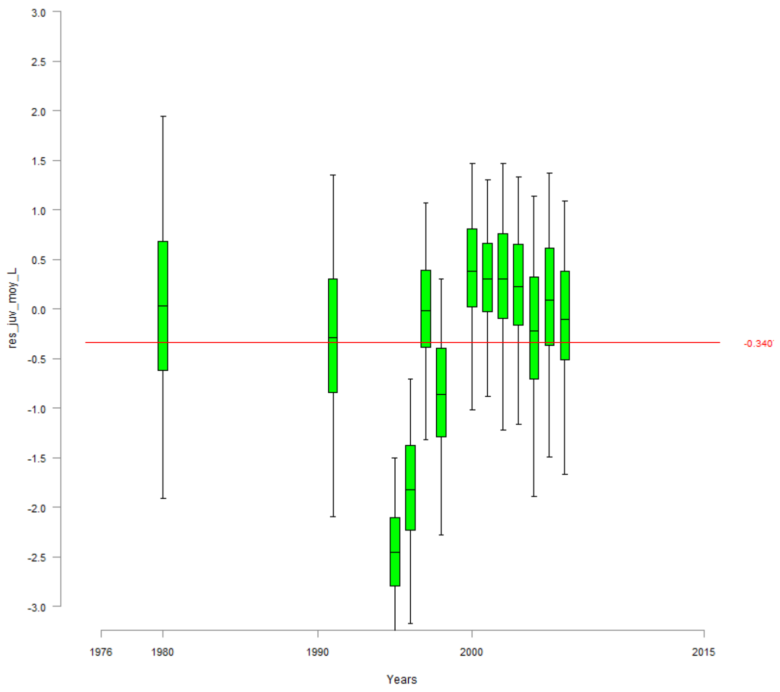


FIGURE 52 – res_juv_moy_L

### 37.4 res_juv_moy_Poutes

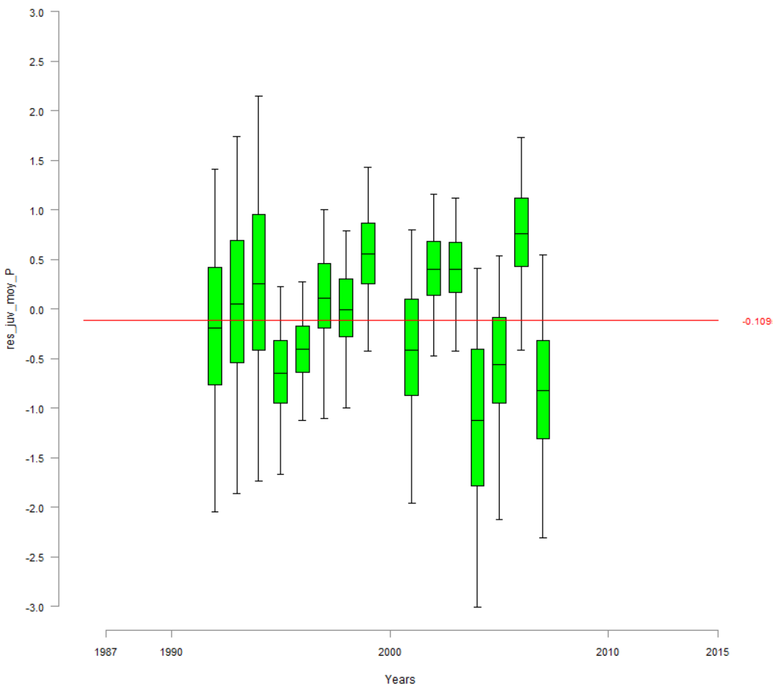


Figure 53 – res_juv_moy_P

# 5 Code du modÃ¨le 2016.12.19

```
Model{

#$$$$$$$$$$$$$$$#
#  FIRST PART           #
#$$$$$$$$$$$$$$$#

 #################
 # 1.CALIBRATION #
 #################

 d_moy~dgamma(1,0.001)
 beta_d~dgamma(0.001,0.001)

 inv_kappa ~ dgamma(0.001,0.001)
 kappa <-1/ inv_kappa

 #eta  ~ dgamma(0.001,0.001)

 L_sigma_p~dunif(0.0001,10)
 L_tau_p<-pow(L_sigma_p,-2)
 L_var_p<-1/L_tau_p

 L_mu_p ~dnorm(0,0.001)
 logit(mu_p)<-L_mu_p

 alpha_d<-d_moy * beta_d

 for (g in 1:calib){

  #======================
  # 1.1 Density part
  #======================
  d[g]~dgamma(alpha_d,beta_d)
  lambda_IA[g]<-kappa*d[g]     #kappa*pow(d[g],eta)
  EF_IA[g]~dpois(lambda_IA[g])
  lambda_N[g]<-(d[g]*S[g])-EF_IA[g]

  #Abundance follows a Poisson distribution
  N_tot[g]~dpois(lambda_N[g])I(,2000)

  L_p[g]~dnorm(L_mu_p,L_tau_p)
  logit(p[g]) <- L_p[g]

  # depletion pass part
  C_1[g]~dbin(p[g],N_tot[g])
  N_1[g]<-N_tot[g]-C_1[g]
  C_2[g]~dbin(p[g],N_1[g])

  #======================
  # 1.2 Posterior check
  #======================
  rep_lambda_IA[g]<-kappa*d[g]          #kappa*pow(d[g],eta)
  rep_EF_IA[g]~dpois(rep_lambda_IA[g])
  res_IA_EF[g]<-EF_IA[g] - rep_EF_IA[g]

  # calculation of the residuals for the predicted C1,C2,C3 conditionally to densities
  rep_C_1[g] ~ dbin(p[g],N_tot[g])
  rep_C_2[g] ~ dbin(p[g],N_1[g])

  res_C_1[g] <- C_1[g]-rep_C_1[g]
  res_C_2[g] <- C_2[g]-rep_C_2[g]
  res_C1_N_tot[g] <- (C_1[g]-rep_C_1[g])/N_tot[g]
  res_C_2_N_1[g] <- (C_2[g]-rep_C_2[g])/N_1[g]

}

 #==================================================
 # 1.3 cut of all the parameters of the calibration
 #==================================================
 L_mu_p_cut<-cut(L_mu_p)
 L_tau_p_cut<-cut(L_tau_p)
 kappa_cut<-cut(kappa)
 #eta_cut<-cut(eta)
 # on suppose que le parametre d'echelle est le meme au fil des années
 beta_d_cut<-cut(beta_d)


####################
# 2. REDD/SPAWNERS #
####################
 #=======================================================
 # 2.1 Parameters of the Redd/spawner relationship model
 #=======================================================
 mu_zone[1]~dgamma(1,0.001)
 mu_zone[2]<-1          #~dgamma(1,0.001)

 beta_zone~dgamma(0.01,0.01)

 alpha_zone[1]<- mu_zone[1]*beta_zone

 alpha_zone[2]<- mu_zone[2]*beta_zone

 #====================================
 # 2.2 Methodology and spatial effect
 #====================================
  #--------------------------
  # 2.2.1 Methodology effect
  #--------------------------
  hel_effect[1]<-1
  hel_effect[2]~dgamma(1,1)
  #-----------------------
  # 2.2.2 Spatial effect
  #-----------------------
  for (t in 1:T){
   zone_effect[t,1]~dgamma(alpha_zone[1],beta_zone)I(0.001,)
                zone_effect[t,2]~dgamma(alpha_zone[1],beta_zone)I(0.001,)
   zone_effect[t,3]~dgamma(alpha_zone[1],beta_zone)I(0.001,)

  }
  for (t in 12:T){
   zone_effect[t,4]~dgamma(alpha_zone[2],beta_zone)I(0.001,)
  }
```

```
  #------------------------------------
  # 2.2.3 Verification of both effects
  #------------------------------------
  diff_hel_effect<-1-hel_effect[2]
  P_diff_hel<-step(diff_hel_effect)

  diff_zone1_2<-mu_zone[1]-mu_zone[2]
  p_diff_zone1_2<-step(diff_zone1_2 )

 #======================
 # 2.3 Area prospected
 #======================
 #loops for proportion of area prospected
 for (t in 1:T){
  for (k in 1:3){
   logit(p_area[t,k])<-  L_p_area[t,k]
  }
 }

 for (t in 12:T){
  logit(p_area[t,4])<- L_p_area[t,4]
 }

 #======================
 # 2.4 Hyperparameters
 #======================
 sigma_Vichy <- sqrt( 1 / tau_vichy)    #~dunif(0.001,5)
 tau_vichy~dgamma(0.001,0.001)       #<-pow(sigma_Vichy,-2)
 L_mu_vichy~dnorm(0,0.01)

        sigma_p_alagnon<-sqrt(1/tau_p_alagnon)
        tau_p_alagnon~dgamma(0.01,0.01)

 sigma_p_langeac<-sqrt(1/tau_p_langeac)
 tau_p_langeac~dgamma(0.01,0.01)

 sigma_p_poutes<-sqrt(1/tau_p_poutes)
 tau_p_poutes~dgamma(0.01,0.01)

 I_surv[7] <- 1
 I_surv_prim[7] <- 1

 level_s~dnorm(0,1)

 for (t in 8:T){
  I_surv_prim[t] ~ dbern(0.5)
  I_surv[t] <-  I_surv[t-1] * I_surv_prim[t]
 }

 for (t in 7:11){
  min_N_1[t]<-tot_C[t]  + S_stocking[t]+2
  pool_juv[t]<-s_juv2ad* Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 * smolts_tot[t] )
         L_mu_Vichy_nm[t]<-log( s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 *
smolts_tot[t] )) + level_s * I_surv[t]
  mean_y_surv[t] <- s_juv2ad * exp(level_s * I_surv[t])
```

```
  N[t,1]~dlnorm(L_mu_Vichy_nm[t],tau_vichy)I(min_N_1[t],15000)
  res_Vichy[t] <- log(N[t,1]) - L_mu_Vichy_nm[t]
 }


############################
# 3. JUVENILE PRODUCTION   #
############################
 #=================================
 # 3.1 Beverthon & Holt parameters
 #=================================
 # BH slope parameter
 #not sure about the beta parameters ...
 zt~dbeta(1,9)#(1,2)
 a<-zt *8000   #I~dunif(1,8000)           #~dgamma(0.01,0.01)I(,8000)#(0.1,)     #~dlnorm(0,0.01)          #
 alpha_dd<- 1/a

 a_juv ~dbeta(2,2)
 alpha_dd_juv<- 1/a_juv

 Rmax~dunif(0,2)
 beta_dd<- 1 / Rmax

 s_egg~dbeta(2,2)
 s_juv2ad~dbeta(2,2)

 #=====================
 # 3.2 Hyperparameters
 #=====================
 alpha_tau <- mu_tau +1
        mu_tau ~ dgamma(0.1,0.1)I(0.000001,)
 beta_tau ~ dgamma(0.1,0.1)I(0.001,)

 tau_wild_moy~dgamma(alpha_tau,beta_tau)
 tau_wild_site~dgamma(alpha_tau,beta_tau)

 tau_juv_moy[1]~dgamma(0.01,0.01)
 tau_juv_site[1]~dgamma(0.01,0.01)
 tau_juv_moy[2]~dgamma(alpha_tau,beta_tau)
 tau_juv_site[2]~dgamma(alpha_tau,beta_tau)

 tau_egg_moy[1]~dgamma(0.01,0.01)I(0.01,)
 tau_egg_site[1]~dgamma(0.01,0.01)I(0.01,)
 tau_egg_moy[2]~dgamma(alpha_tau,beta_tau)#I(,50)
 tau_egg_site[2]~dgamma(alpha_tau,beta_tau)

 sigma_wild_moy <- sqrt( 1 / tau_wild_moy)
 sigma_wild_site <- sqrt( 1 / tau_wild_site)
 sigma_juv_moy <- sqrt( 1 / tau_juv_moy[2])
 sigma_juv_site <- sqrt( 1 / tau_juv_site[2])
 sigma_egg_moy <- sqrt( 1 / tau_egg_moy[2])
 sigma_egg_site <- sqrt( 1 / tau_egg_site[2])

 nu_wild_avg~dnorm(0,0.01)
 nu_wild[1] <-  -nu_wild_avg
        nu_wild[2] <- nu_wild_avg
```

```
   nu_wild[3] <- nu_wild_avg
   nu_wild[4] <- nu_wild_avg

   rho_poutes~dbeta(2,2)
   #=============================================================================
   # 3.3 Number of juveniles 0+ returning in the Allier river for a given year
   #=============================================================================
   # 0+ Juvenile returning in the Allier for a given year
   # and originating from the 3 areas of interest
   for (t in 7:T){
    Juv_tot[t,1] <- (1/3) * Juv[t-3,1] + (1/3) * Juv[t-4,1] + (1/3) * Juv[t-5,1]
        Juv_tot[t,2] <- (1/3) * Juv[t-3,2] + (1/3) * Juv[t-4,2] + (1/3) * Juv[t-5,2]
    Juv_tot[t,3] <- (1/3) * Juv[t-3,3] + (1/3) * Juv[t-4,3] + (1/3) * Juv[t-5,3]
   }
       for (t in 1:15){
             Juv_tot[t,4]<-0
       }
   for (t in 16:16){
    Juv_tot[t,4] <- (1/3) * Juv[t-3,4]
   }
   for (t in 17:17){
    Juv_tot[t,4] <- (1/3) * Juv[t-3,4] + (1/3) * Juv[t-4,4]
   }
   for (t in 18:T){
    Juv_tot[t,4] <- (1/3) * Juv[t-3,4] + (1/3) * Juv[t-4,4] + (1/3) * Juv[t-5,4]
   }
#      for (t in 30:30){
#       Juv_tot[t,2]<-1/3*Juv[t-3,2]
#      }
#      for (t in 31:31){
#       Juv_tot[t,2]<-1/3*Juv[t-3,2] + 1/3*Juv[t-4,2]
#      }
#      for (t in 32:T){
#       Juv_tot[t,2]<-1/3*Juv[t-3,2] + 1/3*Juv[t-4,2] + 1/3*Juv[t-5,2]
#      }
   for (t in 7:15){
    Juv_tot_system[t] <- Juv_tot[t,1]+ Juv_tot[t,2] +Juv_tot[t,3]
   }
   for (t in 16:T){
    Juv_tot_system[t] <- Juv_tot[t,1]+Juv_tot[t,2] +Juv_tot[t,3] +rho_poutes*Juv_tot[t,4]
   }
 #  for (t in 30:T){
 #     Juv_tot_system[t] <- Juv_tot[t,1] + Juv_tot[t,2] +Juv_tot[t,3] + rho_poutes*Juv_tot[t,4]
 #   }


   #=========================================================
   # 3.4 Probability of passing at Vichy, Alagnon, Langeac and Poutes
   #=========================================================
   # incorporating the effect that probability of passing at Langeac and Poutes is conditioned by the
amount of juvenile produced

   #Probability to reach Vichy if not catch downstream
   p_reach_V~dbeta(2,1)
   for (t in 1:T){
    C_dwn_reach[t] <- p_reach_V * C_dwn[t]
    tot_C[t] <-round( C_dwn_reach[t] + C_up[t])
   }
   for (t in 1:6){
    min_N_1[t]<- tot_C[t] + S_stocking[t]+2
    N[t,1]~dlnorm(6.9,0.0453)I(min_N_1[t],15000)
   }

   # For Alagnon, Langeac et Poutes : filter:
    # if negative : fish returning in smaller proportion than what was expected regarding juvenile
production
    # if positive : fish returning in higher proportion than what was expected regarding juvenile production
   adjust_p_L ~ dnorm(0,0.01)
   adjust_p_P ~ dnorm(0,0.01)
       adjust_p_A ~ dnorm(0,0.01)
   rho_station~dbeta(2,2)


   for (t in 1:4){

        ratio_juv_prod_A[t]~dbeta(1,3)
        ratio_juv_A[t]<-rho_station * (S_juv_JP[t,2]/(S_juv_JP[t,1]+S_juv_JP[t,2]+S_juv_JP[t,3])) + (1-
rho_station) * ratio_juv_prod_A[t]
        L_mu_p_alagnon[t]<-logit(ratio_juv_A[t]) + adjust_p_A
        L_p_alagnon[t]~dnorm(L_mu_p_alagnon[t],tau_p_alagnon)
        res_p_alagnon[t]<-L_p_alagnon[t]-L_mu_p_alagnon[t]

    ratio_juv_prod_L[t]~dbeta(3,1)
    ratio_juv_L[t]<- rho_station *(S_juv_JP[t,3]/(S_juv_JP[t,1]+S_juv_JP[t,3])) + (1 -  rho_station) *
ratio_juv_prod_L[t]
    L_ratio_juv_L[t] <- logit(ratio_juv_L[t])
    L_mu_p_langeac[t]<-L_ratio_juv_L[t] + adjust_p_L
    L_p_langeac[t]~dnorm(L_mu_p_langeac[t],tau_p_langeac)
    res_p_langeac[t] <- L_p_langeac[t] - L_mu_p_langeac[t]
   }
   for (t in 5:5){

    ratio_juv_prod_A[t]<- Juv[t-3,2] / ( Juv[t-3,1] + Juv[t-3,2]+ Juv[t-3,3])
        ratio_juv_A[t]<-rho_station * (S_juv_JP[t,2]/(S_juv_JP[t,1]+S_juv_JP[t,2]+S_juv_JP[t,3])) + (1-
rho_station) * ratio_juv_prod_A[t]
        L_mu_p_alagnon[t]<-logit(ratio_juv_A[t]) + adjust_p_A
        L_p_alagnon[t]~dnorm(L_mu_p_alagnon[t],tau_p_alagnon)
        res_p_alagnon[t]<-L_p_alagnon[t]-L_mu_p_alagnon[t]

    ratio_juv_prod_L[t] <-  Juv[t-3,3] / ( Juv[t-3,1] + Juv[t-3,3] )
    ratio_juv_L[t]<- rho_station * (S_juv_JP[t,3]/(S_juv_JP[t,1]+S_juv_JP[t,3])) + (1 -  rho_station) *
ratio_juv_prod_L[t]
    L_ratio_juv_L[t] <- logit(ratio_juv_L[t])
    L_mu_p_langeac[t]<-L_ratio_juv_L[t]+ adjust_p_L
    L_p_langeac[t]~dnorm(L_mu_p_langeac[t],tau_p_langeac)
    res_p_langeac[t] <- L_p_langeac[t] - L_mu_p_langeac[t]
   }
   for (t in 6:6){

        ratio_juv_prod_A[t]<- (Juv[t-3,2]+Juv[t-4,2]) / ( Juv[t-3,1] +Juv[t-4,1]+ Juv[t-3,2] +Juv[t-4,2]+ Juv[t-
3,3]+Juv[t-4,3])
```

```
            ratio_juv_A[t]<-rho_station * (S_juv_JP[t,2]/(S_juv_JP[t,1]+S_juv_JP[t,2]+S_juv_JP[t,3])) + (1-
rho_station) * ratio_juv_prod_A[t]
            L_mu_p_alagnon[t]<-logit(ratio_juv_A[t]) + adjust_p_A
            L_p_alagnon[t]~dnorm(L_mu_p_alagnon[t],tau_p_alagnon)
            res_p_alagnon[t]<-L_p_alagnon[t]-L_mu_p_alagnon[t]

     ratio_juv_prod_L[t] <-  (Juv[t-3,3] + Juv[t-4,3] ) / ( Juv[t-3,1] + Juv[t-4,1] + Juv[t-3,3] +Juv[t-4,3]  )
     ratio_juv_L[t]<- rho_station * (S_juv_JP[t,3]/(S_juv_JP[t,1]+S_juv_JP[t,3])) + (1 -  rho_station) *
ratio_juv_prod_L[t]
     L_ratio_juv_L[t] <- logit(ratio_juv_L[t])
     L_mu_p_langeac[t]<-L_ratio_juv_L[t]+ adjust_p_L
     L_p_langeac[t]~dnorm(L_mu_p_langeac[t],tau_p_langeac)
     res_p_langeac[t] <- L_p_langeac[t] - L_mu_p_langeac[t]
   }
   for (t in 7:11){

            ratio_juv_prod_A[t]<- Juv_tot[t,2] / ( Juv_tot[t,1] + Juv_tot[t,2]+ Juv_tot[t,3] )
            ratio_juv_A[t]<-rho_station * (S_juv_JP[t,2]/(S_juv_JP[t,1]+S_juv_JP[t,2]+S_juv_JP[t,3] ))+ (1-
rho_station) * ratio_juv_prod_A[t]
            L_mu_p_alagnon[t]<-logit(ratio_juv_A[t]) + adjust_p_A
            L_p_alagnon[t]~dnorm(L_mu_p_alagnon[t],tau_p_alagnon)
            res_p_alagnon[t]<-L_p_alagnon[t]-L_mu_p_alagnon[t]

     ratio_juv_prod_L[t] <-  Juv_tot[t,3] / ( Juv_tot[t,1] + Juv_tot[t,3] )
     ratio_juv_L[t]<- rho_station * (S_juv_JP[t,3]/(S_juv_JP[t,1]+S_juv_JP[t,3])) + (1 -  rho_station) *
ratio_juv_prod_L[t]
     L_ratio_juv_L[t] <- logit(ratio_juv_L[t])
     L_mu_p_langeac[t]<-L_ratio_juv_L[t]+ adjust_p_L
     L_p_langeac[t]~dnorm(L_mu_p_langeac[t],tau_p_langeac)
     res_p_langeac[t] <- L_p_langeac[t] - L_mu_p_langeac[t]
   }
   for (t in 12:15){

            ratio_juv_prod_A[t]<- Juv_tot[t,2] / ( Juv_tot[t,1] + Juv_tot[t,2]+ Juv_tot[t,3]+ Juv_tot[t,4]
*rho_poutes)
            ratio_juv_A[t]<-rho_station * (S_juv_JP[t,2]/(S_juv_JP[t,1]+S_juv_JP[t,2]+S_juv_JP[t,3]+
S_juv_JP[t,4])) + (1-rho_station) * ratio_juv_prod_A[t]
            L_mu_p_alagnon[t]<-logit(ratio_juv_A[t]) + adjust_p_A
            L_p_alagnon[t]~dnorm(L_mu_p_alagnon[t],tau_p_alagnon)
            res_p_alagnon[t]<-L_p_alagnon[t]-L_mu_p_alagnon[t]

     ratio_juv_prod_L[t] <-  Juv_tot[t,3] / ( Juv_tot[t,1] + Juv_tot[t,3]+ Juv_tot[t,4] *rho_poutes )
     ratio_juv_L[t]<- rho_station *
((S_juv_JP[t,3]+S_juv_JP[t,4])/(S_juv_JP[t,1]+S_juv_JP[t,3]+S_juv_JP[t,4])) + (1 -  rho_station) *
ratio_juv_prod_L[t]
     L_ratio_juv_L[t] <- logit(ratio_juv_L[t])
     L_mu_p_langeac[t]<-L_ratio_juv_L[t]+ adjust_p_L
     L_p_langeac[t]~dnorm(L_mu_p_langeac[t],tau_p_langeac)
     res_p_langeac[t] <- L_p_langeac[t] - L_mu_p_langeac[t]

     ratio_juv_prod_P[t] <- 0
     ratio_juv_P[t]<- rho_station * (S_juv_JP[t,4] / (S_juv_JP[t,3] + S_juv_JP[t,4]) ) + (1 -  rho_station) *
ratio_juv_prod_P[t]
     L_ratio_juv_P[t] <-  logit(ratio_juv_P[t])
     L_mu_p_poutes[t]<-L_ratio_juv_P[t] + adjust_p_P
     L_p_poutes[t]~dnorm(L_mu_p_poutes[t],tau_p_poutes)
     res_p_poutes[t] <- L_p_poutes[t] - L_mu_p_poutes[t]
   }
   for (t in 16:T){

            ratio_juv_prod_A[t]<- Juv_tot[t,2] / ( Juv_tot[t,1] + Juv_tot[t,2]+ Juv_tot[t,3]+ Juv_tot[t,4]
*rho_poutes)
            ratio_juv_A[t]<-rho_station * (S_juv_JP[t,2]/(S_juv_JP[t,1]+S_juv_JP[t,2]+S_juv_JP[t,3]+
S_juv_JP[t,4])) + (1-rho_station) * ratio_juv_prod_A[t]
            L_mu_p_alagnon[t]<-logit(ratio_juv_A[t]) + adjust_p_A
            L_p_alagnon[t]~dnorm(L_mu_p_alagnon[t],tau_p_alagnon)
            res_p_alagnon[t]<-L_p_alagnon[t]-L_mu_p_alagnon[t]

     ratio_juv_prod_L[t] <-  (Juv_tot[t,3] + Juv_tot[t,4]*rho_poutes) / ( Juv_tot[t,1] + Juv_tot[t,3] +
Juv_tot[t,4]*rho_poutes )
     ratio_juv_L[t]<-
rho_station*((S_juv_JP[t,3]+S_juv_JP[t,4])/(S_juv_JP[t,1]+S_juv_JP[t,3]+S_juv_JP[t,4])) + (1 -
rho_station) * ratio_juv_prod_L[t]
     L_ratio_juv_L[t] <- logit(ratio_juv_L[t])
     L_mu_p_langeac[t]<-L_ratio_juv_L[t]+ adjust_p_L
     L_p_langeac[t]~dnorm(L_mu_p_langeac[t],tau_p_langeac)
     res_p_langeac[t] <- L_p_langeac[t] - L_mu_p_langeac[t]

     ratio_juv_prod_P[t] <-  Juv_tot[t,4] *rho_poutes/ ( Juv_tot[t,3] + Juv_tot[t,4] *rho_poutes)
     ratio_juv_P[t]<- rho_station * (S_juv_JP[t,4] / (S_juv_JP[t,3] + S_juv_JP[t,4])) + (1 -  rho_station) *
ratio_juv_prod_P[t]
     L_ratio_juv_P[t] <- logit(ratio_juv_P[t])
     L_mu_p_poutes[t]<-L_ratio_juv_P[t]+ adjust_p_P
     L_p_poutes[t]~dnorm(L_mu_p_poutes[t],tau_p_poutes)
     res_p_poutes[t] <- L_p_poutes[t] - L_mu_p_poutes[t]
   }


#$$$$$$$$$$$$$$$$$#
#  SECONDE PART #
#$$$$$$$$$$$$$$$$$#

 ##############################################
 # 1. LOOP FOR YEARS (only downstream Poutès)     #
 ##############################################
 for (t in 1:11){
   #=========================
   # 1.1 Redd/Spawners part
   #=========================
   logit(p_alagnon[t])<-L_p_alagnon[t]
          logit(p_langeac[t])<- L_p_langeac[t]
   max_N_langeac[t]<- N_corr[t] - 1

          #without fish caught for breeeding or rod catches
   N_corrected[t] <- N[t,1] - tot_C[t] - S_stocking[t]
          N_corr[t]<-N_corrected[t]-N[t,2]
   N[t,2]~dbin(p_alagnon[t],N_corrected[t])
   N[t,3]~dbin(p_langeac[t],N_corr[t])
     #-----------------------------------
     #1.1.1 Number of potential spawners
     #-----------------------------------
     S_ts[t,1]<- max( N[t,1] - N[t,2] - N[t,3] - tot_C[t] - S_stocking[t],1)
```

```
    S_ts[t,2]<- max(N[t,2],1)
        S_ts[t,3]<-max(N[t,3],1)

        #ratio_S[t,1] <- S_ts[t,1] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3])
    #ratio_S[t,2] <- S_ts[t,2] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3])
    #ratio_S[t,3] <- S_ts[t,3] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3])


#==============================================================================
  # 1.2 Loop for zones (1= Vichy-Langeac, 2= Alagnon, 3= Langeac-Poutès, 4= upstream Poutès)

#==============================================================================
  for (i in 1:3){
    #-------------------------
    # 1.2.1 Redd/Spawners part
    #-------------------------
      #....................................
      # 1.2.1.1 estimation of the spawners
      #....................................
      R[t,i] ~dpois(lambda[t,i])
      lambda[t,i] <- S_ts[t,i]  *zone_effect[t,i] * hel_effect[1] *p_area[t,i]

      #residus calculés pour êtres centrés sur 0 avec varaince homogene
      res_R[t,i]<-(R[t,i]-lambda[t,i])/sqrt(lambda[t,i])
      #..............................
      # 1.2.1.2 Cut of all parameters
      #..............................
      lambda_cut[t,i]<-cut(lambda[t,i])
      R_rep[t,i]~dpois(lambda_cut[t,i])
    #--------------------------
    # 1.2.2 Juvenile production
    #--------------------------
    # I_juv_moy = indicator for stocking of 0+ or not
    # I_egg_moy = indicator for stocking of eggs or not
        #d_tot_moy without taking into acount area for the stocked juveniles (data only from year 31)
    d_tot_moy[t+1,i] <- d_wild_moy[t+1,i] + I_juv_moy[t+1,i] * d_juv_moy[t+1,i]
    Juv[t+1,i] <- d_tot_moy[t+1,i]*S_juv_JP[t+1,i]
      #.......................
      # 1.2.2.1 Wild component
      #.......................
      log(d_wild_moy[t+1,i]) <- L_d_wild_moy[t+1,i]
      L_d_wild_moy[t+1,i] ~ dnorm(L_mu_d_wild[t+1,i],tau_wild_moy)I(-6.91,1.09)
      L_mu_d_wild[t+1,i] <-  log((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd * (S_ts[t,i]/S_juv_JP[t,i]))) +
nu_wild[i]
      res_wild_moy[t+1,i] <-  L_d_wild_moy[t+1,i] - L_mu_d_wild[t+1,i]
      #...................................
      # 1.2.2.2 stocked juvenile component
      #...................................
      log(d_juv_moy[t+1,i]) <- L_d_juv_moy[t+1,i]
      L_d_juv_moy[t+1,i] ~ dnorm(L_mu_d_juv[t+1,i],tau_juv_moy[ I_juv_moy[t+1,i]+1])I(-6.91,1.09)
        #<- L_mu_d_juv[t+1,i]   #
      Rmax_juv_temp[t+1,i] <- (  Rmax - ((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd *
(S_ts[t,i]/S_juv_JP[t,i]))) ) * exp(nu_wild[i])
      Rmax_juv[t+1,i] <- max(Rmax_juv_temp[t+1,i] ,0.000001)
      beta_dd_juv[t+1,i] <- 1 / Rmax_juv[t+1,i]

      L_mu_d_juv[t+1,i] <-  I_juv_moy[t+1,i] * log( (stock_juv[t+1,i]/S_juv_JP[t+1,i]) /
(alpha_dd_juv/exp(nu_wild[i]) +  beta_dd_juv[t+1,i] * (stock_juv[t+1,i]/S_juv_JP[t+1,i])))
      res_juv_moy[t+1,i] <- L_d_juv_moy[t+1,i] - L_mu_d_juv[t+1,i]
      # getting out of the zone loop, one loop for each zones and the local densitie
      # to avoid using 3 dimensions matrix
  }
    #-----------------------------------
    # 1.2.3 Successive removal fisheries
    #-----------------------------------
    #  I_site_juv_V/L/P = indicator for presence/absence of stocking on the site
    # loop for sites with successive removal EF (DE LURY)
        # Pas de pêche Delury pour l'Alagnon
      #................................
      # 1.2.3.1 zone 1 : Vichy Langeac
      #................................
      for (k in 1:J[t+1,1]){
        d_V[t+1,k]<- d_wild_V[t+1,k] + I_site_juv_V[t+1,k] * d_juv_V[t+1,k]
        log(d_wild_V[t+1,k])<-L_d_wild_V[t+1,k]
        L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)I(-6.91,1.09)
        log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
        L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[ I_site_juv_V[t+1,k] + 1])I(-6.91,1.09)


        lambda_N_V[t+1,k]<-d_V[t+1,k]*S_depl_V[t+1,k]
        #Abundance follows a Poisson distribution
        N_tot_V[t+1,k]~dpois(lambda_N_V[t+1,k])

        L_p_V[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
        logit(p_V[t+1,k]) <-L_p_V[t+1,k]

        C_1_V[t+1,k]~dbin(p_V[t+1,k],N_tot_V[t+1,k])
        N_1_V[t+1,k]<-N_tot_V[t+1,k]-C_1_V[t+1,k]
        #not all sites have 2 pass, this vector show which sites does
          for (h in 1:pass_2_V[t+1,k]){
            C_2_V[t+1,k]~dbin(p_V[t+1,k],N_1_V[t+1,k])
            N_2_V[t+1,k]<-N_1_V[t+1,k]-C_2_V[t+1,k]
          }
      }

      #.................................
      # 1.2.3.3 zone 3 : Langeac Poutes
      #................................
      for (k in 1:J[t+1,3]){
        d_L[t+1,k]<- d_wild_L[t+1,k] + I_site_juv_L[t+1,k] * d_juv_L[t+1,k]
        log(d_wild_L[t+1,k])<-L_d_wild_L[t+1,k]
        L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)I(-6.91,1.09)
        log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
        L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ I_site_juv_L[t+1,k] + 1])I(-6.91,1.09)

        lambda_N_L[t+1,k]<-d_L[t+1,k]*S_depl_L[t+1,k]
        #Abundance follows a Poisson distribution
        N_tot_L[t+1,k]~dpois(lambda_N_L[t+1,k])

        L_p_L[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
        logit(p_L[t+1,k]) <-L_p_L[t+1,k]
```

```
        C_1_L[t+1,k]~dbin(p_L[t+1,k],N_tot_L[t+1,k])
        N_1_L[t+1,k]<-N_tot_L[t+1,k]-C_1_L[t+1,k]
        #not all sites have 2 pass, this vector show which sites does
          for (h in 1:pass_2_L[t+1,k]){
            C_2_L[t+1,k]~dbin(p_L[t+1,k],N_1_L[t+1,k])
            N_2_L[t+1,k]<-N_1_L[t+1,k]-C_2_L[t+1,k]
            #not all sites have 3 pass, this vector show which sites does
            for (m in 1:pass_3_L[t+1,k]){
              C_3_L[t+1,k]~dbin(p_L[t+1,k],N_2_L[t+1,k])
            }
          }
      }
          }



####################################################################################
####
  # 2. LOOP FOR YEARS (all zones mais pas encore de juvéniles à Poutès - seulement en année T=16)
          #

####################################################################################
####
  for (t in 12:22){
    #=========================
    # 2.1 Redd/Spawners part
    #=========================
    logit(p_alagnon[t])<-L_p_alagnon[t]
          logit(p_langeac[t])<- L_p_langeac[t]
          logit(p_poutes[t])<- L_p_poutes[t]
          pool_juv[t]<-s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 * smolts_tot[t] )
          L_mu_Vichy_nm[t]<-log(s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 *
smolts_tot[t] )) + level_s *I_surv[t]
          mean_y_surv[t] <- s_juv2ad * exp(level_s * I_surv[t])

    min_N_1[t]<-max(N[t,4]+2,tot_C[t] +2)+S_stocking[t]
    N[t,1]~dlnorm(L_mu_Vichy_nm[t],tau_vichy)I(min_N_1[t],15000)
          #without fish caught for breeeding or rod catches
    N_corrected[t] <- N[t,1] - tot_C[t] - S_stocking[t]
          N_corr[t]<-N_corrected[t] - N[t,2]
          res_Vichy[t] <- log(N[t,1]) - L_mu_Vichy_nm[t]

    max_N_langeac[t]<- N_corr[t] - 1
    min_L_P[t]<-max(min_L[t], N[t,4]+1)
          N[t,2]~dbin(p_alagnon[t],N_corrected[t]) #Pas de I car on pourrait avoir 0 sur alagnon et tout sur
allier une année donnée
    N[t,3]~dbin(p_langeac[t],N_corr[t])I(min_L_P[t],)

    max_N_poutes[t]<-N[t,3]-1
    N[t,4]~dbin(p_poutes[t],N[t,3])
           #-----------------------------------
      # 2.1.1 Number of potential spawners
      #-----------------------------------
      #mu_S_ts[t,1]<- N[t,1] - N[t,2] - N[t,3] - S_stocking[t]
      #mu_S_ts[t,2]<- N[t,2]
      #mu_S_ts[t,3]<- N[t,3]-N[t,4]
          #test[t]<-mu_S_ts[t,1]-S_ts[t,1]

      S_ts[t,1]<- max(N[t,1] - N[t,2] - N[t,3] - S_stocking[t] - tot_C[t] ,1)
      S_ts[t,2]<- max( N[t,2],1)
      S_ts[t,3]<- max( N[t,3]-N[t,4],1)
         S_ts[t,4]<-max( N[t,4],1)
      #ratio_S[t,1] <- S_ts[t,1] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
      #ratio_S[t,2] <- S_ts[t,2] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
      #ratio_S[t,3] <- S_ts[t,3] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
          #ratio_S[t,4] <- S_ts[t,4] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])

#===================================================================================
    # 2.2 Loop for zones (1= Vichy-Langeac, 2= Alagnon, 3= Langeac-Poutès, 4= upstream Poutès)

#===================================================================================
    for (i in 1:4){
      #--------------------------
      # 2.2.1 Redd/Spawners part
      #--------------------------
        #....................................
        # 2.2.1.1 estimation of the spawners
        #....................................
        R[t,i]~dpois(lambda[t,i])
        lambda[t,i] <- S_ts[t,i]  *zone_effect[t,i]* hel_effect[1] *p_area[t,i]
                        res_R[t,i]<-(R[t,i]-lambda[t,i])/sqrt(lambda[t,i])
        #..............................
        # 2.2.1.2 Cut of all parameters
        #..............................
        lambda_cut[t,i]<-cut(lambda[t,i])
        R_rep[t,i]~dpois(lambda_cut[t,i])

      #---------------------------
      # 2.2.2 Juvenile production
      #---------------------------
      # I_juv_moy = indicator for stocking of 0+ or not
      # I_egg_moy = indicator for stocking of eggs or not
          #d_tot_moy without taking into acount area for the stocked juveniles (data only from year 31)
          d_tot_moy[t+1,i] <- d_wild_moy[t+1,i] + I_juv_moy[t+1,i] * d_juv_moy[t+1,i] + I_egg_moy[t+1,i] *
d_egg_moy_surf[t+1,i]
      Juv[t+1,i] <- d_tot_moy[t+1,i]*S_juv_JP[t+1,i]
      d_egg_moy_surf[t+1,i] <-  d_egg_moy[t+1,i]
              #.......................
        # 2.2.2.1 Wild component
        #.......................
        log(d_wild_moy[t+1,i]) <- L_d_wild_moy[t+1,i]
              L_d_wild_moy[t+1,i] ~ dnorm(L_mu_d_wild[t+1,i],tau_wild_moy)I(-6.91,1.09) #<-
L_mu_d_wild[t+1,i]        #
        L_mu_d_wild[t+1,i] <-  log((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd * (S_ts[t,i]/S_juv_JP[t,i]))) +
nu_wild[i]
              res_wild_moy[t+1,i] <-  L_d_wild_moy[t+1,i] -  L_mu_d_wild[t+1,i]
              #....................................
        # 2.2.2.2 stocked juvenile component
        #....................................
        log(d_juv_moy[t+1,i]) <- L_d_juv_moy[t+1,i]
        L_d_juv_moy[t+1,i] ~ dnorm(L_mu_d_juv[t+1,i],tau_juv_moy[ I_juv_moy[t+1,i]+1])I(,1.09)
```

```
        # We recaculate the Rmax "available" to stocked 0+  by substracting wild 0+ density and stocked
eggs density
        # to the total Rmax of the density dependence relationship
        Rmax_juv_temp[t+1,i] <- (  Rmax - ((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd *
(S_ts[t,i]/S_juv_JP[t,i]))) ) * exp(nu_wild[i])
        Rmax_juv[t+1,i]<-max( Rmax_juv_temp[t+1,i] , 0.000001)
        beta_dd_juv[t+1,i] <- 1 / Rmax_juv[t+1,i]
        L_mu_d_juv[t+1,i] <- I_juv_moy[t+1,i] * log( (stock_juv[t+1,i]/S_juv_JP[t+1,i]) /
(alpha_dd_juv/exp(nu_wild[i]) +  beta_dd_juv[t+1,i]*(stock_juv[t+1,i]/S_juv_JP[t+1,i])))
        res_juv_moy[t+1,i] <- L_d_juv_moy[t+1,i] - L_mu_d_juv[t+1,i]
        #................................
        # 2.2.2.3 stocked egg component
        #................................
        log(d_egg_moy[t+1,i]) <- L_d_egg_moy[t+1,i]
        L_d_egg_moy[t+1,i]~  dnorm(L_mu_d_egg[t+1,i],tau_egg_moy[ I_egg_moy[t+1,i] +1])I(-6.91,1.09) #
        res_egg_moy[t+1,i] <- L_d_egg_moy[t+1,i] - L_mu_d_egg[t+1,i]

        # I_egg_unit = indicator of presence of incubators or not: only zone 1 and 3 concerned
        # I_egg_VL = indicator for incubators in zone 1
        # I_egg_LP = indicator for incubators in zone 2
        # I_list_inc = indicator for each incabutors loaded or not
        L_mu_d_egg[t+1,i] <-  equals(i,1) *
                    log(
                    (1- I_egg_moy[t+1,1]) +

                    (s_egg * ((stock_egg[t+1,1] + stock_egg[t+1,2] + stock_egg[t+1,3] + stock_egg[t+1,4]) /
S_juv_JP[t+1,1] ))
                    )
                   +
                   equals(i,3) *
                    log(
                    (1- I_egg_moy[t+1,3]) +
                    (s_egg * ((stock_egg[t+1,5] +stock_egg[t+1,6]) / S_juv_JP[t+1,3] ))
                    )

        # getting out of the zone loop, one loop for each zones and the local densitie
        # to avoid using 3 dimensions matrix
   }
           #------------------------------------
     # 2.2.3 Successive removal fisheries
     #------------------------------------
     #  I_site_juv_V/L/P = indicator for presence/absence of stocking on the site
           # loop for  sites with successive removal EF

      #................................
      # 2.2.3.1 zone 1 : Vichy Langeac
      #................................
      for (k in 1:J[t+1,1]){
        d_V[t+1,k]<- d_wild_V[t+1,k] + I_site_juv_V[t+1,k] * d_juv_V[t+1,k]
        log(d_wild_V[t+1,k])<-L_d_wild_V[t+1,k]
        L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)I(-6.91,1.09)

        log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
        L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[I_site_juv_V[t+1,k] + 1])I(-6.91,1.09)

        lambda_N_V[t+1,k]<-d_V[t+1,k]*S_depl_V[t+1,k]
        #Abundance follows a Poisson distribution
        N_tot_V[t+1,k]~dpois(lambda_N_V[t+1,k])

        L_p_V[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
        logit(p_V[t+1,k]) <-L_p_V[t+1,k]

        C_1_V[t+1,k]~dbin(p_V[t+1,k],N_tot_V[t+1,k])
        N_1_V[t+1,k]<-N_tot_V[t+1,k]-C_1_V[t+1,k]
        #not all sites have 2 pass, this vector show which sites does
          for (h in 1:pass_2_V[t+1,k]){
            C_2_V[t+1,k]~dbin(p_V[t+1,k],N_1_V[t+1,k])
            N_2_V[t+1,k]<-N_1_V[t+1,k]-C_2_V[t+1,k]
          }
      }

      #................................
      # 2.2.3.3 zone 3 : Langeac Poutes
      #................................
      for (k in 1:J[t+1,3]){
        d_L[t+1,k]<- d_wild_L[t+1,k] + I_site_juv_L[t+1,k] * d_juv_L[t+1,k]
        log(d_wild_L[t+1,k])<-L_d_wild_L[t+1,k]
        L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)I(-6.91,3)

        log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
        L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ I_site_juv_L[t+1,k] + 1])I(-6.91,1.09)

        lambda_N_L[t+1,k]<-d_L[t+1,k]*S_depl_L[t+1,k]
        #Abundance follows a Poisson distribution
        N_tot_L[t+1,k]~dpois(lambda_N_L[t+1,k])

        L_p_L[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
        logit(p_L[t+1,k]) <-L_p_L[t+1,k]

        C_1_L[t+1,k]~dbin(p_L[t+1,k],N_tot_L[t+1,k])
        N_1_L[t+1,k]<-N_tot_L[t+1,k]-C_1_L[t+1,k]
        #not all sites have 2 pass, this vector show which sites does
          for (h in 1:pass_2_L[t+1,k]){
            C_2_L[t+1,k]~dbin(p_L[t+1,k],N_1_L[t+1,k])
            N_2_L[t+1,k]<-N_1_L[t+1,k]-C_2_L[t+1,k]
            #not all sites have 2 pass, this vector show which sites does
              for (m in 1:pass_3_L[t+1,k]){
                C_3_L[t+1,k]~dbin(p_L[t+1,k],N_2_L[t+1,k])
              }
          }
      }
      #..................................
      # 2.2.3.4 zone 4 : upstream Poutes
      #..................................
      for (k in 1:J[t+1,4]){
        d_P[t+1,k]<- d_wild_P[t+1,k] + I_site_juv_P[t+1,k] * d_juv_P[t+1,k]
        log(d_wild_P[t+1,k])<-L_d_wild_P[t+1,k]
        L_d_wild_P[t+1,k] ~ dnorm( L_d_wild_moy[t+1,4] , tau_wild_site)I(-6.91,1.09)

        log(d_juv_P[t+1,k]) <- L_d_juv_P[t+1,k]
        L_d_juv_P[t+1,k] ~ dnorm( L_d_juv_moy[t+1,4] , tau_juv_site[ I_site_juv_P[t+1,k] + 1])I(-6.91,1.09)
```

```
      lambda_N_P[t+1,k]<-d_P[t+1,k]*S_depl_P[t+1,k]
      #Abundance follows a Poisson distribution
      N_tot_P[t+1,k]~dpois(lambda_N_P[t+1,k])

      L_p_P[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
      logit(p_P[t+1,k]) <-L_p_P[t+1,k]

      C_1_P[t+1,k]~dbin(p_P[t+1,k],N_tot_P[t+1,k])
      N_1_P[t+1,k]<-N_tot_P[t+1,k]-C_1_P[t+1,k]
      #not all sites have 2 pass, this vector show which sites does
        for (h in 1:pass_2_P[t+1,k]){
          C_2_P[t+1,k]~dbin(p_P[t+1,k],N_1_P[t+1,k])
          N_2_P[t+1,k]<-N_1_P[t+1,k]-C_2_P[t+1,k]
        }
    }
  #--------------------------
  # 2.2.4 5 min IA fisheries
  #--------------------------
    #................................
    # 2.2.4.1 zone 1 : Vichy Langeac
    #................................
    for (k in 1:K[t+1,1]){
      d_V[t+1,k]<- d_wild_V[t+1,k] + I_site_juv_V[t+1,k] * d_juv_V[t+1,k] +  I_site_egg_V[t+1,k] *
d_egg_V[t+1,k]
      log(d_wild_V[t+1,k])<-L_d_wild_V[t+1,k]
      L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)I(-6.91,3)

      log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
      L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[ I_site_juv_V[t+1,k] + 1])I(-6.91,1.09)

      log(d_egg_V[t+1,k]) <- L_d_egg_V[t+1,k]
      L_d_egg_moy_V_inc[t+1,k]<- I_site_egg_V[t+1,k] * ( L_d_egg_moy[t+1,1] +  log( S_juv_JP[t+1,1]) -
log(S_inc_JP[t+1,1]) )
      L_d_egg_V[t+1,k] ~  dnorm( L_d_egg_moy_V_inc[t+1,k] , tau_egg_site[I_site_egg_V[t+1,k] + 1])I(-
6.91,1.09)

      #5minute EF part
      lambda_IA_V[t+1,k]<-kappa_cut*d_V[t+1,k]
              EF_IA_V[t+1,k]~dpois(lambda_IA_V[t+1,k])
    }

        #.................................
    # 2.2.4.2 zone 2 : Alagnon
    #.................................
    for (k in 1:K[t+1,2]){
      d_A[t+1,k]<- d_wild_A[t+1,k] + I_site_juv_A[t+1,k] * d_juv_A[t+1,k]
      log(d_wild_A[t+1,k])<-L_d_wild_A[t+1,k]
      L_d_wild_A[t+1,k] ~ dnorm( L_d_wild_moy[t+1,2] , tau_wild_site)I(-6.91,1.09)

      log(d_juv_A[t+1,k]) <- L_d_juv_A[t+1,k]
      L_d_juv_A[t+1,k] ~ dnorm( L_d_juv_moy[t+1,2] , tau_juv_site[ I_site_juv_A[t+1,k] + 1])I(-6.91,1.09)

      #5minute EF part
      lambda_IA_A[t+1,k]<-kappa_cut*d_A[t+1,k]
      EF_IA_A[t+1,k]~dpois(lambda_IA_A[t+1,k])
    }
    #.................................
    # 2.2.4.3 zone 3 : Langeac Poutes
    #.................................
    for (k in 1:K[t+1,3]){
      d_L[t+1,k]<- d_wild_L[t+1,k] + I_site_juv_L[t+1,k] * d_juv_L[t+1,k] +  I_site_egg_L[t+1,k] *
d_egg_L[t+1,k]
      log(d_wild_L[t+1,k])<-L_d_wild_L[t+1,k]
      L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)I(-6.91,1.09)

      log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
      L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ I_site_juv_L[t+1,k] + 1])I(-6.91,1.09)

      log(d_egg_L[t+1,k]) <- L_d_egg_L[t+1,k]
      L_d_egg_moy_L_inc[t+1,k]<-  I_site_egg_L[t+1,k] * ( L_d_egg_moy[t+1,3] +  log( S_juv_JP[t+1,3]) -
log(S_inc_JP[t+1,3]) )
      L_d_egg_L[t+1,k] ~  dnorm( L_d_egg_moy_L_inc[t+1,k] , tau_egg_site[I_site_egg_L[t+1,k] + 1])I(-
6.91,1.09)

      #5minute EF part
      lambda_IA_L[t+1,k]<-kappa_cut*d_L[t+1,k]
      EF_IA_L[t+1,k]~dpois(lambda_IA_L[t+1,k])
    }
    #...................................
    # 2.2.4.3 zone 4 : upstream Poutes
    #...................................
    for (k in 1:K[t+1,4]){
      d_P[t+1,k]<- d_wild_P[t+1,k] + I_site_juv_P[t+1,k] * d_juv_P[t+1,k]
      log(d_wild_P[t+1,k])<-L_d_wild_P[t+1,k]
      L_d_wild_P[t+1,k] ~ dnorm( L_d_wild_moy[t+1,4] , tau_wild_site)I(-6.91,1.09)

      log(d_juv_P[t+1,k]) <- L_d_juv_P[t+1,k]
      L_d_juv_P[t+1,k] ~ dnorm( L_d_juv_moy[t+1,4] , tau_juv_site[ I_site_juv_P[t+1,k] + 1])I(-6.91,1.09)

      #5minute EF part
      lambda_IA_P[t+1,k]<-kappa_cut*d_P[t+1,k]
      EF_IA_P[t+1,k]~dpois(lambda_IA_P[t+1,k])
    }
  }


  ##############################################
  # 3. Change in redd count methodology #
  ##############################################
  for (t in 23:30){
    #========================
    # 3.1 Redd/Spawners part
    #========================
    logit(p_alagnon[t])<-L_p_alagnon[t]
          logit(p_langeac[t])<- L_p_langeac[t]
    logit(p_poutes[t])<- L_p_poutes[t]
    pool_juv[t]<-s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 * smolts_tot[t] )
    L_mu_Vichy_nm[t]<-log(s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 *
smolts_tot[t] )) + level_s *I_surv[t]
    mean_y_surv[t] <- s_juv2ad * exp(level_s * I_surv[t])
```

```
    # max is only added for the year we only have a minimum figure at vichy
    temp[t]<-max(tot_C[t]  + S_stocking[t]+2,min_N_V[t])

    min_N_1[t]<-max(N[t,4]+2,temp[t] +2)+S_stocking[t]
    N[t,1]~dlnorm(L_mu_Vichy_nm[t],tau_vichy)I(min_N_1[t],15000)
    #without fish caught for breeeding or rod catches
    N_corrected[t] <- N[t,1] - tot_C[t] - S_stocking[t]
          N_corr[t]<-N_corrected[t] - N[t,2]
    res_Vichy[t] <- log(N[t,1]) - L_mu_Vichy_nm[t]

          max_N_langeac[t]<- N_corr[t] - 1
    min_L_P[t]<-max(min_L[t], N[t,4]+1)
    N[t,2]~dbin(p_alagnon[t],N_corrected[t]) #Pas de I car on pourrait avoir 0 sur alagnon et tout sur allier
une année donnée
    N[t,3]~dbin(p_langeac[t],N_corr[t])I(min_L_P[t],)

    max_N_poutes[t]<-N[t,3]-1
    N[t,4]~dbin(p_poutes[t],N[t,3])
           #-------------------------------------
      # 3.1.1 Number of potential spawners
      #-------------------------------------
      #mu_S_ts[t,1]<- N[t,1] - N[t,2] - S_stocking[t]
      #mu_S_ts[t,2]<- N[t,2]-N[t,3]
           #test[t]<-mu_S_ts[t,1]-S_ts[t,1]

          S_ts[t,1]<- max(N[t,1] - N[t,2] - N[t,3] - S_stocking[t] - tot_C[t] ,1)
     S_ts[t,2]<- max( N[t,2],1)
     S_ts[t,3]<- max( N[t,3]-N[t,4],1)
          S_ts[t,4]<-max( N[t,4],1)

          #ratio_S[t,1] <- S_ts[t,1] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
     #ratio_S[t,2] <- S_ts[t,2] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
     #ratio_S[t,3] <- S_ts[t,3] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
           #ratio_S[t,4] <- S_ts[t,4] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])

#===============================================================================
    # 3.2 Loop for zones (1= Vichy-Langeac, 2= Alagnon, 3= Langeac-Poutès, 4= upstream Poutès)

#===============================================================================
    for (i in 1:4){
     #--------------------------
     # 3.2.1 Redd/Spawners part
     #--------------------------
      #....................................
      # 3.2.1.1 Estimation of the spawners
      #....................................
      R[t,i]~dpois(lambda[t,i])
      lambda[t,i] <- S_ts[t,i]  *zone_effect[t,i]* hel_effect[2] *p_area[t,i]
      res_R[t,i]<-(R[t,i]-lambda[t,i])/sqrt(lambda[t,i])
      #.............................
      # 3.2.1.2 Cut of all parameters
      #.............................
      lambda_cut[t,i]<-cut(lambda[t,i])
      R_rep[t,i]~dpois(lambda_cut[t,i])

      #S_counter[t,i]<-R[t,i] / (zone_effect[t,i] *p_area[t,i])

      #chisq_disc_R[t,i]<- (R[t,i]-lambda[t,i]) * (R[t,i]-lambda[t,i]) / (lambda[t,i])
      #chisq_disc_R_rep[t,i]<- (R_rep[t,i]-lambda[t,i]) * (R_rep[t,i]-lambda[t,i]) / (lambda[t,i])
     #--------------------------
     # 3.2.2 Juvenile production
     #--------------------------
     # I_juv_moy = indicator for stocking of 0+ or not
     # I_egg_moy = indicator for stocking of eggs or not
          #d_tot_moy without taking into acount area for the stocked juveniles (data only from year 31)
          d_tot_moy[t+1,i] <- d_wild_moy[t+1,i] + I_juv_moy[t+1,i] * d_juv_moy[t+1,i] + I_egg_moy[t+1,i] *
d_egg_moy_surf[t+1,i]
          Juv[t+1,i] <- d_tot_moy[t+1,i]*S_juv_JP[t+1,i]
          d_egg_moy_surf[t+1,i] <- d_egg_moy[t+1,i]
      #........................
      # 3.2.2.1 Wild component
      #........................
      log(d_wild_moy[t+1,i]) <- L_d_wild_moy[t+1,i]
      L_d_wild_moy[t+1,i] ~ dnorm(L_mu_d_wild[t+1,i],tau_wild_moy)I(-6.91,1.09)       #<-
L_mu_d_wild[t+1,i]      #
      L_mu_d_wild[t+1,i] <-  log((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd * (S_ts[t,i]/S_juv_JP[t,i]))) +
nu_wild[i]
          res_wild_moy[t+1,i] <-  L_d_wild_moy[t+1,i] -  L_mu_d_wild[t+1,i]
          #..................................
      # 3.2.2.2 Stocked juvenile component
      #..................................
      log(d_juv_moy[t+1,i]) <- L_d_juv_moy[t+1,i]
      L_d_juv_moy[t+1,i] ~ dnorm(L_mu_d_juv[t+1,i],tau_juv_moy[ I_juv_moy[t+1,i]+1])I(,1.09)

      # We recaculate the Rmax "available" to stocked 0+  by substracting wild 0+ density and stocked
eggs density
      # to the total Rmax of the density dependence relationship
      Rmax_juv_temp[t+1,i] <- (  Rmax - ((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd *
(S_ts[t,i]/S_juv_JP[t,i]))) ) * exp(nu_wild[i])
      Rmax_juv[t+1,i] <- max(Rmax_juv_temp[t+1,i] ,0.000001)
          beta_dd_juv[t+1,i] <- 1 / Rmax_juv[t+1,i]
          L_mu_d_juv[t+1,i] <-  I_juv_moy[t+1,i] * log( (stock_juv[t+1,i]/S_juv_JP[t+1,i]) /
(alpha_dd_juv/exp(nu_wild[i]) +  beta_dd_juv[t+1,i] * (stock_juv[t+1,i]/S_juv_JP[t+1,i])))
      res_juv_moy[t+1,i] <- L_d_juv_moy[t+1,i] - L_mu_d_juv[t+1,i]
          #...............................
      # 3.2.2.3 Stocked egg component
      #...............................
      log(d_egg_moy[t+1,i]) <- L_d_egg_moy[t+1,i]
      L_d_egg_moy[t+1,i]~  dnorm(L_mu_d_egg[t+1,i],tau_egg_moy[ I_egg_moy[t+1,i] +1])I(-6.91,1.09)  #
      res_egg_moy[t+1,i] <- L_d_egg_moy[t+1,i] - L_mu_d_egg[t+1,i]

      # I_egg_unit = indicator of presence of incubators or not: only zone 1 and 2 concerned
      # I_egg_VL = indicator for incubators in zone 1
      # I_egg_LP = indicator for incubators in zone 2
      # I_list_inc = indicator for each incabutors loaded or not
      L_mu_d_egg[t+1,i] <-  equals(i,1) *
                  log(
                  (1- I_egg_moy[t+1,1]) +

                  (s_egg * ((stock_egg[t+1,1] + stock_egg[t+1,2] + stock_egg[t+1,3] + stock_egg[t+1,4]) /
S_juv_JP[t+1,1] ))
                  )
                 +
```

```
                   equals(i,3) *
                    log(
                    (1- I_egg_moy[t+1,3]) +
                    (s_egg * ((stock_egg[t+1,5] +stock_egg[t+1,6]) / S_juv_JP[t+1,2] ))
                    )
  # getting out of the zone loop, one loop for each zones and the local densitie
  # to avoid using 3 dimensions matrix
}
     #-----------------------------------
 # 3.2.3 Successive removal fisheries
 #-----------------------------------
 #  I_site_juv_V/L/P = indicator for presence/absence of stocking on the site
     # loop for  sites with successive removal EF

  #................................
  # 3.2.3.1 zone 1 : Vichy Langeac
  #................................
  for (k in 1:J[t+1,1]){
    d_V[t+1,k]<- d_wild_V[t+1,k] + I_site_juv_V[t+1,k] * d_juv_V[t+1,k]
    log(d_wild_V[t+1,k])<-L_d_wild_V[t+1,k]
    L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)I(-6.91,1.09)

    log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
    L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[I_site_juv_V[t+1,k] + 1])I(-6.91,1.09)

    lambda_N_V[t+1,k]<-d_V[t+1,k]*S_depl_V[t+1,k]
    #Abundance follows a Poisson distribution
    N_tot_V[t+1,k]~dpois(lambda_N_V[t+1,k])

    L_p_V[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
    logit(p_V[t+1,k]) <-L_p_V[t+1,k]

    C_1_V[t+1,k]~dbin(p_V[t+1,k],N_tot_V[t+1,k])
    N_1_V[t+1,k]<-N_tot_V[t+1,k]-C_1_V[t+1,k]
    #not all sites have 2 pass, this vector show which sites does
      for (h in 1:pass_2_V[t+1,k]){
        C_2_V[t+1,k]~dbin(p_V[t+1,k],N_1_V[t+1,k])
        N_2_V[t+1,k]<-N_1_V[t+1,k]-C_2_V[t+1,k]
      }
  }


  #................................
  # 3.2.3.3 zone 3 : Langeac Poutes
  #................................
  for (k in 1:J[t+1,3]){
    d_L[t+1,k]<- d_wild_L[t+1,k] + I_site_juv_L[t+1,k] * d_juv_L[t+1,k]
    log(d_wild_L[t+1,k])<-L_d_wild_L[t+1,k]
    L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)I(-6.91,3)

    log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
    L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ I_site_juv_L[t+1,k] + 1])I(-6.91,1.09)

    lambda_N_L[t+1,k]<-d_L[t+1,k]*S_depl_L[t+1,k]
    #Abundance follows a Poisson distribution
    N_tot_L[t+1,k]~dpois(lambda_N_L[t+1,k])

    L_p_L[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
    logit(p_L[t+1,k]) <-L_p_L[t+1,k]

     C_1_L[t+1,k]~dbin(p_L[t+1,k],N_tot_L[t+1,k])
     N_1_L[t+1,k]<-N_tot_L[t+1,k]-C_1_L[t+1,k]
     #not all sites have 2 pass, this vector show which sites does
      for (h in 1:pass_2_L[t+1,k]){
        C_2_L[t+1,k]~dbin(p_L[t+1,k],N_1_L[t+1,k])
        N_2_L[t+1,k]<-N_1_L[t+1,k]-C_2_L[t+1,k]
        #not all sites have 2 pass, this vector show which sites does
          for (m in 1:pass_3_L[t+1,k]){
            C_3_L[t+1,k]~dbin(p_L[t+1,k],N_2_L[t+1,k])
          }
      }
  }
  #................................
  # 3.2.3.4 zone 4 : upstream Poutes
  #................................
  for (k in 1:J[t+1,4]){
    d_P[t+1,k]<- d_wild_P[t+1,k] + I_site_juv_P[t+1,k] * d_juv_P[t+1,k]
    log(d_wild_P[t+1,k])<-L_d_wild_P[t+1,k]
    L_d_wild_P[t+1,k] ~ dnorm( L_d_wild_moy[t+1,4] , tau_wild_site)I(-6.91,1.09)

    log(d_juv_P[t+1,k]) <- L_d_juv_P[t+1,k]
    L_d_juv_P[t+1,k] ~ dnorm( L_d_juv_moy[t+1,4] , tau_juv_site[ I_site_juv_P[t+1,k] + 1])I(-6.91,1.09)

    lambda_N_P[t+1,k]<-d_P[t+1,k]*S_depl_P[t+1,k]
    #Abundance follows a Poisson distribution
    N_tot_P[t+1,k]~dpois(lambda_N_P[t+1,k])

    L_p_P[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
    logit(p_P[t+1,k]) <-L_p_P[t+1,k]

    C_1_P[t+1,k]~dbin(p_P[t+1,k],N_tot_P[t+1,k])
    N_1_P[t+1,k]<-N_tot_P[t+1,k]-C_1_P[t+1,k]
    #not all sites have 2 pass, this vector show which sites does
      for (h in 1:pass_2_P[t+1,k]){
        C_2_P[t+1,k]~dbin(p_P[t+1,k],N_1_P[t+1,k])
        N_2_P[t+1,k]<-N_1_P[t+1,k]-C_2_P[t+1,k]
      }
  }
 #-----------------------------------
 # 3.2.4 5 min IA fisheries
 #-----------------------------------
  #................................
  # 3.2.4.1 zone 1 : Vichy Langeac
  #................................
  for (k in 1:K[t+1,1]){
    d_V[t+1,k]<- d_wild_V[t+1,k] + I_site_juv_V[t+1,k] * d_juv_V[t+1,k] +  I_site_egg_V[t+1,k] *
d_egg_V[t+1,k]
    log(d_wild_V[t+1,k])<-L_d_wild_V[t+1,k]
    L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)I(-6.91,3)

    log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
    L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[ I_site_juv_V[t+1,k] + 1])I(-6.91,1.09)
```

```
    log(d_egg_V[t+1,k]) <- L_d_egg_V[t+1,k]
    L_d_egg_moy_V_inc[t+1,k]<- I_site_egg_V[t+1,k] * ( L_d_egg_moy[t+1,1] +  log( S_juv_JP[t+1,1]) -
log(S_inc_JP[t+1,1]) )
    L_d_egg_V[t+1,k] ~  dnorm( L_d_egg_moy_V_inc[t+1,k] , tau_egg_site[I_site_egg_V[t+1,k] + 1])I(-
6.91,1.09)

    #5minute EF part
    lambda_IA_V[t+1,k]<-kappa_cut*d_V[t+1,k]          #kappa_cut*pow(d_V[t+1,k],eta_cut)
    EF_IA_V[t+1,k]~dpois(lambda_IA_V[t+1,k])
   }

     #.................................
   # 3.2.4.2 zone 2 : Alagnon
   #.................................
   for (k in 1:K[t+1,2]){
    d_A[t+1,k]<- d_wild_A[t+1,k] + I_site_juv_A[t+1,k] * d_juv_A[t+1,k]
    log(d_wild_A[t+1,k])<-L_d_wild_A[t+1,k]
    L_d_wild_A[t+1,k] ~ dnorm( L_d_wild_moy[t+1,2] , tau_wild_site)I(-6.91,1.09)

    log(d_juv_A[t+1,k]) <- L_d_juv_A[t+1,k]
    L_d_juv_A[t+1,k] ~ dnorm( L_d_juv_moy[t+1,2] , tau_juv_site[ I_site_juv_A[t+1,k] + 1])I(-6.91,1.09)

    #5minute EF part
    lambda_IA_A[t+1,k]<-kappa_cut*d_A[t+1,k]
    EF_IA_A[t+1,k]~dpois(lambda_IA_A[t+1,k])
   }

   #.................................
   # 3.2.4.3 zone 3 : Langeac Poutes
   #.................................
   for (k in 1:K[t+1,3]){
    d_L[t+1,k]<- d_wild_L[t+1,k] + I_site_juv_L[t+1,k] * d_juv_L[t+1,k] +  I_site_egg_L[t+1,k] *
d_egg_L[t+1,k]
    log(d_wild_L[t+1,k])<-L_d_wild_L[t+1,k]
    L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)I(-6.91,1.09)

    log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
    L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ I_site_juv_L[t+1,k] + 1])I(-6.91,1.09)

    log(d_egg_L[t+1,k]) <- L_d_egg_L[t+1,k]
    L_d_egg_moy_L_inc[t+1,k]<-  I_site_egg_L[t+1,k] * ( L_d_egg_moy[t+1,3] +  log( S_juv_JP[t+1,3]) -
log(S_inc_JP[t+1,3]) )
    L_d_egg_L[t+1,k] ~  dnorm( L_d_egg_moy_L_inc[t+1,k] , tau_egg_site[I_site_egg_L[t+1,k] + 1])I(-
6.91,1.09)

    #5minute EF part
    lambda_IA_L[t+1,k]<-kappa_cut*d_L[t+1,k]
    EF_IA_L[t+1,k]~dpois(lambda_IA_L[t+1,k])
   }
   #..................................
   # 3.2.4.4 zone 4 : upstream Poutes
   #..................................
   for (k in 1:K[t+1,4]){
    d_P[t+1,k]<- d_wild_P[t+1,k] + I_site_juv_P[t+1,k] * d_juv_P[t+1,k]
    log(d_wild_P[t+1,k])<-L_d_wild_P[t+1,k]
    L_d_wild_P[t+1,k] ~ dnorm( L_d_wild_moy[t+1,4] , tau_wild_site)I(-6.91,1.09)

    log(d_juv_P[t+1,k]) <- L_d_juv_P[t+1,k]
    L_d_juv_P[t+1,k] ~ dnorm( L_d_juv_moy[t+1,4] , tau_juv_site[ I_site_juv_P[t+1,k] + 1])I(-6.91,1.09)

    #5minute EF part
    lambda_IA_P[t+1,k]<-kappa_cut*d_P[t+1,k]
    EF_IA_P[t+1,k]~dpois(lambda_IA_P[t+1,k])
   }
 }

 ##############################################
 # 4. Take into acount the area used for stocked juveniles #
 ##############################################
 for (t in 31:T-1){
  #========================
  # 4.1 Redd/Spawners part
  #========================
  logit(p_alagnon[t])<-L_p_alagnon[t]
      logit(p_langeac[t])<- L_p_langeac[t]
  logit(p_poutes[t])<- L_p_poutes[t]
  pool_juv[t]<-s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 * smolts_tot[t] )
   L_mu_Vichy_nm[t]<-log(s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 *
smolts_tot[t] )) + level_s *I_surv[t]
  mean_y_surv[t] <- s_juv2ad * exp(level_s * I_surv[t])

  # max is only added for the year we only have a minimum figure at vichy
  temp[t]<-max(tot_C[t]  + S_stocking[t]+2,min_N_V[t])

  min_N_1[t]<-max(N[t,4]+2,temp[t] +2)+S_stocking[t]
  N[t,1]~dlnorm(L_mu_Vichy_nm[t],tau_vichy)I(min_N_1[t],15000)
  #without fish caught for breeeding or rod catches
  N_corrected[t] <- N[t,1] - tot_C[t] - S_stocking[t]
      N_corr[t]<-N_corrected[t] - N[t,2]
  res_Vichy[t] <- log(N[t,1]) - L_mu_Vichy_nm[t]

      max_N_langeac[t]<- N_corr[t] - 1
  min_L_P[t]<-max(min_L[t], N[t,4]+1)
  N[t,2]~dbin(p_alagnon[t],N_corrected[t]) #Pas de I car on pourrait avoir 0 sur alagnon et tout sur allier
une année donnée
      N[t,3]~dbin(p_langeac[t],N_corr[t])I(min_L_P[t],)

  max_N_poutes[t]<-N[t,3]-1
  N[t,4]~dbin(p_poutes[t],N[t,3])
       #------------------------------------
   # 4.1.1 Number of potential spawners
   #------------------------------------
   #mu_S_ts[t,1]<- N[t,1] - N[t,2] - S_stocking[t]
   #mu_S_ts[t,2]<- N[t,2]-N[t,3]
       #test[t]<-mu_S_ts[t,1]-S_ts[t,1]

   S_ts[t,1]<- max(N[t,1] - N[t,2] - N[t,3] - S_stocking[t] - tot_C[t] ,1)
   S_ts[t,2]<- max( N[t,2],1)
   S_ts[t,3]<- max( N[t,3]-N[t,4],1)
      S_ts[t,4]<-max( N[t,4],1)
```

```
      #ratio_S[t,1] <- S_ts[t,1] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
      #ratio_S[t,2] <- S_ts[t,2] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
      #ratio_S[t,3] <- S_ts[t,3] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
            #ratio_S[t,4] <- S_ts[t,4] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])

#==============================================================================
   # 4.2 Loop for zones (1= Vichy-Langeac, 2= Alagnon, 3= Langeac-Poutès, 4= upstream Poutès)

#==============================================================================
   for (i in 1:4){
      #--------------------------
      # 4.2.1 Redd/Spawners part
      #--------------------------
       #....................................
       # 4.2.1.1 Estimation of the spawners
       #....................................
       R[t,i]~dpois(lambda[t,i])
       lambda[t,i] <- S_ts[t,i]  *zone_effect[t,i]* hel_effect[2] *p_area[t,i]
       res_R[t,i]<-(R[t,i]-lambda[t,i])/sqrt(lambda[t,i])
       #.............................
       # 4.2.1.2 Cut of all parameters
       #.............................
       lambda_cut[t,i]<-cut(lambda[t,i])
       R_rep[t,i]~dpois(lambda_cut[t,i])

       #S_counter[t,i]<-R[t,i] / (zone_effect[t,i] *p_area[t,i])
       #chisq_disc_R[t,i]<- (R[t,i]-lambda[t,i]) * (R[t,i]-lambda[t,i]) / (lambda[t,i])
       #chisq_disc_R_rep[t,i]<- (R_rep[t,i]-lambda[t,i]) * (R_rep[t,i]-lambda[t,i]) / (lambda[t,i])
      #--------------------------
      # 4.2.2 Juvenile production
      #--------------------------
      # I_juv_moy = indicator for stocking of 0+ or not
      # I_egg_moy = indicator for stocking of eggs or not
            #d_tot_moy with taking into acount area for the stocked juveniles (data only from year 31)
            d_tot_moy[t+1,i] <- d_wild_moy[t+1,i] + I_juv_moy[t+1,i] *
d_juv_moy[t+1,i]*S_juv_JP_dev[t+1,i]/S_juv_JP[t+1,i]  + I_egg_moy[t+1,i] * d_egg_moy_surf[t+1,i]
            Juv[t+1,i] <- d_tot_moy[t+1,i]*S_juv_JP[t+1,i]
            d_egg_moy_surf[t+1,i] <- d_egg_moy[t+1,i]
       #.........................
       # 4.2.2.1 Wild component
       #.........................
       log(d_wild_moy[t+1,i]) <- L_d_wild_moy[t+1,i]
       L_d_wild_moy[t+1,i] ~ dnorm(L_mu_d_wild[t+1,i],tau_wild_moy)I(-6.91,1.09)          #<-
L_mu_d_wild[t+1,i]        #
       L_mu_d_wild[t+1,i] <-  log((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd * (S_ts[t,i]/S_juv_JP[t,i]))) +
nu_wild[i]
              res_wild_moy[t+1,i] <-  L_d_wild_moy[t+1,i] -  L_mu_d_wild[t+1,i]
              #...................................
       # 4.2.2.2 Stocked juvenile component
       #...................................
       log(d_juv_moy[t+1,i]) <- L_d_juv_moy[t+1,i]
       L_d_juv_moy[t+1,i] ~ dnorm(L_mu_d_juv[t+1,i],tau_juv_moy[ I_juv_moy[t+1,i]+1])I(,1.09)

       # We recaculate the Rmax "available" to stocked 0+  by substracting wild 0+ density and stocked
eggs density
       # to the total Rmax of the density dependence relationship
       Rmax_juv_temp[t+1,i] <- (  Rmax - ((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd *
(S_ts[t,i]/S_juv_JP[t,i]))) ) * exp(nu_wild[i])
       Rmax_juv[t+1,i] <- max(Rmax_juv_temp[t+1,i] ,0.000001)
            beta_dd_juv[t+1,i] <- 1 / Rmax_juv[t+1,i]
            L_mu_d_juv[t+1,i] <-  I_juv_moy[t+1,i] * log( (stock_juv[t+1,i]/S_juv_JP[t+1,i]) /
(alpha_dd_juv/exp(nu_wild[i]) +  beta_dd_juv[t+1,i] * (stock_juv[t+1,i]/S_juv_JP[t+1,i])))
       res_juv_moy[t+1,i] <- L_d_juv_moy[t+1,i] - L_mu_d_juv[t+1,i]
            #...............................
       # 4.2.2.3 Stocked egg component
       #...............................
       log(d_egg_moy[t+1,i]) <- L_d_egg_moy[t+1,i]
       L_d_egg_moy[t+1,i]~  dnorm(L_mu_d_egg[t+1,i],tau_egg_moy[ I_egg_moy[t+1,i] +1])I(-6.91,1.09)  #
       res_egg_moy[t+1,i] <- L_d_egg_moy[t+1,i] - L_mu_d_egg[t+1,i]

       # I_egg_unit = indicator of presence of incubators or not: only zone 1 and 2 concerned
       # I_egg_VL = indicator for incubators in zone 1
       # I_egg_LP = indicator for incubators in zone 2
       # I_list_inc = indicator for each incabutors loaded or not
       L_mu_d_egg[t+1,i] <-  equals(i,1) *
                   log(
                   (1- I_egg_moy[t+1,1]) +

                   (s_egg * ((stock_egg[t+1,1] + stock_egg[t+1,2] + stock_egg[t+1,3] + stock_egg[t+1,4]) /
S_juv_JP[t+1,1] ))
                   )
                  +
                  equals(i,3) *
                   log(
                   (1- I_egg_moy[t+1,3]) +
                   (s_egg * ((stock_egg[t+1,5] +stock_egg[t+1,6]) / S_juv_JP[t+1,2] ))
                   )
       # getting out of the zone loop, one loop for each zones and the local densitie
       # to avoid using 3 dimensions matrix
   }
          #-----------------------------------
     # 4.2.3 Successive removal fisheries
     #-----------------------------------
     #  I_site_juv_V/L/P = indicator for presence/absence of stocking on the site
          # loop for  sites with successive removal EF

       #................................
       # 4.2.3.1 zone 1 : Vichy Langeac
       #................................
       for (k in 1:J[t+1,1]){
        d_V[t+1,k]<- d_wild_V[t+1,k] + I_site_juv_V[t+1,k] * d_juv_V[t+1,k]
        log(d_wild_V[t+1,k])<-L_d_wild_V[t+1,k]
        L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)I(-6.91,1.09)

        log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
        L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[I_site_juv_V[t+1,k] + 1])I(-6.91,1.09)

        lambda_N_V[t+1,k]<-d_V[t+1,k]*S_depl_V[t+1,k]
        #Abundance follows a Poisson distribution
        N_tot_V[t+1,k]~dpois(lambda_N_V[t+1,k])

        L_p_V[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
```

```
  logit(p_V[t+1,k]) <-L_p_V[t+1,k]

  C_1_V[t+1,k]~dbin(p_V[t+1,k],N_tot_V[t+1,k])
  N_1_V[t+1,k]<-N_tot_V[t+1,k]-C_1_V[t+1,k]
  #not all sites have 2 pass, this vector show which sites does
   for (h in 1:pass_2_V[t+1,k]){
     C_2_V[t+1,k]~dbin(p_V[t+1,k],N_1_V[t+1,k])
     N_2_V[t+1,k]<-N_1_V[t+1,k]-C_2_V[t+1,k]
   }
}

 #................................
# 4.2.3.3 zone 3 : Langeac Poutes
#................................
for (k in 1:J[t+1,3]){
  d_L[t+1,k]<- d_wild_L[t+1,k] + I_site_juv_L[t+1,k] * d_juv_L[t+1,k]
  log(d_wild_L[t+1,k])<-L_d_wild_L[t+1,k]
  L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)I(-6.91,3)

  log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
  L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ I_site_juv_L[t+1,k] + 1])I(-6.91,1.09)

  lambda_N_L[t+1,k]<-d_L[t+1,k]*S_depl_L[t+1,k]
  #Abundance follows a Poisson distribution
  N_tot_L[t+1,k]~dpois(lambda_N_L[t+1,k])

  L_p_L[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
  logit(p_L[t+1,k]) <-L_p_L[t+1,k]

   C_1_L[t+1,k]~dbin(p_L[t+1,k],N_tot_L[t+1,k])
   N_1_L[t+1,k]<-N_tot_L[t+1,k]-C_1_L[t+1,k]
   #not all sites have 2 pass, this vector show which sites does
    for (h in 1:pass_2_L[t+1,k]){
      C_2_L[t+1,k]~dbin(p_L[t+1,k],N_1_L[t+1,k])
      N_2_L[t+1,k]<-N_1_L[t+1,k]-C_2_L[t+1,k]
      #not all sites have 2 pass, this vector show which sites does
        for (m in 1:pass_3_L[t+1,k]){
          C_3_L[t+1,k]~dbin(p_L[t+1,k],N_2_L[t+1,k])
        }
    }
}
#................................
# 4.2.3.4 zone 4 : upstream Poutes
#................................
for (k in 1:J[t+1,4]){
  d_P[t+1,k]<- d_wild_P[t+1,k] + I_site_juv_P[t+1,k] * d_juv_P[t+1,k]
  log(d_wild_P[t+1,k])<-L_d_wild_P[t+1,k]
  L_d_wild_P[t+1,k] ~ dnorm( L_d_wild_moy[t+1,4] , tau_wild_site)I(-6.91,1.09)

  log(d_juv_P[t+1,k]) <- L_d_juv_P[t+1,k]
  L_d_juv_P[t+1,k] ~ dnorm( L_d_juv_moy[t+1,4] , tau_juv_site[ I_site_juv_P[t+1,k] + 1])I(-6.91,1.09)

  lambda_N_P[t+1,k]<-d_P[t+1,k]*S_depl_P[t+1,k]
  #Abundance follows a Poisson distribution
  N_tot_P[t+1,k]~dpois(lambda_N_P[t+1,k])
```

```
    L_p_P[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
    logit(p_P[t+1,k]) <-L_p_P[t+1,k]

    C_1_P[t+1,k]~dbin(p_P[t+1,k],N_tot_P[t+1,k])
    N_1_P[t+1,k]<-N_tot_P[t+1,k]-C_1_P[t+1,k]
    #not all sites have 2 pass, this vector show which sites does
     for (h in 1:pass_2_P[t+1,k]){
       C_2_P[t+1,k]~dbin(p_P[t+1,k],N_1_P[t+1,k])
       N_2_P[t+1,k]<-N_1_P[t+1,k]-C_2_P[t+1,k]
     }
   }
  #-----------------------------------
  # 4.2.4 5 min IA fisheries
  #-----------------------------------
   #................................
   # 4.2.4.1 zone 1 : Vichy Langeac
   #................................
   for (k in 1:K[t+1,1]){
    d_V[t+1,k]<- d_wild_V[t+1,k] + I_site_juv_V[t+1,k] * d_juv_V[t+1,k] +  I_site_egg_V[t+1,k] *
d_egg_V[t+1,k]
    log(d_wild_V[t+1,k])<-L_d_wild_V[t+1,k]
    L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)I(-6.91,3)

    log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
    L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[ I_site_juv_V[t+1,k] + 1])I(-6.91,1.09)

    log(d_egg_V[t+1,k]) <- L_d_egg_V[t+1,k]
    L_d_egg_moy_V_inc[t+1,k]<- I_site_egg_V[t+1,k] * ( L_d_egg_moy[t+1,1] +  log( S_juv_JP[t+1,1]) -
log(S_inc_JP[t+1,1]) )
    L_d_egg_V[t+1,k] ~  dnorm( L_d_egg_moy_V_inc[t+1,k] , tau_egg_site[I_site_egg_V[t+1,k] + 1])I(-
6.91,1.09)

    #5minute EF part
    lambda_IA_V[t+1,k]<-kappa_cut*d_V[t+1,k]          #kappa_cut*pow(d_V[t+1,k],eta_cut)
    EF_IA_V[t+1,k]~dpois(lambda_IA_V[t+1,k])
   }

       #................................
   # 4.2.4.2 zone 2 : Alagnon
   #................................
   for (k in 1:K[t+1,2]){
    d_A[t+1,k]<- d_wild_A[t+1,k] + I_site_juv_A[t+1,k] * d_juv_A[t+1,k]
    log(d_wild_A[t+1,k])<-L_d_wild_A[t+1,k]
    L_d_wild_A[t+1,k] ~ dnorm( L_d_wild_moy[t+1,2] , tau_wild_site)I(-6.91,1.09)

    log(d_juv_A[t+1,k]) <- L_d_juv_A[t+1,k]
    L_d_juv_A[t+1,k] ~ dnorm( L_d_juv_moy[t+1,2] , tau_juv_site[ I_site_juv_A[t+1,k] + 1])I(-6.91,1.09)

    #5minute EF part
    lambda_IA_A[t+1,k]<-kappa_cut*d_A[t+1,k]
    EF_IA_A[t+1,k]~dpois(lambda_IA_A[t+1,k])
   }


   #................................
   # 4.2.4.3 zone 3 : Langeac Poutes
```

```
        #.................................
        for (k in 1:K[t+1,3]){
          d_L[t+1,k]<- d_wild_L[t+1,k] + I_site_juv_L[t+1,k] * d_juv_L[t+1,k] +  I_site_egg_L[t+1,k] *
d_egg_L[t+1,k]
          log(d_wild_L[t+1,k])<-L_d_wild_L[t+1,k]
          L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)I(-6.91,1.09)

          log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
          L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ I_site_juv_L[t+1,k] + 1])I(-6.91,1.09)

          log(d_egg_L[t+1,k]) <- L_d_egg_L[t+1,k]
          L_d_egg_moy_L_inc[t+1,k]<-  I_site_egg_L[t+1,k] * ( L_d_egg_moy[t+1,3] +  log( S_juv_JP[t+1,3]) -
log(S_inc_JP[t+1,3]) )
          L_d_egg_L[t+1,k] ~  dnorm( L_d_egg_moy_L_inc[t+1,k] , tau_egg_site[I_site_egg_L[t+1,k] + 1])I(-
6.91,1.09)

          #5minute EF part
          lambda_IA_L[t+1,k]<-kappa_cut*d_L[t+1,k]
          EF_IA_L[t+1,k]~dpois(lambda_IA_L[t+1,k])
        }

        #..................................
        # 4.2.4.4 zone 4 : upstream Poutes
        #..................................
        for (k in 1:K[t+1,4]){
          d_P[t+1,k]<- d_wild_P[t+1,k] + I_site_juv_P[t+1,k] * d_juv_P[t+1,k]
          log(d_wild_P[t+1,k])<-L_d_wild_P[t+1,k]
          L_d_wild_P[t+1,k] ~ dnorm( L_d_wild_moy[t+1,4] , tau_wild_site)I(-6.91,1.09)

          log(d_juv_P[t+1,k]) <- L_d_juv_P[t+1,k]
          L_d_juv_P[t+1,k] ~ dnorm( L_d_juv_moy[t+1,4] , tau_juv_site[ I_site_juv_P[t+1,k] + 1])I(-6.91,1.09)

          #5minute EF part
          lambda_IA_P[t+1,k]<-kappa_cut*d_P[t+1,k]
          EF_IA_P[t+1,k]~dpois(lambda_IA_P[t+1,k])
        }
  }

  ##########################################
  # 5. Just the last year to estimate spawners       #
  ##########################################
  for (t in T:T){
    #=========================
    # 5.1 Redd/Spawners part
    #=========================
     logit(p_alagnon[t])<-L_p_alagnon[t]
             logit(p_langeac[t])<- L_p_langeac[t]
     logit(p_poutes[t])<- L_p_poutes[t]
     pool_juv[t]<-s_juv2ad* Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 * smolts_tot[t] )
     L_mu_Vichy_nm[t]<-log(s_juv2ad *Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 *
smolts_tot[t] )) + level_s *I_surv[t]

     min_N_1[t]<-max(N[t,4]+2,tot_C[t] +2)+S_stocking[t]
     N[t,1]~dlnorm(L_mu_Vichy_nm[t],tau_vichy)I(min_N_1[t],15000)
     #without fish caught for breeeding
             N_corrected[t]<-N[t,1]-S_stocking[t]
             N_corr[t]<-N_corrected[t] - N[t,2]
             res_Vichy[t] <- log(N[t,1]) - L_mu_Vichy_nm[t]

             max_N_langeac[t]<- N_corr[t] -1
     min_L_P[t]<-N[t,4]+1

             N[t,2]~dbin(p_alagnon[t],N_corrected[t]) #Pas de I car on pourrait avoir 0 sur alagnon et tout sur
allier une année donnée
            N[t,3]~dbin(p_langeac[t],N_corr[t])I(min_L_P[t],)

             max_N_poutes[t]<-N[t,3]-1
             N[t,4]~dbin(p_poutes[t],N[t,3])

             #-------------------------------------
      # 5.1.1 Number of potential spawners
      #-------------------------------------
      S_ts[t,1]<- max(N[t,1] - N[t,2] - N[t,3] - S_stocking[t] - tot_C[t] ,1)
      S_ts[t,2]<- max( N[t,2],1)
      S_ts[t,3]<- max( N[t,3]-N[t,4],1)
            S_ts[t,4]<-max( N[t,4],1)

      #ratio_S[t,1] <- S_ts[t,1] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
      #ratio_S[t,2] <- S_ts[t,2] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
      #ratio_S[t,3] <- S_ts[t,3] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
              #ratio_S[t,4] <- S_ts[t,4] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])

             #===========================================================================
     # 5.2 Loop for zones (1= Vichy-Langeac, 2= Langeac-Poutès, 3= upstream Poutès)
     #===========================================================================
     for (i in 1:4){
       #---------------------------
       # 5.2.1 Redd/Spawners part
       #---------------------------
         #.....................................
         # 5.2.1.1 Estimation of the spawners
         #.....................................
         R[t,i]~dpois(lambda[t,i])
         lambda[t,i] <- S_ts[t,i]  *zone_effect[t,i] * hel_effect[2]*p_area[t,i]
         res_R[t,i]<-(R[t,i]-lambda[t,i])/sqrt(lambda[t,i])
         #................................
         # 5.2.1.2 Cut of all parameters
         #................................
         lambda_cut[t,i]<-cut(lambda[t,i])
         R_rep[t,i]~dpois(lambda_cut[t,i])

         #S_counter[t,i]<-R[t,i] / (zone_effect[t,i] *p_area[t,i])
                 #chisq_disc_R[t,i]<- (R[t,i]-lambda[t,i]) * (R[t,i]-lambda[t,i]) / (lambda[t,i])
         #chisq_disc_R_rep[t,i]<- (R_rep[t,i]-lambda[t,i]) * (R_rep[t,i]-lambda[t,i]) / (lambda[t,i])
      #---------------------------------------
      # 5.2.2 Juvenile production (wild only)
      #---------------------------------------
      d_tot_moy[t+1,i] <- d_wild_moy[t+1,i]
      Juv[t+1,i] <- d_tot_moy[t+1,i]*S_juv_JP[t+1,i]
        #........................
        # 5.2.2.1 Wild component
        #........................
```

```
                log(d_wild_moy[t+1,i]) <- L_d_wild_moy[t+1,i]
                L_d_wild_moy[t+1,i] ~ dnorm(L_mu_d_wild[t+1,i],tau_wild_moy)I(-6.91,1.09)
        L_mu_d_wild[t+1,i] <-  log((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd * (S_ts[t,i]/S_juv_JP[t,i]))) +
nu_wild[i]
                res_wild_moy[t+1,i] <-  L_d_wild_moy[t+1,i] -  L_mu_d_wild[t+1,i]
    }
  }


### END MODEL BRACKET
}
```

L'Europe c'est ici.
L'Europe c'est maintenant.

# PROGRAMME D'ACTIONS EN FAVEUR DES POISSONS MIGRATEURS

## Tableaux de bord Migrateurs du Bassin Loire


Le programme Tableaux de bord Migrateurs en faveur des poissons migrateurs est cofinancé par l'Union européenne.
L'Europe s'engage dans le bassin de la Loire avec le Fonds européen de développement régional.